1.2 The Basic Principle of Counting

The basic principle of counting will be fundamental to all our work. Loosely put, it states that if one experiment can result in any of m possible outcomes and if another experiment can result in any of n possible outcomes, then there are *mn* possible outcomes of the two experiments.

> The basic principle of counting
>
> Suppose that two experiments are to be performed. Then if experiment 1 can result in any one of m possible outcomes and if, for each outcome of experiment 1, there are n possible outcomes of experiment 2, then together there are *mn* possible outcomes of the two experiments.

Proof of the Basic Principle: The basic principle may be proven by enumerating all the possible outcomes of the two experiments; that is,

$$(1,1),(1,2),\ldots,(1,n)(2,1),(2,2),\ldots,(2,n):(m,1),(m,2),\ldots,(m,n)$$

where we say that the outcome is (i, j) if experiment 1 results in its ith possible outcome and experiment 2 then results in its jth possible outcome. Hence, the set of possible outcomes consists of m rows, each containing n elements. This proves the result.

Example 2a

A small community consists of 10 women, each of whom has 3 children. If one woman and one of her children are to be chosen as mother and child of the year, how many different choices are possible?

Solution

By regarding the choice of the woman as the outcome of the first experiment and the subsequent choice of one of her children as the outcome of the second experiment, we see from the basic principle that there are $10\times3=30$ possible choices.

When there are more than two experiments to be performed, the basic principle can be generalized.

> The generalized basic principle of counting
>
> If r experiments that are to be performed are such that the first one may result in any of n1 possible outcomes; and if, for each of these n1 possible outcomes, there are n2 possible outcomes of the second experiment; and if, for each of the possible outcomes of the first two experiments, there are n3 possible outcomes of the third experiment; and if, then there is a total of $n1\cdot n2\cdots nr$ possible outcomes of the r experiments.

Example 2b

A college planning committee consists of 3 freshmen, 4 sophomores, 5 juniors, and 2 seniors. A subcommittee of 4, consisting of 1 person from each class, is to be chosen. How many different subcommittees are possible?

Solution

We may regard the choice of a subcommittee as the combined outcome of the four separate experiments of choosing a single representative from each of the classes. It then follows from the generalized version of the basic principle that there are $3\times4\times5\times2=120$ possible subcommittees.

Example 2c

How many different 7-place license plates are possible if the first 3 places are to be occupied by letters and the final 4 by numbers?

Solution

By the generalized version of the basic principle, the answer is $26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 \cdot 10 = 175,760,000$.

Example 2d

How many functions defined on n points are possible if each functional value is either 0 or 1?

Solution

Let the points be 1,2,. . .,n. Since f(i) must be either 0 or 1 for each i=1,2,. . .,n, it follows that there are 2n possible functions.

Example 2e

In **Example 2c**, how many license plates would be possible if repetition among letters or numbers were prohibited?

Solution

In this case, there would be $26 \cdot 25 \cdot 24 \cdot 10 \cdot 9 \cdot 8 \cdot 7 = 78,624,000$ possible license plates.

1.3 Permutations

How many different ordered arrangements of the letters a, b, and c are possible? By direct enumeration we see that there are 6, namely, *abc, acb, bac, bca, cab*, and *cba*. Each arrangement is known as a *permutation*. Thus, there are 6 possible permutations of a set of 3 objects. This result could also have been obtained from the basic principle, since the first object in the permutation can be any of the 3, the second object in the permutation can then be chosen from any of the remaining 2, and the third object in the permutation is then the remaining 1. Thus, there are $3 \cdot 2 \cdot 1 = 6$ possible permutations.

Suppose now that we have n objects. Reasoning similar to that we have just used for the 3 letters then shows that there are

$$n(n-1)(n-2)\cdots 3 \cdot 2 \cdot 1 = n!$$

different permutations of the n objects.

Whereas n! (read as "n factorial") is defined to equal $1 \cdot 2 \cdots n$ when n is a positive integer, it is convenient to define 0! to equal 1.

Example 3a

How many different batting orders are possible for a baseball team consisting of 9 players?

Solution

There are $9! = 362{,}880$ possible batting orders.

Example 3b

A class in probability theory consists of 6 men and 4 women. An examination is given, and the students are ranked according to their performance. Assume that no two students obtain the same score.

a. How many different rankings are possible?
b. If the men are ranked just among themselves and the women just among themselves, how many different rankings are possible?

Solution

a. (a) Because each ranking corresponds to a particular ordered arrangement of the 10 people, the answer to this part is $10! = 3{,}628{,}800$.
b. (b) Since there are 6! possible rankings of the men among themselves and 4! possible rankings of the women among themselves, it follows from the basic principle that there are $(6!)(4!) = (720)(24) = 17{,}280$ possible rankings in this case.

Example 3c

Ms. Jones has 10 books that she is going to put on her bookshelf. Of these, 4 are mathematics books, 3 are chemistry books, 2 are history books, and 1 is a language book. Ms. Jones wants to arrange her books so that all the books dealing with the same subject are together on the shelf. How many different arrangements are possible?

Solution

There are $4!\,3!\,2!\,1!$ arrangements such that the mathematics books are first in line, then the chemistry books, then the history books, and then the language book. Similarly, for each possible ordering of the subjects, there are $4!\,3!\,2!\,1!$ possible arrangements. Hence, as there are 4! possible orderings of the subjects, the desired answer is $4!\,4!\,3!\,2!\,1! = 6912$.

We shall now determine the number of permutations of a set of n objects when certain of the objects are indistinguishable from one another. To set this situation straight in our minds, consider the following example.

Example 3d

How many different letter arrangements can be formed from the letters PEPPER?

Solution

We first note that there are 6! permutations of the letters P1E1P2P3E2R when the 3P's and the 2E's are distinguished from one another. However, consider any one of these permutations for instance, P1P2E1P3E2R. If we now permute the P's among themselves and the E's among themselves, then the resultant arrangement would still be of the form PPEPER. That is, all $3!\,2!$ permutations

P1P2E1P3E2RP1P2E2P3E1RP1P3E1P2E2RP1P3E2P2E1RP2P1E1P3E2RP2P1E2P3E1RP2P3E1P1E2RP2P3E2P1E1RP3P1E1P2E2RP3P1E2P2E1RP3P2E1P1E2RP3P2E2P1E1R

are of the form PPEPER. Hence, there are $6!/(3!\,2!) = 60$ possible letter arrangements of the letters PEPPER.

In general, the same reasoning as that used in **Example 3d** shows that there are

$$n!n1!\,n2!\cdots nr!$$

different permutations of n objects, of which n1 are alike, n2 are alike, . . .,nr are alike.

Example 3e

A chess tournament has 10 competitors, of which 4 are Russian, 3 are from the United States, 2 are from Great Britain, and 1 is from Brazil. If the tournament result lists just the nationalities of the players in the order in which they placed, how many outcomes are possible?

Solution

There are

$$10!4!\,3!\,2!\,1! = 12{,}600$$

possible outcomes.

Example 3f

How many different signals, each consisting of 9 flags hung in a line, can be made from a set of 4 white flags, 3 red flags, and 2 blue flags if all flags of the same color are identical?

Solution

There are

$9!4!\,3!\,2!=1260$

different signals.

1.4 Combinations

We are often interested in determining the number of different groups of r objects that could be formed from a total of n objects. For instance, how many different groups of 3 could be selected from the 5 items A, B, C, D, and E? To answer this question, reason as follows: Since there are 5 ways to select the initial item, 4 ways to then select the next item, and 3 ways to select the final item, there are thus 5·4·3 ways of selecting the group of 3 when the order in which the items are selected is relevant. However, since every group of 3–say, the group consisting of items A, B, and C will be counted 6 times (that is, all of the permutations *ABC, ACB, BAC, BCA, CAB*, and *CBA* will be counted when the order of selection is relevant), it follows that the total number of groups that can be formed is

$5 \cdot 4 \cdot 33 \cdot 2 \cdot 1 = 10$

In general, as $n(n-1)\cdots(n-r+1)$ represents the number of different ways that a group of r items could be selected from n items when the order of selection is relevant, and as each group of r items will be counted r! times in this count, it follows that the number of different groups of r items that could be formed from a set of n items is

$n(n-1)\cdots(n-r+1)r! = n!(n-r)!\, r!$

> Notation and terminology
>
> We define $\binom{n}{r}$, for $r \le n$, by
>
> $\binom{n}{r} = n!(n-r)!\, r!$
>
> and say that $\binom{n}{r}$ (read as “n choose r“) represents the number of possible combinations of n objects taken r at a time.

Thus, $\binom{n}{r}$ represents the number of different groups of size r that could be selected from a set of n objects when the order of selection is not considered relevant.

Equivalently, $\binom{n}{r}$ is the number of subsets of size r that can be chosen from a set of size n. Using that $0! = 1$, note that $\binom{n}{n} = \binom{n}{0} = n!0!n! = 1$, which is consistent with the preceding interpretation because in a set of size n there is exactly 1 subset of size n (namely, the entire set), and exactly one subset of size 0 (namely the empty set). A useful convention is to define $\binom{n}{r}$ equal to 0 when either $r > n$ or $r < 0$.

Example 4a

A committee of 3 is to be formed from a group of 20 people. How many different committees are possible?

Solution

There are $\binom{20}{3} = 20 \cdot 19 \cdot 183 \cdot 2 \cdot 1 = 1140$ possible committees.

Example 4b

From a group of 5 women and 7 men, how many different committees consisting of 2 women and 3 men can be formed? What if 2 of the men are feuding and refuse to serve on the committee together?

Solution

As there are $\binom{5}{2}$ possible groups of 2 women, and $\binom{7}{3}$ possible groups of 3 men, it follows from the basic principle that there are $\binom{5}{2}\binom{7}{3} = 5 \cdot 42 \cdot 1.\ 7 \cdot 6 \cdot 53 \cdot 2 \cdot 1 = 350$ possible committees consisting of 2 women and 3 men.

Now suppose that 2 of the men refuse to serve together. Because a total of $\binom{2}{2}\binom{5}{1} = 5$ out of the $\binom{7}{3} = 35$ possible groups of 3 men contain both of the feuding men, it follows that there are $35 - 5 = 30$ groups that do not contain both of the feuding men. Because there are still $\binom{5}{2} = 10$ ways to choose the 2 women, there are $30 \cdot 10 = 300$ possible committees in this case.

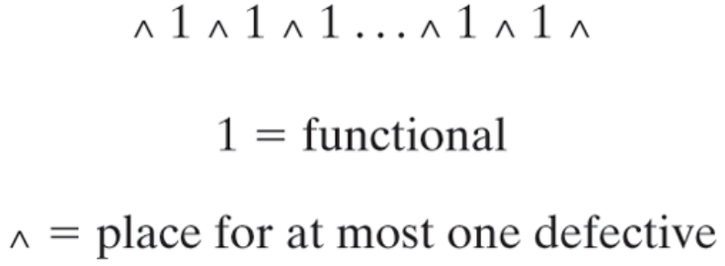
Example 4c

Consider a set of n antennas of which m are defective and $n-m$ are functional and assume that all of the defectives and all of the functionals are considered indistinguishable. How many linear orderings are there in which no two defectives are consecutive?

Solution

Imagine that the $n-m$ functional antennas are lined up among themselves. Now, if no two defectives are to be consecutive, then the spaces between the functional antennas must each contain at most one defective antenna. That is, in the $n-m+1$ possible positions–represented in **Figure 1.1** by carets–between the $n-m$ functional antennas, we must select m of these in which to put the defective antennas. Hence, there are $\binom{n-m+1}{m}$ possible orderings in which there is at least one functional antenna between any two defective ones.

Figure 1.1 No consecutive defectives.



A useful combinatorial identity, known as *Pascal's identity*, is

$$\binom{n}{r}=\binom{n-1}{r-1}+\binom{n-1}{r} \quad 1\le r\le n \tag{4.1}$$

Equation (4.1) may be proved analytically or by the following combinatorial argument: Consider a group of n objects, and fix attention on some particular one of these objects–call it object 1. Now, there are $\binom{n-1}{r-1}$ groups of size r that contain object 1 (since each such group is formed by selecting r−1 from the remaining n−1 objects). Also, there are $\binom{n-1}{r}$ groups of size r that do not contain object 1. As there is a total of $\binom{n}{r}$ groups of size r, **Equation (4.1)** follows.

The values $\binom{n}{r}$ are often referred to as *binomial coefficients* because of their prominence in the binomial theorem.

> The binomial theorem
>
> $$(x+y)^n=\sum_{k=0}^{n}\binom{n}{k}x^k y^{n-k} \tag{4.2}$$

We shall present two proofs of the binomial theorem. The first is a proof by mathematical induction, and the second is a proof based on combinatorial considerations.

Proof of the Binomial Theorem by Induction: When n=1, **Equation (4.2)** reduces to

$$x+y=\binom{1}{0}x^0y^1+\binom{1}{1}x^1y^0=y+x$$

Assume **Equation (4.2)** for n−1. Now,

$$(x+y)^n=(x+y)(x+y)^{n-1}=(x+y)\sum_{k=0}^{n-1}\binom{n-1}{k}x^k y^{n-1-k}=\sum_{k=0}^{n-1}\binom{n-1}{k}x^{k+1}y^{n-1-k}+\sum_{k=0}^{n-1}\binom{n-1}{k}x^k y^{n-k}$$

Letting i=k+1 in the first sum and i=k in the second sum, we find that

$$(x+y)^n=\sum_{i=1}^{n}\binom{n-1}{i-1}x^i y^{n-i}+\sum_{i=0}^{n-1}\binom{n-1}{i}x^i y^{n-i}=\sum_{i=1}^{n-1}\binom{n-1}{i-1}x^i y^{n-i}+x^n+y^n+\sum_{i=1}^{n-1}\binom{n-1}{i}x^i y^{n-i}=x^n+\sum_{i=1}^{n-1}\left[\binom{n-1}{i-1}+\binom{n-1}{i}\right]x^i y^{n-i}+y^n$$
$$=x^n+\sum_{i=1}^{n-1}\binom{n}{i}x^i y^{n-i}+y^n=\sum_{i=0}^{n}\binom{n}{i}x^i y^{n-i}$$

where the next-to-last equality follows by **Equation (4.1)**. By induction, the theorem is now proved.

Combinatorial Proof of the Binomial Theorem: Consider the product

$$(x_1+y_1)(x_2+y_2)\cdots(x_n+y_n)$$

Its expansion consists of the sum of 2n terms, each term being the product of n factors. Furthermore, each of the 2n terms in the sum will contain as a factor either x_i or y_i for each i=1, 2,. . .,n. For example,

$$(x_1+y_1)(x_2+y_2)=x_1x_2+x_1y_2+y_1x_2+y_1y_2$$

Now, how many of the 2n terms in the sum will have k of the x_i's and (n−k) of the y_i's as factors? As each term consisting of k of the x_i's and (n−k) of the y_i's corresponds to a choice of a group of k from the n values $x_1, x_2\ldots, x_n$, there are $\binom{n}{k}$ such terms. Thus, letting $x_i=x$, $y_i=y$, i=1,. . .,n, we see that

$$(x+y)^n=\sum_{k=0}^{n}\binom{n}{k}x^k y^{n-k}$$

Example 4d

Expand $(x+y)^3$.

Solution

$$(x+y)^3=\binom{3}{0}x^0y^3+\binom{3}{1}x^1y^2+\binom{3}{2}x^2y^1+\binom{3}{3}x^3y^0=y^3+3xy^2+3x^2y+x^3$$

Example 4e

How many subsets are there of a set consisting of n elements?

Solution

Since there are (nk) subsets of size k, the desired answer is

$\sum_{k=0}^{n}\binom{n}{k}=(1+1)^n=2^n$

This result could also have been obtained by assigning either the number 0 or the number 1 to each element in the set. To each assignment of numbers, there corresponds, in a one-to-one fashion, a subset, namely, that subset consisting of all elements that were assigned the value 1. As there are 2n possible assignments, the result follows.

Note that we have included the set consisting of 0 elements (that is, the null set) as a subset of the original set. Hence, the number of subsets that contain at least 1 element is 2^n-1.

1.5 Multinomial Coefficients

In this section, we consider the following problem: A set of n distinct items is to be divided into r distinct groups of respective sizes n1,n2,. . .,nr, where ∑i=1rni=n. How many different divisions are possible? To answer this question, we note that there are (nn1) possible choices for the first group; for each choice of the first group, there are (n−n1n2) possible choices for the second group; for each choice of the first two groups, there are (n−n1−n2n3) possible choices for the third group; and so on. It then follows from the generalized version of the basic counting principle that there are

(nn1)(n−n1n2)⋯(n−n1−n2−⋯−nr−1nr)=n!(n−n1)! n1!(n−n1)!(n−n1−n2)! n2!⋯(n−n1−n2−⋯−nr−1)!0! nr!=n!n1! n2!⋯nr!

possible divisions.

Another way to see this result is to consider the n values 1,1,. . .,1,2,. . .,2,. . ., r,. . .,r, where i appears ni times, for i=1,. . .,r. Every permutation of these values corresponds to a division of the n items into the r groups in the following manner: Let the permutation i1,i2,. . .,in correspond to assigning item 1 to group i1, item 2 to group i2, and so on. For instance, if n=8 and if n1=4,n2=3, and n3=1, then the permutation 1, 1, 2, 3, 2, 1, 2, 1 corresponds to assigning items 1, 2, 6, 8 to the first group, items 3, 5, 7 to the second group, and item 4 to the third group. Because every permutation yields a division of the items and every possible division results from some permutation, it follows that the number of divisions of n items into r distinct groups of sizes n1,n2,. . .,nr is the same as the number of permutations of n items of which n1 are alike, and n2 are alike, . . ., and nr are alike, which was shown in **Section 1.3** to equal n!n1!n2!⋯nr!.

> Notation
>
> If n1+n2+⋯+nr=n, we define (nn1,n2,. . .,nr) by
>
> (nn1,n2,. . .,nr)=n!n1! n2!⋯nr!
>
> Thus, (nn1,n2,. . .,nr) represents the number of possible divisions of n distinct objects into r distinct groups of respective sizes n1,n2,. . .,nr.

Example 5a

A police department in a small city consists of 10 officers. If the department policy is to have 5 of the officers patrolling the streets, 2 of the officers working full time at the station, and 3 of the officers on reserve at the station, how many different divisions of the 10 officers into the 3 groups are possible?

Solution

There are 10!5! 2! 3!=2520 possible divisions.

Example 5b

Ten children are to be divided into an A team and a B team of 5 each. The A team will play in one league and the B team in another. How many different divisions are possible?

Solution

There are 10!5! 5!=252 possible divisions.

Example 5c

In order to play a game of basketball, 10 children at a playground divide themselves into two teams of 5 each. How many different divisions are possible?

Solution

Note that this example is different from **Example 5b** because now the order of the two teams is irrelevant. That is, there is no A or B team, but just a division consisting of 2 groups of 5 each. Hence, the desired answer is

10!/(5! 5!)2!=126

The proof of the following theorem, which generalizes the binomial theorem, is left as an exercise.

> The multinomial theorem
>
> $$(x1+x2+\cdots+xr)n=\sum_{(n1,\ldots,nr):n1+\cdots+nr=n} (nn1,n2,\ldots,nr)\, x1n1x2n2\cdots xr \quad nr$$
>
> That is, the sum is over all nonnegative integer-valued vectors (n1,n2,. . .,nr) such that n1+n2+⋯+nr=n.

The numbers (nn1,n2,. . .,nr) are known as *multinomial coefficients*.

Example 5d

In the first round of a knockout tournament involving n=2m players, the n players are divided into n/2 pairs, with each of these pairs then playing a game. The losers of the games are eliminated while the winners go on to the next round, where the process is repeated until only a single player remains. Suppose we have a knockout tournament of 8 players.

a. How many possible outcomes are there for the initial round? (For instance, one outcome is that 1 beats 2, 3 beats 4, 5 beats 6, and 7 beats 8.)
b. How many outcomes of the tournament are possible, where an outcome gives complete information for all rounds?

Solution

One way to determine the number of possible outcomes for the initial round is to first determine the number of possible pairings for that round. To do so, note that the number of ways to divide the 8 players into a *first* pair, a *second* pair, a *third* pair, and a *fourth* pair is (82, 2, 2, 2)=8!24. Thus, the number of possible pairings when there is no ordering of the 4 pairs is 8!244!. For each such pairing, there are 2 possible choices from each pair as to the winner of that game, showing that there are 8!24244!=8!4! possible results of round 1. [Another way to see this is to note that there are (84) possible choices of the 4 winners and, for each such choice, there are 4! ways to pair the 4 winners with the 4 losers, showing that there are 4!(84)=8!4! possible results for the first round.]

Similarly, for each result of round 1, there are 4!2! possible outcomes of round 2, and for each of the outcomes of the first two rounds, there are 2!1! possible outcomes of round 3. Consequently, by the generalized basic principle of counting, there are 8!4!4!2!2!1!=8! possible outcomes of the tournament. Indeed, the same argument can be used to show that a knockout tournament of n=2m players has n! possible outcomes.

Knowing the preceding result, it is not difficult to come up with a more direct argument by showing that there is a one-to-one correspondence between the set of possible tournament results and the set of permutations of 1,. . .,n. To obtain such a correspondence, rank the players as follows for any tournament result: Give the tournament winner rank 1, and give the final-round loser rank 2. For the two players who lost in the next-to-last round, give rank 3 to the one who lost to the player ranked 1 and give rank 4 to the one who lost to the player ranked 2. For the four players who lost in the second-to-last round, give rank 5 to the one who lost to player ranked 1, rank 6 to the one who lost to the player ranked 2, rank 7 to the one who lost to the player ranked 3, and rank 8 to the one who lost to the player ranked 4. Continuing on in this manner gives a rank to each player. (A more succinct description is to give the winner of the tournament rank 1 and let the rank of a player who lost in a round having 2k matches be 2k plus the rank of the player who beat him, for k=0, . . ., m−1.) In this manner, the result of the tournament can be represented by a permutation i1,i2,. . .,in, where ij is the player who was given rank j. Because different tournament results give rise to different permutations, and because there is a tournament result for each permutation, it follows that there are the same number of possible tournament results as there are permutations of 1,. . .,n.

Example 5e

(x1+x2+x3)2=(22, 0, 0) x12x20x30+(20, 2, 0) x10x22x30+(20, 0, 2) x10x20x32+(21, 1, 0) x11x21x30+(21, 0, 1) x11x20x31+(20, 1, 1) x10x21x31
=x12+x22+x32+2x1x2+2x1x3+2x2x3

Problems

1.
   a. How many different 7-place license plates are possible if the first 2 places are for letters and the other 5 for numbers?
   b. Repeat part (a) under the assumption that no letter or number can be repeated in a single license plate.

2. How many outcome sequences are possible when a die is rolled four times, where we say, for instance, that the outcome is 3, 4, 3, 1 if the first roll landed on 3, the second on 4, the third on 3, and the fourth on 1?
3. Twenty workers are to be assigned to 20 different jobs, one to each job. How many different assignments are possible?
4. John, Jim, Jay, and Jack have formed a band consisting of 4 instruments. If each of the boys can play all 4 instruments, how many different arrangements are possible? What if John and Jim can play all 4 instruments, but Jay and Jack can each play only piano and drums?
5. For years, telephone area codes in the United States and Canada consisted of a sequence of three digits. The first digit was an integer between 2 and 9, the second digit was either 0 or 1, and the third digit was any integer from 1 to 9. How many area codes were possible? How many area codes starting with a 4 were possible?
6. A well-known nursery rhyme starts as follows:
   "As I was going to St. Ives
   I met a man with 7 wives.
   Each wife had 7 sacks.
   Each sack had 7 cats.
   Each cat had 7 kittens. . ."
   How many kittens did the traveler meet?

7.
   a. In how many ways can 3 boys and 3 girls sit in a row?
   b. In how many ways can 3 boys and 3 girls sit in a row if the boys and the girls are each to sit together?
   c. In how many ways if only the boys must sit together?
   d. In how many ways if no two people of the same sex are allowed to sit together?

8. When all letters are used, how many different letter arrangements can be made from the letters
   a. Fluke?
   b. Propose?
   c. Mississippi?
   d. Arrange?

9. A child has 12 blocks, of which 6 are black, 4 are red, 1 is white, and 1 is blue. If the child puts the blocks in a line, how many arrangements are possible?
10. In how many ways can 8 people be seated in a row if
   a. there are no restrictions on the seating arrangement?
   b. persons A and B must sit next to each other?
   c. there are 4 men and 4 women and no 2 men or 2 women can sit next to each other?
   d. there are 5 men and they must sit next to one another?
   e. there are 4 married couples and each couple must sit together?

11. In how many ways can 3 novels, 2 mathematics books, and 1 chemistry book be arranged on a bookshelf if

a. the books can be arranged in any order?
b. the mathematics books must be together and the novels must be together?
c. the novels must be together, but the other books can be arranged in any order?

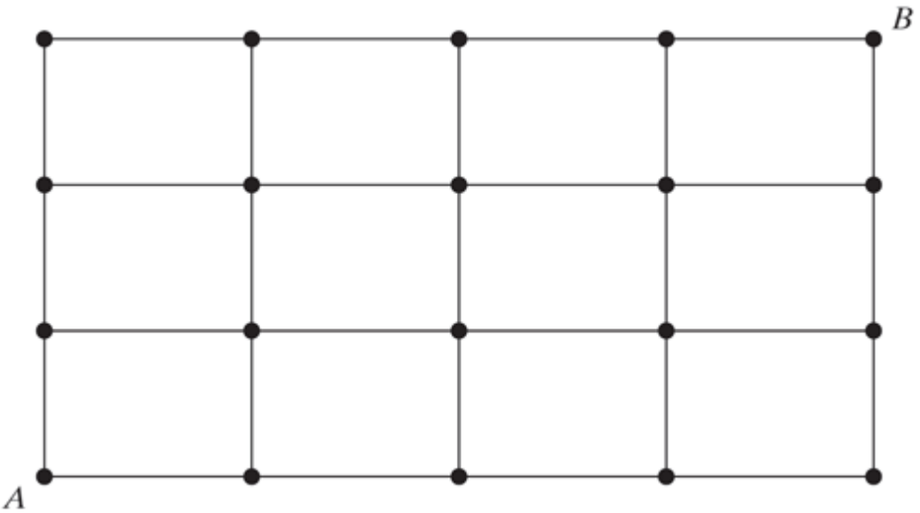
12. How many 3 digit numbers xyz, with x, y, z all ranging from 0 to 9 have at least 2 of their digits equal. How many have exactly 2 equal digits.
13. How many different letter permutations, of any length, can be made using the letters M O T T O. (For instance, there are 3 possible permutations of length 1.)
14. Five separate awards (best scholarship, best leadership qualities, and so on) are to be presented to selected students from a class of 30. How many different outcomes are possible if
a. a student can receive any number of awards?
b. each student can receive at most 1 award?

15. Consider a group of 20 people. If everyone shakes hands with everyone else, how many handshakes take place?
16. How many 5-card poker hands are there?
17. A dance class consists of 22 students, of which 10 are women and 12 are men. If 5 men and 5 women are to be chosen and then paired off, how many results are possible?
18. A student has to sell 2 books from a collection of 6 math, 7 science, and 4 economics books. How many choices are possible if
a. both books are to be on the same subject?
b. the books are to be on different subjects?

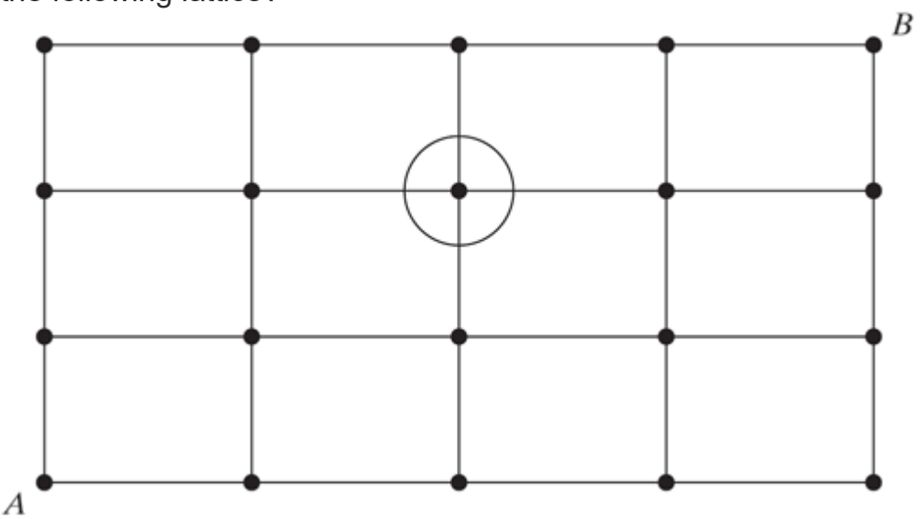
19. Seven different gifts are to be distributed among 10 children. How many distinct results are possible if no child is to receive more than one gift?
20. A committee of 7, consisting of 2 Republicans, 2 Democrats, and 3 Independents, is to be chosen from a group of 5 Republicans, 6 Democrats, and 4 Independents. How many committees are possible?
21. From a group of 8 women and 6 men, a committee consisting of 3 men and 3 women is to be formed. How many different committees are possible if
a. 2 of the men refuse to serve together?
b. 2 of the women refuse to serve together?
c. 1 man and 1 woman refuse to serve together?

22. A person has 8 friends, of whom 5 will be invited to a party.
a. How many choices are there if 2 of the friends are feuding and will not attend together?
b. How many choices if 2 of the friends will only attend together?

23. Consider the grid of points shown at the top of the next column. Suppose that, starting at the point labeled A, you can go one step up or one step to the right at each move. This procedure is continued until the point labeled B is reached. How many different paths from A to B are possible?
Hint: Note that to reach B from A, you must take 4 steps to the right and 3 steps upward.



24. In **Problem 23**, how many different paths are there from A to B that go through the point circled in the following lattice?



25. A psychology laboratory conducting dream research contains 3 rooms, with 2 beds in each room. If 3 sets of identical twins are to be assigned to these 6 beds so that each set of twins sleeps in different beds in the same room, how many assignments are possible?

26.

a. Show $\sum_{k=0}^{n} \binom{n}{k} 2^k = 3^n$
b. Simplify $\sum_{k=0}^{n} \binom{n}{k} x^k$

27. Expand $(3x^2+y)^5$.

28. The game of bridge is played by 4 players, each of whom is dealt 13 cards. How many bridge deals are possible?

29. Expand $(x_1+2x_2+3x_3)^4$.

30. If 12 people are to be divided into 3 committees of respective sizes 3, 4, and 5, how many divisions are possible?

31. If 8 new teachers are to be divided among 4 schools, how many divisions are possible? What if each school must receive 2 teachers?

32. Ten weight lifters are competing in a team weight-lifting contest. Of the lifters, 3 are from the United States, 4 are from Russia, 2 are from China, and 1 is from Canada. If the scoring takes account of the countries that the lifters represent, but not their individual identities, how many different outcomes are possible from the point of view of scores? How many different outcomes correspond to results in which the United States has 1 competitor in the top three and 2 in the bottom three?

33. Delegates from 10 countries, including Russia, France, England, and the United States, are to be seated in a row. How many different seating arrangements are possible if the French and English delegates are to be seated next to each other and the Russian and U.S. delegates are not to be next to each other?

* 34. If 8 identical blackboards are to be divided among 4 schools, how many divisions are possible? How many if each school must receive at least 1 blackboard?

* 35. An elevator starts at the basement with 8 people (not including the elevator operator) and discharges them all by the time it reaches the top floor, number 6. In how many ways could the operator have perceived the people leaving the elevator if all people look alike to him? What if the 8 people consisted of 5 men and 3 women and the operator could tell a man from a woman?

* 36. We have $20,000 that must be invested among 4 possible opportunities. Each investment must be integral in units of $1000, and there are minimal investments that need to be made if one is to invest in these opportunities. The minimal investments are $2000, $2000, $3000, and $4000. How many different investment strategies are available if

a. an investment must be made in each opportunity?
b. investments must be made in at least 3 of the 4 opportunities?

* 37. Suppose that 10 fish are caught at a lake that contains 5 distinct types of fish.

a. How many different outcomes are possible, where an outcome specifies the numbers of caught fish of each of the 5 types?
b. How many outcomes are possible when 3 of the 10 fish caught are trout?
c. How many when at least 2 of the 10 are trout?

Theoretical Exercises

1. Prove the generalized version of the basic counting principle.
2. Two experiments are to be performed. The first can result in any one of m possible outcomes. If the first experiment results in outcome i, then the second experiment can result in any of ni possible outcomes, i=1, 2,. . .,m. What is the number of possible outcomes of the two experiments?
3. In how many ways can r objects be selected from a set of n objects if the order of selection is considered relevant?
4. There are (nr) different linear arrangements of n balls of which r are black and n−r are white. Give a combinatorial explanation of this fact.
5. Determine the number of vectors (x1,. . .,xn), such that each xi is either 0 or 1 and $\sum_{i=1}^{n} x_i \geq k$

6. How many vectors x1,. . .,xk are there for which each xi is a positive integer such that $1 \leq x_i \leq n$ and $x_1 < x_2 < \cdots < x_k$?
7. Give an analytic proof of **Equation (4.1)**.
8. Prove that
$\binom{n+m}{r} = \binom{n}{0}\binom{m}{r} + \binom{n}{1}\binom{m}{r-1} + \cdots + \binom{n}{r}\binom{m}{0}$

Hint: Consider a group of n men and m women. How many groups of size r are possible?
9. Use **Theoretical Exercise 8** to prove that
$\binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}^2$

10. From a group of n people, suppose that we want to choose a committee of k, $k \leq n$, one of whom is to be designated as chairperson.
    a. By focusing first on the choice of the committee and then on the choice of the chair, argue that there are $\binom{n}{k}$ k possible choices.
    b. By focusing first on the choice of the nonchair committee members and then on the choice of the chair, argue that there are $\binom{n}{k-1}$ $(n-k+1)$ possible choices.
    c. By focusing first on the choice of the chair and then on the choice of the other committee members, argue that there are $n\binom{n-1}{k-1}$ possible choices.
    d. Conclude from parts (a), (b), and (c) that
    $k\binom{n}{k} = (n-k+1)\binom{n}{k-1} = n\binom{n-1}{k-1}$

    e. Use the factorial definition of $\binom{n}{r}$ to verify the identity in part (d).

11. The following identity is known as Fermat's combinatorial identity:
$\binom{n}{k} = \sum_{i=k}^{n} \binom{i-1}{k-1} \quad n \geq k$

Give a combinatorial argument (no computations are needed) to establish this identity.
Hint: Consider the set of numbers 1 through n. How many subsets of size k have i as their highest numbered member?
12. Consider the following combinatorial identity:
$\sum_{k=1}^{n} k\binom{n}{k} = n \cdot 2^{n-1}$

    a. Present a combinatorial argument for this identity by considering a set of n people and determining, in two ways, the number of possible selections of a committee of any size and a chairperson for the committee.
    Hint:
        i. How many possible selections are there of a committee of size k and its chairperson?
        ii. How many possible selections are there of a chairperson and the other committee members?

b. Verify the following identity for n=1, 2, 3, 4, 5:
$\sum_{k=1}^{n}(nk)\, k2=2n-2n(n+1)$

For a combinatorial proof of the preceding, consider a set of n people and argue that both sides of the identity represent the number of different selections of a committee, its chairperson, and its secretary (possibly the same as the chairperson).
Hint:
i. How many different selections result in the committee containing exactly k people?
ii. How many different selections are there in which the chairperson and the secretary are the same? (ANSWER: n2n−1.)
iii. How many different selections result in the chairperson and the secretary being different?

c. Now argue that
$\sum_{k=1}^{n}(nk)\, k3=2n-3n2(n+3)$

13. Show that, for n> 0,
$\sum_{i=0}^{n}(-1)i(ni)=0$

Hint: Use the binomial theorem.
14. From a set of n people, a committee of size j is to be chosen, and from this committee, a subcommittee of size i,i≤j, is also to be chosen.
a. Derive a combinatorial identity by computing, in two ways, the number of possible choices of the committee and subcommittee–first by supposing that the committee is chosen first and then the subcommittee is chosen, and second by supposing that the subcommittee is chosen first and then the remaining members of the committee are chosen.
b. Use part (a) to prove the following combinatorial identity:
$\sum_{j=i}^{n}(nj)(ji)=(ni)\, 2n-i \quad i\leq n$

c. Use part (a) and **Theoretical Exercise 13** to show that
$\sum_{j=i}^{n}(nj)(ji)\,(-1)n-j=0 \quad i<n$

15. Let Hk(n) be the number of vectors x1,. . .,xk for which each xi is a positive integer satisfying 1≤xi≤n and x1≤x2≤⋯≤xk.
a. Without any computations, argue that
$H1(n)=nHk(n)=\sum_{j=1}^{n}Hk-1(j) \quad k>1$

Hint: How many vectors are there in which xk=j?
b. Use the preceding recursion to compute H3(5).
Hint: First compute H2(n) for n=1, 2, 3, 4, 5.

16. Consider a tournament of n contestants in which the outcome is an ordering of these contestants, with ties allowed. That is, the outcome partitions the players into groups, with the first group consisting of the players who tied for first place, the next group being those who tied for the next-best position, and so on. Let N(n) denote the number of different possible outcomes. For instance, N(2)=3, since, in a tournament with 2 contestants, player 1 could be uniquely first, player 2 could be uniquely first, or they could tie for first
a. List all the possible outcomes when n=3.
b. With N(0) defined to equal 1, argue, without any computations, that
$N(n)=\sum_{i=1}^{n}(ni)\, N(n-i)$

Hint: How many outcomes are there in which i players tie for last place?

c. Show that the formula of part (b) is equivalent to the following:
$N(n)=\sum_{i=0}^{n-1}\binom{n}{i}N(i)$

d. Use the recursion to find N(3) and N(4).

17. Present a combinatorial explanation of why $\binom{n}{r}=\binom{n}{r, n-r}$
18. Argue that
$\binom{n}{n_1,n_2,\ldots,n_r}=\binom{n-1}{n_1-1,n_2,\ldots,n_r}+\binom{n-1}{n_1,n_2-1,\ldots,n_r}+\cdots+\binom{n-1}{n_1,n_2,\ldots,n_r-1}$

Hint: Use an argument similar to the one used to establish **Equation (4.1)**.
19. Prove the multinomial theorem.
* 20. In how many ways can n identical balls be distributed into r urns so that the ith urn contains at least m_i balls, for each $i=1,\ldots,r$? Assume that $n\geq\sum_{i=1}^{r}m_i$.
* 21. Argue that there are exactly $\binom{k}{n-r+k}$ solutions of
$x_1+x_2+\cdots+x_r=n$

for which exactly k of the x_i are equal to 0.
* 22. Consider a function $f(x_1,\ldots,x_n)$ of n variables. How many different partial derivatives of order r does f possess?
* 23. Determine the nu mber of vectors $(x_1,\ldots,x_n)$ such that each x_i is a nonnegative integer and $\sum_{i=1}^{n}x_i\leq k$

Self-Test Problems and Exercises

1. How many different linear arrangements are there of the letters A, B, C, D, E, F for which
   a. A and B are next to each other?
   b. A is before B?
   c. A is before B and B is before C?
   d. A is before B and C is before D?
   e. A and B are next to each other and C and D are also next to each other?
   f. E is not last in line?

2. If 4 Americans, 3 French people, and 3 British people are to be seated in a row, how many seating arrangements are possible when people of the same nationality must sit next to each other?
3. A president, treasurer, and secretary, all different, are to be chosen from a club consisting of 10 people. How many different choices of officers are possible if
   a. there are no restrictions?
   b. A and B will not serve together?
   c. C and D will serve together or not at all?
   d. E must be an officer?
   e. F will serve only if he is president?

4. A student is to answer 7 out of 10 questions in an examination. How many choices has she? How many if she must answer at least 3 of the first 5 questions?
5. In how many ways can a man divide 7 gifts among his 3 children if the eldest is to receive 3 gifts and the others 2 each?
6. How many different 7-place license plates are possible when 3 of the entries are letters and 4 are digits? Assume that repetition of letters and numbers is allowed and that there is no restriction on where the letters or numbers can be placed.
7. Give a combinatorial explanation of the identity
$(nr)=(nn-r)$

8. Consider n-digit numbers where each digit is one of the 10 integers 0,1,. . .,9. How many such numbers are there for which
   a. no two consecutive digits are equal?
   b. 0 appears as a digit a total of i times, i=0,. . .,n?

9. Consider three classes, each consisting of n students. From this group of 3n students, a group of 3 students is to be chosen.
   a. How many choices are possible?
   b. How many choices are there in which all 3 students are in the same class?
   c. How many choices are there in which 2 of the 3 students are in the same class and the other student is in a different class?
   d. How many choices are there in which all 3 students are in different classes?
   e. Using the results of parts (a) through (d), write a combinatorial identity.

10. How many 5-digit numbers can be formed from the integers 1,2,. . .,9 if no digit can appear more than twice? (For instance, 41434 is not allowed.)
11. From 10 married couples, we want to select a group of 6 people that is not allowed to contain a married couple.
   a. How many choices are there?
   b. How many choices are there if the group must also consist of 3 men and 3 women?

12. A committee of 6 people is to be chosen from a group consisting of 7 men and 8 women. If the

2.2 Sample Space and Events

Consider an experiment whose outcome is not predictable with certainty. However, although the outcome of the experiment will not be known in advance, let us suppose that the set of all possible outcomes is known. This set of all possible outcomes of an experiment is known as the *sample space* of the experiment and is denoted by *S*. Following are some examples:

1. If the outcome of an experiment consists of the determination of the sex of a newborn child, then
   $S=\{g, b\}$

   where the outcome g means that the child is a girl and b that it is a boy.
2. If the outcome of an experiment is the order of finish in a race among the 7 horses having post positions 1, 2, 3, 4, 5, 6, and 7, then
   $S=\{\text{all } 7! \text{ permutations of } (1, 2, 3, 4, 5, 6, 7)\}$

   The outcome (2, 3, 1, 6, 5, 4, 7) means, for instance, that the number 2 horse comes in first, then the number 3 horse, then the number 1 horse, and so on.
3. If the experiment consists of flipping two coins, then the sample space consists of the following four points:
   $S=\{(h, h), (h, t), (t, h), (t, t)\}$

   The outcome will be (h, h) if both coins are heads, (h, t) if the first coin is heads and the second tails, (t, h) if the first is tails and the second heads, and (t, t) if both coins are tails.
4. If the experiment consists of tossing two dice, then the sample space consists of the 36 points
   $S=\{(i, j): i, j=1, 2, 3, 4, 5, 6\}$

   where the outcome (i,j) is said to occur if i appears on the leftmost die and j on the other die.
5. If the experiment consists of measuring (in hours) the lifetime of a transistor, then the sample space consists of all nonnegative real numbers; that is,
   $S=\{x: 0\leq x<\infty\}$

Any subset E of the sample space is known as an *event*. In other words, an event is a set consisting of possible outcomes of the experiment. If the outcome of the experiment is contained in E, then we say that E has occurred. Following are some examples of events.

In the preceding Example 1, if $E=\{g\}$, then E is the event that the child is a girl. Similarly, if $F=\{b\}$, then F is the event that the child is a boy.

In Example 2, if

$E=\{\text{all outcomes in S starting with a 3}\}$

then E is the event that horse 3 wins the race.

In Example 3, if $E=\{(h, h), (h, t)\}$, then E is the event that a head appears on the first coin.

In Example 4, if $E=\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$, then E is the event that the sum of the dice equals 7.

In Example 5, if $E=\{x: 0\leq x\leq 5\}$, then E is the event that the transistor does not last longer than 5 hours.

For any two events E and F of a sample space S, we define the new event $E\cup F$ to consist of all outcomes that are either in E or in F or in both E and F. That is, the event $E\cup F$ will occur if *either E* or F occurs. For

instance, in Example 1, if E={g} is the event that the child is a girl and F={b} is the event that the child is a boy, then

$E \cup F = \{g, b\}$

is the whole sample space S. In Example 3, if E={(h, h), (h, t)} is the event that the first coin lands heads, and F={(t, h), (h, h)} is the event that the second coin lands heads, then

$E \cup F = \{(h, h), (h, t), (t, h)\}$

is the event that at least one of the coins lands heads and thus will occur provided that both coins do not land tails.

The event $E \cup F$ is called the *union* of the event E and the event F.

Similarly, for any two events E and F, we may also define the new event *EF*, called the *intersection* of E and F, to consist of all outcomes that are both in E and in F. That is, the event *EF* (sometimes written $E \cap F$) will occur only if both E and F occur. For instance, in Example 3, if E={(h, h), (h, t), (t, h)} is the event that at least 1 head occurs and F={(h, t), (t, h), (t, t)} is the event that at least 1 tail occurs, then

$EF = \{(h, t), (t, h)\}$

is the event that exactly 1 head and 1 tail occur. In Example 4, if E={(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)} is the event that the sum of the dice is 7 and F={(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)} is the event that the sum is 6, then the event *EF* does not contain any outcomes and hence could not occur. To give such an event a name, we shall refer to it as the null event and denote it by Ø. (That is, Ø refers to the event consisting of no outcomes.) If EF=Ø, then E and F are said to be *mutually exclusive*.

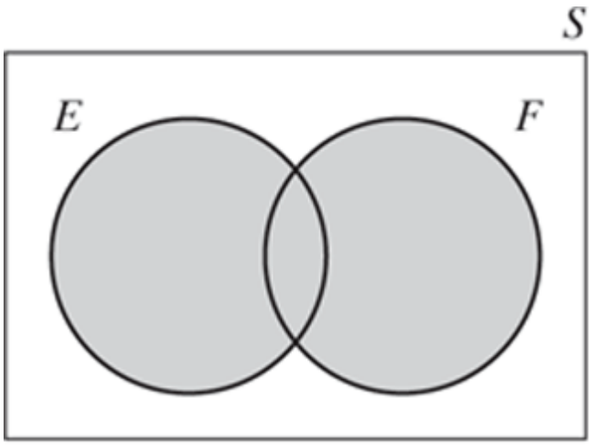
We define unions and intersections of more than two events in a similar manner. If E1, E2, ... are events, then the union of these events, denoted by $\cup_{n=1}^{\infty} E_n$, is defined to be that event that consists of all outcomes that are in En for at least one value of n=1, 2, ... Similarly, the intersection of the events En, denoted by $\cap_{n=1}^{\infty} E_n$, is defined to be the event consisting of those outcomes that are in all of the events En, n=1, 2, ...

Finally, for any event E, we define the new event Ec, referred to as the *complement* of E, to consist of all outcomes in the sample space S that are not in E. That is, Ec will occur if and only if E does not occur. In Example 4, if event E={(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)}, then Ec will occur when the sum of the dice does not equal 7. Note that because the experiment must result in some outcome, it follows that $S^c = \emptyset$.

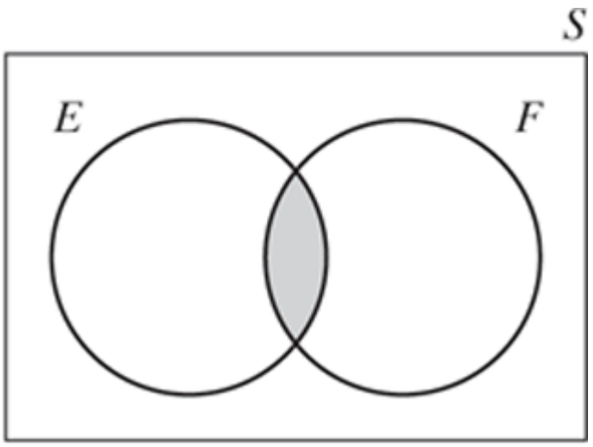
For any two events E and F, if all of the outcomes in E are also in F, then we say that E is *contained* in F, or E is a *subset* of F, and write $E \subset F$ (or equivalently, $F \supset E$, which we sometimes say as F is a *superset* of E). Thus, if $E \subset F$, then the occurrence of E implies the occurrence of F. If $E \subset F$ and $F \subset E$, we say that E and F are equal and write E=F.

A graphical representation that is useful for illustrating logical relations among events is the Venn diagram. The sample space S is represented as consisting of all the outcomes in a large rectangle, and the events E, F, G, ... are represented as consisting of all the outcomes in given circles within the rectangle. Events of interest can then be indicated by shading appropriate regions of the diagram. For instance, in the three Venn diagrams shown in **Figure 2.1**, the shaded areas represent, respectively, the events $E \cup F$, *EF*, and Ec. The Venn diagram in **Figure 2.2** indicates that $E \subset F$.

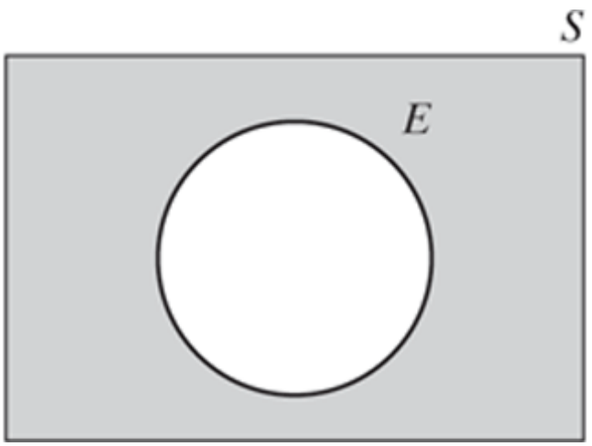
Figure 2.1 Venn diagrams.



(a) Shaded region: E∪F.

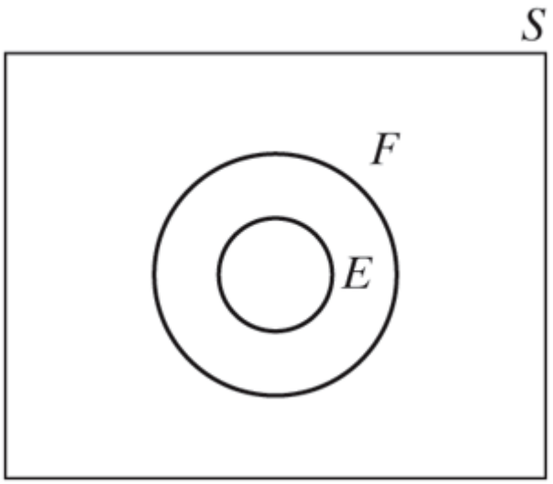


(b) Shaded region: *FF*



(b) Shaded region: E^c

Figure 2.2 E⊂F.

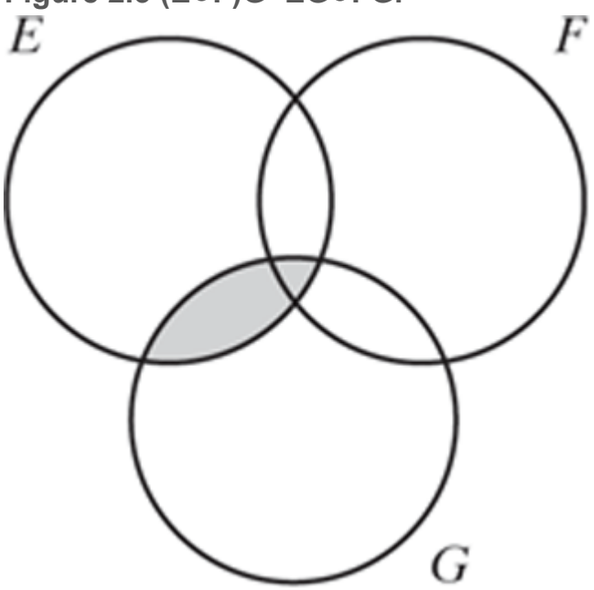


The operations of forming unions, intersections, and complements of events obey certain rules similar to the rules of algebra. We list a few of these rules:

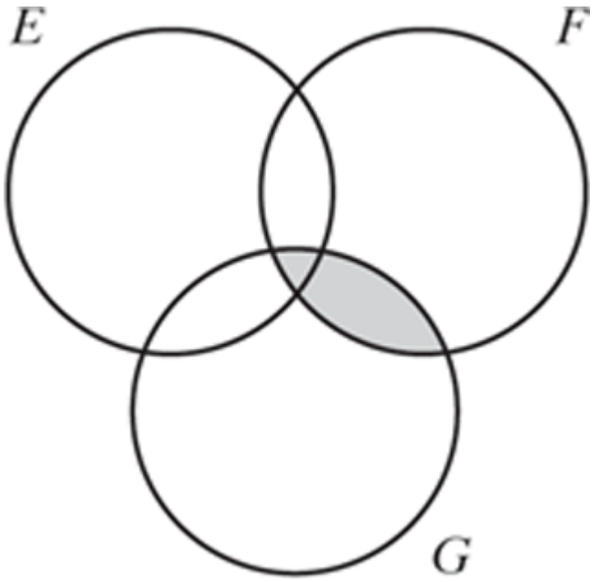
Commutative laws E∪F =F∪E EF =FE Associative laws (E∪F)∪G =E∪(F∪G) (EF)G =E(FG)
Distributive laws (E∪F)G =EG∪FG EF∪G =(E∪G)(F∪G)

These relations are verified by showing that any outcome that is contained in the event on the left side of the equality sign is also contained in the event on the right side, and vice versa. One way of showing this is by means of Venn diagrams. For instance, the distributive law may be verified by the sequence of diagrams in **Figure 2.3**.

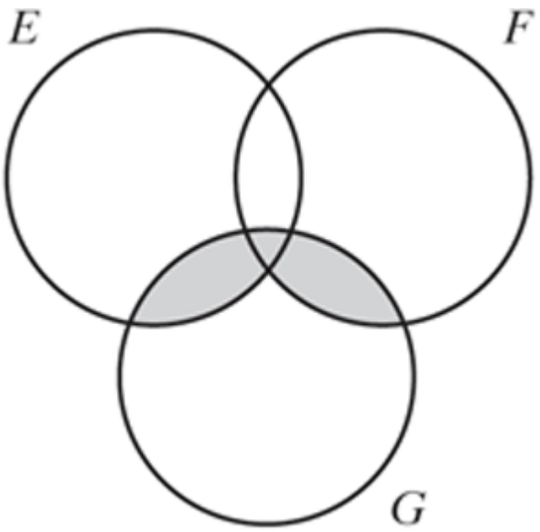
Figure 2.3 (E∪F)G=EG∪FG.



(a) Shaded region: *EG*.



(b) Shaded region: *FG*.



(c) Shaded region: (E∪F)G.

The following useful relationships among the three basic operations of forming unions, intersections, and complements are known as *DeMorgan's laws*:

(∪ i=1 n E i) c = ∩ i=1 n E i c (∩ i=1 n E i) c = ∪ i=1 n E i c

For instance, for two events *E* and *F*, DeMorgan's laws state that

(E∪F)c=EcFc and (EF)c=Ec∪Fc

which can be easily proven by using Venn diagrams (see **Theoretical Exercise 7**).

To prove DeMorgan's laws for general n, suppose first that x is an outcome of (∪i=1nEi)c. Then x is not contained in ∪i=1nEi, which means that *x* is not contained in any of the events Ei, i=1, 2, ... , n, implying that x is contained in Eic for all i=1, 2, . . . , and thus is contained in ∩i=1nEic. To go the other way, suppose that *x* is an outcome of ∩ i=1 n E i c . Then x is contained in Eic for all i=1, 2, . . . , n, which means that x is not contained in Ei for any i=1, 2, . . ., n, implying that *x* is not contained in ∪ i n E i , in turn implying that x

is contained in (∪1nEi)c. This proves the first of DeMorgan’s laws.

To prove the second of DeMorgan’s laws, we use the first law to obtain

(∪ i=1 n E i c) c = ∩ i=1 n (E i c) c

which, since (Ec)c=E, is equivalent to

(∪1nEic)c=∩1nEi

Taking complements of both sides of the preceding equation yields the result we seek, namely,

∪ 1 n E i c = (∩ 1 n E i) c

2.3 Axioms of Probability

One way of defining the probability of an event is in terms of its long run *relative frequency*. Such a definition usually goes as follows: We suppose that an experiment, whose sample space is S, is repeatedly performed under exactly the same conditions. For each event E of the sample space S, we define n(E) to be the number of times in the first n repetitions of the experiment that the event E occurs. Then P(E), the *probability* of the event *E*, is defined as

$$P(E)=\lim_{n\to\infty} n(E)n$$

That is, P(E) is defined as the (limiting) proportion of time that E occurs. It is thus the limiting relative frequency of E.

Although the preceding definition is certainly intuitively pleasing and should always be kept in mind by the reader, it possesses a serious drawback: How do we know that n(E)/n will converge to some constant limiting value that will be the same for each possible sequence of repetitions of the experiment? For example, suppose that the experiment to be repeatedly performed consists of flipping a coin. How do we know that the proportion of heads obtained in the first n flips will converge to some value as n gets large? Also, even if it does converge to some value, how do we know that, if the experiment is repeatedly performed a second time, we shall obtain the same limiting proportion of heads?

Proponents of the relative frequency definition of probability usually answer this objection by stating that the convergence of n(E)/n to a constant limiting value is an assumption, or an *axiom*, of the system. However, to assume that n(E)/n will necessarily converge to some constant value seems to be an extraordinarily complicated assumption. For, although we might indeed hope that such a constant limiting frequency exists, it does not at all seem to be a priori evident that this need be the case. In fact, would it not be more reasonable to assume a set of simpler and more self-evident axioms about probability and then attempt to prove that such a constant limiting frequency does in some sense exist? The latter approach is the modern axiomatic approach to probability theory that we shall adopt in this text. In particular, we shall assume that, for each event E in the sample space S, there exists a value P(E), referred to as the probability of E. We shall then assume that all these probabilities satisfy a certain set of axioms, which, we hope the reader will agree, is in accordance with our intuitive notion of probability.

Consider an experiment whose sample space is S. For each event E of the sample space S, we assume that a number P(E) is defined and satisfies the following three axioms:

> The three axioms of probability
>
> **Axiom 1**
>
> $0\le P(E)\le 1$
>
> **Axiom 2**
>
> $P(S)=1$
>
> **Axiom 3**
>
> For any sequence of mutually exclusive events E1, E2, ... (that is, events for which $E_i E_j=\emptyset$ when $i\neq j$),
>
> $P(\cup_{i=1}^{\infty} E_i)= \sum_{i=1}^{\infty} P(E_i)$
>
> We refer to P(E) as the *probability* of the event E.

Thus, **Axiom 1** states that the probability that the outcome of the experiment is an outcome in E is some

number between 0 and 1. **Axiom 2** states that, with probability 1, the outcome will be a point in the sample space S. **Axiom 3** states that, for any sequence of mutually exclusive events, the probability of at least one of these events occurring is just the sum of their respective probabilities.

If we consider a sequence of events E 1 , E 2 , . . . , , where E 1 =S and E 1 =∅ for i>1, then, because the events are mutually exclusive and because S=∪i=1∞Ei, we have, from **Axiom 3**,

P(S)=∑i=1∞P(Ei)=P(S)+∑i=2∞P(Ø)

implying that

P(Ø)=0

That is, the null event has probability 0 of occurring.

Note that it follows that, for any finite sequence of mutually exclusive events E1, E2, . . ., En,

P(∪ 1 n E i)= Σ i=1 n P(E i) (3.1)

This equation follows from **Axiom 3** by defining Ei as the null event for all values of i greater than n. **Axiom 3** is equivalent to **Equation (3.1)** when the sample space is finite. (Why?) However, the added generality of **Axiom 3** is necessary when the sample space consists of an infinite number of points.

Example 3a

If our experiment consists of tossing a coin and if we assume that a head is as likely to appear as a tail, then we would have

P({H})=P({T})= 1 2

On the other hand, if the coin were biased and we believed that a head were twice as likely to appear as a tail, then we would have

P({H})=23 P({T})=13

Example 3b

If a die is rolled and we suppose that all six sides are equally likely to appear, then we would have P({1})=P({2})=P({3})=P({4})=P({5})=P({6})=16. From **Axiom 3**, it would thus follow that the probability of rolling an even number would equal

P({2,4,6})=P({2})+P({4})+P({6})=12

The assumption of the existence of a set function P, defined on the events of a sample space S and satisfying **Axioms 1**, **2**, and **3**, constitutes the modern mathematical approach to probability theory. It is hoped that the reader will agree that the axioms are natural and in accordance with our intuitive concept of probability as related to chance and randomness. Furthermore, using these axioms, we shall be able to prove that if an experiment is repeated over and over again, then, with probability 1, the proportion of time during which any specific event *E* occurs will equal P(E). This result, known as the strong law of large numbers, is presented in **Chapter 8**. In addition, we present another possible interpretation of probability– as being a measure of belief–in **Section 2.7**.

Technical Remark. We have supposed that P(E) is defined for all the events E of the sample space. Actually, when the sample space is an uncountably infinite set, P(E) is defined only for a class of events called *measurable*. However, this restriction need not concern us, as all events of any practical interest are

measurable.

2.4 Some Simple Propositions

In this section, we prove some simple propositions regarding probabilities. We first note that since E and Ec are always mutually exclusive and since E∪Ec=S, we have, by **Axioms 2** and **3**,

$$1=P(S)=P(E\cup Ec)=P(E)+P(Ec)$$

Or, equivalently, we have **Proposition 4.1**.

$$P(Ec)=1-P(E)$$

Proposition 4.1

In words, **Proposition 4.1** states that the probability that an event does not occur is 1 minus the probability that it does occur. For instance, if the probability of obtaining a head on the toss of a coin is 38, then the probability of obtaining a tail must be 5 8 .

Our second proposition states that if the event E is contained in the event F, then the probability of E is no greater than the probability of F.

Proposition 4.2

If E⊂F, then P(E)≤P(F).

Proof. Since E⊂F, it follows that we can express F as

$$F=E\cup EcF$$

Hence, because E and EcF are mutually exclusive, we obtain, from **Axiom 3**,

$$P(F)=P(E)+P(E c F)$$

which proves the result, since P(EcF)≥0.

Proposition 4.2 tells us, for instance, that the probability of rolling a 1 with a die is less than or equal to the probability of rolling an odd value with the die.

The next proposition gives the relationship between the probability of the union of two events, expressed in terms of the individual probabilities, and the probability of the intersection of the events.

$$P(E\cup F)=P(E)+P(F)-P(EF)$$

Proposition 4.3

Proof To derive a formula for P(E∪F), we first note that E∪F can be written as the union of the two disjoint events E and EcF. Thus, from **Axiom 3**, we obtain

$$P(E\cup F) =P(E\cup E c F) =P(E)+P(E c F)$$

Furthermore, since F=EF∪EcF, we again obtain from **Axiom 3**

$$P(F)=P(EF)+P(EcF)$$

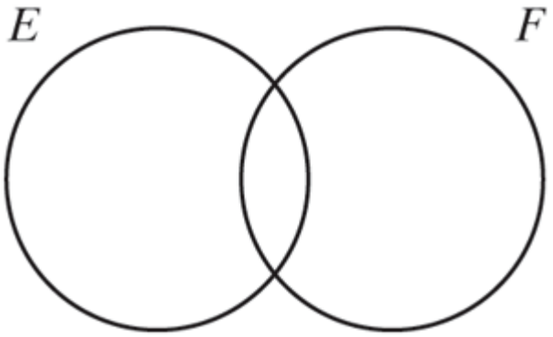
or, equivalently,

$$P(EcF)=P(F)-P(EF)$$

thereby completing the proof.

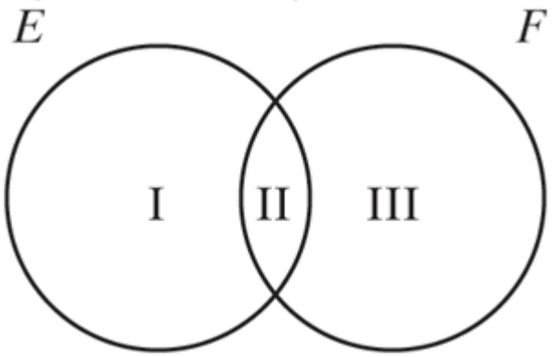
Proposition 4.3 could also have been proved by making use of the Venn diagram in **Figure 2.4**.

Figure 2.4: Venn diagram.



Let us divide E∪F into three mutually exclusive sections, as shown in **Figure 2.5**. In words, section I represents all the points in E that are not in F (that is, EFc), section II represents all points both in E and in F (that is, *EF*), and section III represents all points in F that are not in E (that is, EcF).

Figure 2.5 Venn diagram in sections.



From **Figure 2.5**, we see that

$$E\cup F = I\cup II\cup III \quad E = I\cup II \quad F = II\cup III$$

As I, II, and III are mutually exclusive, it follows from **Axiom 3** that

$$P(E\cup F) = P(I)+P(II)+P(III) \quad P(E) = P(I)+P(II) \quad P(F) = P(II)+P(III)$$

which shows that

$$P(E\cup F)=P(E)+P(F)-P(II)$$

and **Proposition 4.3** is proved, since II = *EF*.

Example 4a

J is taking two books along on her holiday vacation. With probability .5, she will like the first book; with probability .4, she will like the second book; and with probability .3, she will like both books. What is the

probability that she likes neither book?

Solution

Let Bi denote the event that J likes book i, i=1, 2. Then the probability that she likes at least one of the books is

P(B1∪B2)=P(B1)+P(B2)−P(B1B2)=.5+.4−.3=.6

Because the event that J likes neither book is the complement of the event that she likes at least one of them, we obtain the result

P(B1cB2c)=P((B1∪B2)c)=1−P(B1∪B2)=.4

We may also calculate the probability that any one of the three events E, F, and G occurs, namely,

P(E∪F∪G)=P[(E∪F)∪G]

which, by **Proposition 4.3**, equals

P(E∪F)+P(G)−P[(E∪F)G]

Now, it follows from the distributive law that the events (E∪F)G and EG∪FG are equivalent; hence, from the preceding equations, we obtain

P(E∪F∪G) =P(E)+P(F)−P(EF)+P(G)−P(EG∪FG) =P(E)+P(F)−P(EF)+P(G)−P(EG)−P(FG)+P(EGFG) =P(E)+P(F)+P(G)+P(EF)−P(EG)−P(FG)+P(EFG)

In fact, the following proposition, known as the *inclusion–exclusion identity*, can be proved by mathematical induction:

Proposition 4.4

P(E 1 ∪ E 2 ∪⋯∪ E n)= ∑ i=1 n P(E i)− ∑ i 1 < i 2 P(E i 1 E i 2)+⋯ + (−1) r+1 ∑ i 1 < i 2 <⋯< i r P(E i 1 E i 2 ⋯ E i r) +⋯+ (−1) n+1 P(E 1 E 2 ⋯ E n)

The summation ∑i1<i2<⋯<irP(Ei1Ei2⋯Eir) is taken over all of the (nr) possible subsets of size r of the set {1, 2 … ,n}.

In words, **Proposition 4.4** states that the probability of the union of n events equals the sum of the probabilities of these events taken one at a time, minus the sum of the probabilities of these events taken two at a time, plus the sum of the probabilities of these events taken three at a time, and so on.

Remarks 1. For a noninductive argument for **Proposition 4.4**, note first that if an outcome of the sample space is not a member of any of the sets Ei, then its probability does not contribute anything to either side of the equality. Now, suppose that an outcome is in exactly m of the events Ei, where m>0. Then, since it is in ∪iEi, its probability is counted once in P(∪iEi) ; also, as this outcome is contained in (mk) subsets of the type Ei1Ei2⋯Eik, its probability is counted

(m1)−(m2)+(m3)−⋯±(mm)

times on the right of the equality sign in **Proposition 4.4**. Thus, for m>0, we must show that

1=(m 1)−(m 2)+(m 3)− ⋯ ±(m m)

However, since 1=(m0), the preceding equation is equivalent to

$$\sum i=0\ m\ (\ m\ i\ )\ (-1)\ i\ =0$$

and the latter equation follows from the binomial theorem, since

$$0=(-1+1)m=\sum i=0m(mi)(-1)i(1)m-i$$

2. The following is a succinct way of writing the inclusion–exclusion identity:
   $P(\cup i=1nEi)=\sum r=1n(-1)r+1\sum i1<\cdots<irP(Ei1\cdots Eir)$

3. In the inclusion–exclusion identity, going out one term results in an upper bound on the probability of the union, going out two terms results in a lower bound on the probability, going out three terms results in an upper bound on the probability, going out four terms results in a lower bound, and so on. That is, for events E1,...,En, we have
   $$P(\ \cup\ i=1\ n\ E\ i\ )\leq\ \sum\ i=1\ n\ P(\ E\ i\ ) \tag{4.1}$$

   $$P(\cup i=1nEi)\geq\sum i=1nP(Ei)-\sum j<iP(EiEj) \tag{4.2}$$

   $$P(\cup i=1nEi)\leq\sum i=1nP(Ei)-\sum j<iP(EiEj)+\sum k<j<iP(EiEjEk) \tag{4.3}$$

and so on. To prove the validity of these bounds, note the identity

$$\cup i=1nEi=E1\cup E1cE2\cup E1cE2cE3\cup\cdots\cup E1c\cdots En-1cEn$$

That is, at least one of the events Ei occurs if E1 occurs, or if E1 does not occur but E2 does, or if E1 and E2 do not occur but E3 does, and so on. Because the right-hand side is the union of disjoint events, we obtain

$$P(\ \cup\ i=1\ n\ E\ i\ )\ =P(\ E\ 1\ )+P(\ E\ 1\ c\ E\ 2\ )+P(\ E\ 1\ c\ E\ 2\ c\ E\ 3\ )+...+P(\ E\ 1\ c\ \cdots\ E\ n-1\ c\ E\ n\ )\ =P(\ E\ i\ )+\sum\ i=2\ n\ P(\ E\ 1\ c\ \cdots\ E\ i-1\ c\ E\ i\ ) \tag{4.4}$$

Now, let $B\ i\ =\ E\ 1\ c\ \cdots\ E\ i-1\ c\ =\ (\ \cup\ j\ <i\ E\ j\ )\ c$ be the event that none of the first i−1 events occurs. Applying the identity

$$P(Ei)=P(BiEi)+P(BicEi)$$

shows that

$$P(\ E\ i\ )=P(\ E\ 1\ c\ \ldots\ E\ i-1\ c\ E\ i\ )=P(\ E\ i\ )+P(\ \cup\ j<i\ E\ i\ E\ j\ )$$

or, equivalently,

$$P(E1c\cdots Ei-1cEi)=P(Ei)-P(\cup j<iEiEj)$$

Substituting this equation into **(4.4)** yields

$$P(\cup i=1nEi)=\sum iP(Ei)-\sum iP(\cup j<iEiEj) \tag{4.5}$$

Because probabilities are always nonnegative, Inequality (4.1) follows directly from **Equation (4.5)**. Now, fixing i and applying Inequality (1) to $P(\cup j<iEiEj)$ yields

$$P(\cup j<iEiEj)\leq\sum j<iP(EiEj)$$

which, by **Equation (4.5)**, gives Inequality (4.2). Similarly, fixing *i* and applying Inequality (4.2) to $P(\cup_{j<i} E_i E_j)$ yields

$$P(\cup_{j<i} E_i E_j) \geq \sum_{j<i} P(E_i E_j) - \sum_{k<j<i} P(E_i E_j E_i E_k) = \sum_{j<i} P(E_i E_j) - \sum_{k<j<i} P(E_i E_j E_k)$$

which, by **Equation (4.5)**, gives Inequality (4.3). The next inclusion–exclusion inequality is now obtained by fixing i and applying Inequality (4.3) to $P(\cup_{j<i} E_i E_j)$, and so on.

The first inclusion-exclusion inequality, namely that

$$P(\cup_{i=1}^{n} E_i) \leq \sum_{i=1}^{n} P(E_i)$$

is known as *Boole's inequality*.

2.5 Sample Spaces Having Equally Likely Outcomes

In many experiments, it is natural to assume that all outcomes in the sample space are equally likely to occur. That is, consider an experiment whose sample space S is a finite set, say, S={1, 2, ... , N}. Then, it is often natural to assume that

$$P(\{1\})=P(\{2\})=\cdots=P(\{N\})$$

which implies, from **Axioms 2** and **3** (why?), that

$$P(\{i\})=1N \quad i=1, 2, \ldots , N$$

From this equation, it follows from **Axiom 3** that, for any event E,

$$P(E)= \text{number of outcomes in E number of outcomes in S}$$

In words, if we assume that all outcomes of an experiment are equally likely to occur, then the probability of any event E equals the proportion of outcomes in the sample space that are contained in E.

Example 5a

If two dice are rolled, what is the probability that the sum of the upturned faces will equal 7?

Solution

We shall solve this problem under the assumption that all of the 36 possible outcomes are equally likely. Since there are 6 possible outcomes–namely, (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), and (6, 1)–that result in the sum of the dice being equal to 7, the desired probability is 636=16.

Example 5b

If 3 balls are “randomly drawn” from a bowl containing 6 white and 5 black balls, what is the probability that one of the balls is white and the other two black?

Solution

If we regard the balls as being distinguishable and the order in which they are selected as being relevant, then the sample space consists of $11\cdot10\cdot9=990$ outcomes. Furthermore, there are $6\cdot5\cdot4=120$ outcomes in which the first ball selected is white and the other two are black; $5\cdot6\cdot4=120$ outcomes in which the first is black, the second is white, and the third is black; and $5\cdot4\cdot6=120$ in which the first two are black and the third is white. Hence, assuming that “randomly drawn” means that each outcome in the sample space is equally likely to occur, we see that the desired probability is

$$120+120+120990=411$$

This problem could also have been solved by regarding the outcome of the experiment as the unordered set of drawn balls. From this point of view, there are $(113)=165$ outcomes in the sample space. Now, each set of 3 balls corresponds to 3! outcomes when the order of selection is noted. As a result, if all outcomes are assumed equally likely when the order of selection is noted, then it follows that they remain equally likely when the outcome is taken to be the unordered set of selected balls. Hence, using the latter representation of the experiment, we see that the desired probability is

$$(61)\,(52)(113)=411$$

which, of course, agrees with the answer obtained previously.

When the experiment consists of a random selection of k items from a set of n items, we have the flexibility of either letting the outcome of the experiment be the ordered selection of the k items or letting it be the unordered set of items selected. In the former case, we would assume that each new selection is equally likely to be any of the so far unselected items of the set, and in the latter case, we would assume that all (nk) possible subsets of k items are equally likely to be the set selected. For instance, suppose 5 people are to be randomly selected from a group of 20 individuals consisting of 10 married couples, and we want to determine P(N), the probability that the 5 chosen are all unrelated. (That is, no two are married to each other.) If we regard the sample space as the set of 5 people chosen, then there are (20 5) equally likely outcomes. An outcome that does not contain a married couple can be thought of as being the result of a six-stage experiment: In the first stage, 5 of the 10 couples to have a member in the group are chosen; in the next 5 stages, 1 of the 2 members of each of these couples is selected. Thus, there are (105)25 possible outcomes in which the 5 members selected are unrelated, yielding the desired probability of

P(N)=(105)25(205)

In contrast, we could let the outcome of the experiment be the *ordered* selection of the 5 individuals. In this setting, there are 20·19·18·17·16 equally likely outcomes, of which 20·18·16·14·12 outcomes result in a group of 5 unrelated individuals, yielding the result

P(N)=20·18·16·14·1220·19·18·17·16

We leave it for the reader to verify that the two answers are identical.

Example 5c

A committee of 5 is to be selected from a group of 6 men and 9 women. If the selection is made randomly, what is the probability that the committee consists of 3 men and 2 women?

Solution

Because each of the (15 5) possible committees is equally likely to be selected, the desired probability is

(63)(92)(155)=2401001

Example 5d

An urn contains n balls, one of which is special. If k of these balls are withdrawn one at a time, with each selection being equally likely to be any of the balls that remain at the time, what is the probability that the special ball is chosen?

Solution

Since all of the balls are treated in an identical manner, it follows that the set of k balls selected is equally likely to be any of the (nk) sets of k balls. Therefore,

P{special ball is selected}=(11)(n−1k−1)(nk)=kn

We could also have obtained this result by letting Ai denote the event that the special ball is the ith ball to be chosen, i=1, ... , k. Then, since each one of the n balls is equally likely to be the ith ball chosen, it follows that P(Ai)=1/n. Hence, because these events are clearly mutually exclusive, we have

P{special ball is selected}=P(∪ i=1 k A i)= ∑ i=1 k P(A i)= k n

We could also have argued that P(A i)=1/n, by noting that there are n(n−1)⋯(n−k+1)=n!/(n−k)! equally likely outcomes of the experiment, of which (n−1)(n−2)⋯(n−i+1)(1)(n−i)⋯(n−k+1)=(n−1)!/(n−k)! result in the special ball being the ith one chosen. From this reasoning, it follows that

P(Ai)=(n−1)!n!=1n

Example 5e

Suppose that n+m balls, of which n are red and m are blue, are arranged in a linear order in such a way that all (n+m)! possible orderings are equally likely. If we record the result of this experiment by listing only the colors of the successive balls, show that all the possible results remain equally likely.

Solution

Consider any one of the (n+m)! possible orderings, and note that any permutation of the red balls among themselves and of the blue balls among themselves does not change the sequence of colors. As a result, every ordering of colorings corresponds to n! m! different orderings of the n+m balls, so every ordering of the colors has probability n!m! (n+m)! of occurring.

For example, suppose that there are 2 red balls, numbered r1,r2, and 2 blue balls, numbered b1,b2. Then, of the 4! possible orderings, there will be 2! 2! orderings that result in any specified color combination. For instance, the following orderings result in the successive balls alternating in color, with a red ball first:

r 1 , b 1 , r 2 , b 2 r 1 , b 2 , r 2 , b 1 r 2 , b 1 , r 1 , b 2 r 2 , b 2 , r 1 , b 1

Therefore, each of the possible orderings of the colors has probability 424=16 of occurring.

Example 5f

A poker hand consists of 5 cards. If the cards have distinct consecutive values and are not all of the same suit, we say that the hand is a straight. For instance, a hand consisting of the five of spades, six of spades, seven of spades, eight of spades, and nine of hearts is a straight. What is the probability that one is dealt a straight?

Solution

We start by assuming that all (525) possible poker hands are equally likely. To determine the number of outcomes that are straights, let us first determine the number of possible outcomes for which the poker hand consists of an ace, two, three, four, and five (the suits being irrelevant). Since the ace can be any 1 of the 4 possible aces, and similarly for the two, three, four, and five, it follows that there are 45 outcomes leading to exactly one ace, two, three, four, and five. Hence, since in 4 of these outcomes all the cards will be of the same suit (such a hand is called a straight flush), it follows that there are 45−4 hands that make up a straight of the form ace, two, three, four, and five. Similarly, there are 45−4 hands that make up a straight of the form ten, jack, queen, king, and ace. Thus, there are 10(45−4) hands that are straights, and it follows that the desired probability is

10(45−4)(525)≈.0039

Example 5g

A 5-card poker hand is said to be a full house if it consists of 3 cards of the same denomination and 2 other cards of the same denomination (of course, different from the first denomination). Thus, a full house is three of a kind plus a pair. What is the probability that one is dealt a full house?

Solution

Again, we assume that all (525) possible hands are equally likely. To determine the number of possible full houses, we first note that there are (42)(43) different combinations of, say, 2 tens and 3 jacks. Because there are 13 different choices for the kind of pair and, after a pair has been chosen, there are 12 other choices for the denomination of the remaining 3 cards, it follows that the probability of a full house is

$13 \cdot 12 \cdot (42)(43)(525) \approx .0014$

Example 5h

In the game of bridge, the entire deck of 52 cards is dealt out to 4 players. What is the probability that

a. one of the players receives all 13 spades;
b. each player receives 1 ace?

Solution

a. Letting Ei be the event that hand i has all 13 spades, then
$P(E_i) = 1 / \binom{52}{13}, \quad i=1,2,3,4$

   Because the events Ei, i=1, 2, 3, 4, are mutually exclusive, the probability that one of the hands is dealt all 13 spades is
$P(\cup_{i=1}^{4} E_i) = \sum_{i=1}^{4} P(E_i) = 4/\binom{52}{13} \approx 6.3 \times 10^{-12}$

b. Let the outcome of the experiment be the sets of 13 cards of each of the players 1, 2, 3, 4. To determine the number of outcomes in which each of the distinct players receives exactly 1 ace, put aside the aces and note that there are (4812, 12, 12, 12) possible divisions of the other 48 cards when each player is to receive 12. Because there are 4! ways of dividing the 4 aces so that each player receives 1, we see that the number of possible outcomes in which each player receives exactly 1 ace is 4!(4812, 12, 12, 12).

As there are $\binom{52}{13, 13, 13, 13}$ possible hands, the desired probability is thus

$4!(4812,12,12,12)/(5213,13,13,13) \approx .1055$

Some results in probability are quite surprising when initially encountered. Our next two examples illustrate this phenomenon.

Example 5i

If n people are present in a room, what is the probability that no two of them celebrate their birthday on the same day of the year? How large need n be so that this probability is less than 12?

Solution

As each person can celebrate his or her birthday on any one of 365 days, there are a total of $(365)^n$ possible outcomes. (We are ignoring the possibility of someone having been born on February 29.) Assuming that each outcome is equally likely, we see that the desired probability is $(365)(364)(363) \cdots (365-n+1)/(365)^n$. It is a rather surprising fact that when $n \geq 23$, this probability is less than 12. That is, if there are 23 or more people in a room, then the probability that at least two of them have the same birthday exceeds 12. Many people are initially surprised by this result, since 23 seems so small in relation to 365, the number of days of the year. However, every pair of individuals has probability 365 (365) 2 = 1 365 of having the same birthday, and in a group of 23 people, there are

(232)=253 different pairs of individuals. Looked at this way, the result no longer seems so surprising.

When there are 50 people in the room, the probability that at least two share the same birthday is approximately .970, and with 100 persons in the room, the odds are better than 3,000,000:1. (That is, the probability is greater than 3×1063×106+1 that at least two people have the same birthday.)

Example 5j

A deck of 52 playing cards is shuffled, and the cards are turned up one at a time until the first ace appears. Is the next card–that is, the card following the first ace–more likely to be the ace of spades or the two of clubs?

Solution

To determine the probability that the card following the first ace is the ace of spades, we need to calculate how many of the (52)! possible orderings of the cards have the ace of spades immediately following the first ace. To begin, note that each ordering of the 52 cards can be obtained by first ordering the 51 cards different from the ace of spades and then inserting the ace of spades into that ordering. Furthermore, for each of the (51)! orderings of the other cards, there is only one place where the ace of spades can be placed so that it follows the first ace. For instance, if the ordering of the other 51 cards is

4c, 6h, Jd, 5s, Ac, 7d, ... , Kh

then the only insertion of the ace of spades into this ordering that results in its following the first ace is

4c, 6h, Jd, 5s, Ac, As, 7d, ... , Kh

Therefore, there are (51)! orderings that result in the ace of spades following the first ace, so

P{the ace of spades follows the first ace}=(51)!(52)!=152

In fact, by exactly the same argument, it follows that the probability that the two of clubs (or any other specified card) follows the first ace is also 152. In other words, each of the 52 cards of the deck is equally likely to be the one that follows the first ace!

Many people find this result rather surprising. Indeed, a common reaction is to suppose initially that it is more likely that the two of clubs (rather than the ace of spades) follows the first ace, since that first ace might itself be the ace of spades. This reaction is often followed by the realization that the two of clubs might itself appear before the first ace, thus negating its chance of immediately following the first ace. However, as there is one chance in four that the ace of spades will be the first ace (because all 4 aces are equally likely to be first) and only one chance in five that the two of clubs will appear before the first ace (because each of the set of 5 cards consisting of the two of clubs and the 4 aces is equally likely to be the first of this set to appear), it again appears that the two of clubs is more likely. However, this is not the case, and our more complete analysis shows that they are equally likely.

Example 5k

A football team consists of 20 offensive and 20 defensive players. The players are to be paired in groups of 2 for the purpose of determining roommates. If the pairing is done at random, what is the probability that there are no offensive–defensive roommate pairs? What is the probability that there are 2i offensive–defensive roommate pairs, i=1, 2, ... , 10?

Solution

There are

(402, 2, ... , 2)=(40)!(2!)20

ways of dividing the 40 players into 20 *ordered* pairs of two each. (That is, there are (40)!/ 2 20 ways of dividing the players into a *first* pair, a *second* pair, and so on.) Hence, there are (40)!/220(20)! ways of dividing the players into (unordered) pairs of 2 each. Furthermore, since a division will result in no offensive–defensive pairs if the offensive (and defensive) players are paired among themselves, it follows that there are [(20)!/210(10)!]2 such divisions. Hence, the probability of no offensive–defensive roommate pairs, call it P0, is given by

P0=((20)!210(10)!)2(40)!220(20)!=[(20)!]3[(10)!]2(40)!

To determine P2i, the probability that there are 2i offensive–defensive pairs, we first note that there are (202i)2 ways of selecting the 2i offensive players and the 2i defensive players who are to be in the offensive–defensive pairs. These 4i players can then be paired up into (2i)! possible offensive–defensive pairs. (This is so because the first offensive player can be paired with any of the 2i defensive players, the second offensive player with any of the remaining 2i−1 defensive players, and so on.) As the remaining 20−2i offensive (and defensive) players must be paired among themselves, it follows that there are

(202i)2(2i)![(20−2i)!210−i(10−i)!]2

divisions that lead to 2i offensive–defensive pairs. Hence,

P2i=(202i)2(2i)![(20−2i)!210−i(10−i)!]2(40)!220(20)! i=0, 1, ... , 10

The P2i, i=0, 1, ..., 10, can now be computed, or they can be approximated by making use of a result of Stirling, which shows that n! can be approximated by nn+1/2e−n2π. For instance, we obtain

P 0 ≈1.3403× 10 −6 P 10 ≈.345861 P 20 ≈7.6068× 10 −6

Our next three examples illustrate the usefulness of the inclusion–exclusion identity (**Proposition 4.4**). In **Example 51**, the introduction of probability enables us to obtain a quick solution to a counting problem.

Example 5l

A total of 36 members of a club play tennis, 28 play squash, and 18 play badminton. Furthermore, 22 of the members play both tennis and squash, 12 play both tennis and badminton, 9 play both squash and badminton, and 4 play all three sports. How many members of this club play at least one of three sports?

Solution

Let N denote the number of members of the club, and introduce probability by assuming that a member of the club is randomly selected. If, for any subset C of members of the club, we let P(C) denote the probability that the selected member is contained in C, then

P(C)=number of members in CN

Now, with T being the set of members that plays tennis, S being the set that plays squash, and B being the set that plays badminton, we have, from **Proposition 4.4**,

P(T∪S∪B) =P(T)+P(S)+P(B)−P(TS)−P(TB)−P(SB)+P(TSB) = 36+28+18−22−12−9+4 N = 43 N

Hence, we can conclude that 43 members play at least one of the sports.

The next example in this section not only possesses the virtue of giving rise to a somewhat surprising answer, but is also of theoretical interest.

Example 5m The Matching Problem

Suppose that each of N men at a party throws his hat into the center of the room. The hats are first mixed up, and then each man randomly selects a hat. What is the probability that none of the men selects his own hat?

Solution

We first calculate the complementary probability of at least one man selecting his own hat. Let us denote by Ei, i=1, 2, ... , N the event that the ith man selects his own hat. Now, by the inclusion-exclusion identity P(∪i=1NEi), the probability that at least one of the men selects his own hat, is given by

$$P(\cup i=1 N E i)= \sum i=1 N P(E i)- \sum i 1 < i 2 P(E i 1 E i 2)+\cdots + (-1) n+1 \sum i 1 < i 2 \cdots< i n P(E i 1 E i 2 \cdots E i n) +\cdots+ (-1) N+1 P(E 1 E 2 \cdots E N)$$

If we regard the outcome of this experiment as a vector of *N* numbers, where the ith element is the number of the hat drawn by the ith man, then there are N! possible outcomes. [The outcome (1, 2, 3, ... , N) means, for example, that each man selects his own hat.] Furthermore, Ei1Ei2...Ein, the event that each of the *n* men i1, i2, ..., in selects his own hat, can occur in any of $(N-n)(N-n-1)\cdots 3\cdot 2\cdot 1=(N-n)!$ possible ways; for, of the remaining N−n men, the first can select any of N−n hats, the second can then select any of N−n−1 hats, and so on. Hence, assuming that all N! possible outcomes are equally likely, we see that

$$P(Ei1Ei2\cdots Ein)=(N-n)!N!$$

Also, as there are (Nn) terms in $\sum i1<i2\cdots<inP(Ei1Ei2\cdots Ein)$, it follows that

$$\sum i1<i2\cdots<inP(Ei1Ei2\cdots Ein)=N!(N-n)!n!(N-n)!N!=1n!$$

Thus,

$$P(\cup i=1NEi)=1-12!+13!-\cdots+(-1)N+11N!$$

Hence, the probability that none of the men selects his own hat is

$$1-1+12!-13!+...+(-1)NN!=\sum i=0N(-1)i/i!$$

Upon letting x=−1 in the identity $e x = \sum i=0 \infty x i /i!$, the preceding probability when N is large is seen to be approximately equal to $e-1\approx.3679$. In other words, for N large, the probability that none of the men selects his own hat is approximately .37. (How many readers would have incorrectly thought that this probability would go to 1 as $N\rightarrow\infty$?)

For another illustration of the usefulness of the inclusion-exclusion identity, consider the following example.

Example 5n

Compute the probability that if 10 married couples are seated at random at a round table, then no wife sits next to her husband.

Solution

If we let Ei, i=1, 2, ... , 10 denote the event that the ith couple sit next to each other, it follows that the desired probability is 1−P(∪i=110Ei). Now, from the inclusion-exclusion identity,

P(∪ 1 10 E i)= ∑ 1 10 P(E i)−⋯+ (−1) n+1 ∑ i 1 < i 2 <⋯< i n P(E i 1 E i 2 ⋯ E i n) +⋯−P(E 1 E 2 ⋯ E 10)

To compute P(Ei1Ei2⋯Ein), we first note that there are 19! ways of arranging 20 people around a round table. (Why?) The number of arrangements that result in a specified set of n men sitting next to their wives can most easily be obtained by first thinking of each of the n married couples as being single entities. If this were the case, then we would need to arrange 20−n entities around a round table, and there are clearly (20−n−1)! such arrangements. Finally, since each of the n married couples can be arranged next to each other in one of two possible ways, it follows that there are 2n(20−n−1)! arrangements that result in a specified set of n men each sitting next to their wives. Therefore,

P(Ei1Ei2⋯Ein)=2n(19−n)!(19)!

Thus, from **Proposition 4.4**, we obtain that the probability that at least one married couple sits together is

(101)21(18)!(19)!−(102)22(17)!(19)!+(103)23(16)!(19)!−⋯−(1010)2109!(19)!≈.6605

and the desired probability is approximately .3395.

***Example 5o Runs**

Consider an athletic team that had just finished its season with a final record of n wins and m losses. By examining the sequence of wins and losses, we are hoping to determine whether the team had stretches of games in which it was more likely to win than at other times. One way to gain some insight into this question is to count the number of runs of wins and then see how likely that result would be when all (n+m)!/(n! m!) orderings of the n wins and m losses are assumed equally likely. By a run of wins, we mean a consecutive sequence of wins. For instance, if n=10, m=6, and the sequence of outcomes was *WWLLWWWLWLLLWWWW*, then there would be 4 runs of wins–the first run being of size 2, the second of size 3, the third of size 1, and the fourth of size 4.

Suppose now that a team has n wins and m losses. Assuming that all (n+m)!/ (n! m!)=(n+mn) orderings are equally likely, let us determine the probability that there will be exactly r runs of wins. To do so, consider first any vector of positive integers x 1 , x 2 ,…, x r with x1+⋯+xr=n, and let us see how many outcomes result in *r* runs of wins in which the ith run is of size xi, i=1, ... , r. For any such outcome, if we let y1 denote the number of losses before the first run of wins, y2 the number of losses between the first 2 runs of wins, ... , yr+1 the number of losses after the last run of wins, then the yi satisfy

y1+y2+⋯+yr+1=m y1≥0,yr+1≥0,yi>0,i=2, ... , r

and the outcome can be represented schematically as

LL ... L‿y1 WW ... W‿x1 L ... L‿y2 WW ... W‿x2 ⋯ WW‿xr L ... L‿yr+1

Hence, the number of outcomes that result in r runs of wins the ith of size x i ,i=1, … r− is equal to the number of integers y1 , ... , yr+1 that satisfy the foregoing, or, equivalently, to the number of positive integers

y ‾ 1 = y 1 +1 y ‾ 1 = y i , i=2,…,r, y ‾ r+1 = y r+1 +1

that satisfy

$$\bar{y}_1+\bar{y}_2+\cdots+\bar{y}_{r+1}=m+2$$

By **Proposition 6.1** in **Chapter 1**, there are $\binom{m+1}{r}$ such outcomes. Hence, the total number of outcomes that result in r runs of wins is $\binom{m+1}{r}$ multiplied by the number of positive integral solutions of $x_1+\cdots+x_r=n$. Thus, again from **Proposition 6.1**, there are $\binom{n-1}{r-1}\binom{m+1}{r}$ outcomes resulting in r runs of wins. As there are $\binom{n+m}{n}$ equally likely outcomes, it follows that

$$P(\{\text{r runs of wins}\})=\frac{\binom{m+1}{r}\binom{n-1}{r-1}}{\binom{m+n}{n}} \quad r\geq 1$$

For example, if n=8 and m=6, then the probability of 7 runs is $\binom{7}{7}\binom{7}{6}/\binom{14}{8}=1/429$ if all $\binom{14}{8}$ outcomes are equally likely. Hence, if the outcome was *WLWLWLWLWWLWLW*, then we might suspect that the team's probability of winning was changing over time. (In particular, the probability that the team wins seems to be quite high when it lost its last game and quite low when it won its last game.) On the other extreme, if the outcome were *WWWWWWWWWLLLLL*, then there would have been only 1 run, and as $P(\{1\text{ run}\})=\binom{7}{1}\binom{7}{0}/\binom{14}{8}=1/429$, it would thus again seem unlikely that the team's probability of winning remained unchanged over its 14 games.

Summary

Let S denote the set of all possible outcomes of an experiment. S is called the *sample space* of the experiment. An *event* is a subset of S. If Ai,i=1,...,n, are events, then ∪i=1nAi, called the *union* of these events, consists of all outcomes that are in at least one of the events Ai,i=1,...,n. Similarly, ∩i=1nAi, sometimes written as A1⋯An, is called the *intersection* of the events Ai and consists of all outcomes that are in all of the events Ai,i=1,...,n.

For any event A, we define Ac to consist of all outcomes in the sample space that are not in A . We call Ac the *complement* of the event *A*. The event Sc, which is empty of outcomes, is designated by Ø and is called the *null* set. If AB=Ø, then we say that A and B are *mutually exclusive*.

For each event A of the sample space S, we suppose that a number P(A), called the probability of A, is defined and is such that

i. $0 \le P(A) \le 1$
ii. $P(S)=1$
iii. For mutually exclusive events Ai,i≥1,

$$P(\cup_{i=1}^{\infty} A_i)=\sum_{i=1}^{\infty} P(A_i)$$

P(A) represents the *probability* that the outcome of the experiment is in A.

It can be shown that

$$P(A^c)=1-P(A)$$

A useful result is that

$$P(A \cup B)=P(A)+P(B)-P(AB)$$

which can be generalized to give

$$P(\cup_{i=1}^{n} A_i)= \Sigma_{i=1}^{n} P(A_i)- \Sigma\Sigma_{i<j} P(A_i A_j) + \sum\sum\sum_{i<j<k} P(A_i A_j A_k) +\cdots+ (-1)^{n+1} P(A_1 \cdots A_n)$$

This result is known as the *inclusion–exclusion identity*.

If S is finite and each one point set is assumed to have equal probability, then

$$P(A)=|A||S|$$

where |E| denotes the number of outcomes in the event E.

P(A) can be interpreted either as a long-run relative frequency or as a measure of one’s degree of belief.

Problems

1. A box contains 3 marbles: 1 red, 1 green, and 1 blue. Consider an experiment that consists of taking 1 marble from the box and then replacing it in the box and drawing a second marble from the box. Describe the sample space. Repeat when the second marble is drawn without replacing the first marble.
2. In an experiment, die is rolled continually until a 6 appears, at which point the experiment stops. What is the sample space of this experiment? Let En denote the event that n rolls are necessary to complete the experiment. What points of the sample space are contained in En? What is (∪1∞En)c?
3. Two dice are thrown. Let E be the event that the sum of the dice is odd, let F be the event that at least one of the dice lands on 1, and let G be the event that the sum is 5. Describe the events EF,E∪F,FG,EFc, and *EFG*.
4. *A*, *B*, and *C* take turns flipping a coin. The first one to get a head wins. The sample space of this experiment can be defined by
S={1,01,001,0001,...,0000⋯

a. Interpret the sample space.
b. Define the following events in terms of S: 10pt
    i. *A* wins = *A*.
    ii. *B* wins = *B*.
    iii. (A∪B)c.
    Assume that *A* flips first, then *B*, then *C*, then *A*, and so on.

5. A system is composed of 5 components, each of which is either working or failed. Consider an experiment that consists of observing the status of each component, and let the outcome of the experiment be given by the vector (x1, x2, x3, x4, x5), where xi is equal to 1 if component i is working and is equal to 0 if component i is failed.
a. How many outcomes are in the sample space of this experiment?
b. Suppose that the system will work if components 1 and 2 are both working, or if components 3 and 4 are both working, or if components 1, 3, and 5 are all working. Let W be the event that the system will work. Specify all the outcomes in W.
c. Let A be the event that components 4 and 5 are both failed. How many outcomes are contained in the event A?
d. Write out all the outcomes in the event *AW*.

6. A hospital administrator codes incoming patients suffering gunshot wounds according to whether they have insurance (coding 1 if they do and 0 if they do not) and according to their condition, which is rated as good (g), fair (f), or serious (s). Consider an experiment that consists of the coding of such a patient.
a. Give the sample space of this experiment.
b. Let A be the event that the patient is in serious condition. Specify the outcomes in A.
c. Let B be the event that the patient is uninsured. Specify the outcomes in B.
d. Give all the outcomes in the event B c ∪A.

7. Consider an experiment that consists of determining the type of job–either blue collar or white collar–and the political affiliation Republican, Democratic, or Independent–of the 15 members of an adult soccer team. How many outcomes are
a. in the sample space?
b. in the event that at least one of the team members is a blue-collar worker?
c. in the event that none of the team members considers himself or herself an Independent?

8. Suppose that A and B are mutually exclusive events for which P(A)=.3 and P(B)=.5. What is the

probability that

a. either A or B occurs?
b. A occurs but B does not?
c. both A and B occur?

9. A retail establishment accepts either the American Express or the VISA credit card. A total of 24 percent of its customers carry an American Express card, 61 percent carry a VISA card, and 11 percent carry both cards. What percentage of its customers carry a credit card that the establishment will accept?

10. Sixty percent of the students at a certain school wear neither a ring nor a necklace. Twenty percent wear a ring and 30 percent wear a necklace. If one of the students is chosen randomly, what is the probability that this student is wearing

a. a ring or a necklace?
b. a ring and a necklace?

11. A total of 28 percent of American males smoke cigarettes, 7 percent smoke cigars, and 5 percent smoke both cigars and cigarettes.

a. What percentage of males smokes neither cigars nor cigarettes?
b. What percentage smokes cigars but not cigarettes?

12. An elementary school is offering 3 language classes: one in Spanish, one in French, and one in German. The classes are open to any of the 100 students in the school. There are 28 students in the Spanish class, 26 in the French class, and 16 in the German class. There are 12 students who are in both Spanish and French, 4 who are in both Spanish and German, and 6 who are in both French and German. In addition, there are 2 students taking all 3 classes.

a. If a student is chosen randomly, what is the probability that he or she is not in any of the language classes?
b. If a student is chosen randomly, what is the probability that he or she is taking exactly one language class?
c. If 2 students are chosen randomly, what is the probability that at least 1 is taking a language class?

13. A certain town with a population of 100,000 has 3 newspapers: I, II, and III. The proportions of townspeople who read these papers are as follows:

I: 10 percent I and II: 8 percent I and II and III: 1 percent
II: 30 percent I and III: 2 percent
III: 5 percent II and III: 4 percent
(The list tells us, for instance, that 8000 people read newspapers I and II.)

a. Find the number of people who read only one newspaper.
b. How many people read at least two newspapers?
c. If I and III are morning papers and II is an evening paper, how many people read at least one morning paper plus an evening paper?
d. How many people do not read any newspapers?
e. How many people read only one morning paper and one evening paper?

14. The following data were given in a study of a group of 1000 subscribers to a certain magazine: In reference to job, marital status, and education, there were 312 professionals, 470 married persons, 525 college graduates, 42 professional college graduates, 147 married college graduates, 86 married professionals, and 25 married professional college graduates. Show that the numbers reported in the study must be incorrect.

Hint: Let M, W, and G denote, respectively, the set of professionals, married persons, and college graduates. Assume that one of the 1000 persons is chosen at random, and use **Proposition 4.4** to show that if the given numbers are correct, then $P(M \cup W \cup G) > 1$.

15. If it is assumed that all (525) poker hands are equally likely, what is the probability of being dealt
    a. a flush? (A hand is said to be a flush if all 5 cards are of the same suit.)
    b. one pair? (This occurs when the cards have denominations a, a, b, c, d, where a, b, c, and d are all distinct.)
    c. two pairs? (This occurs when the cards have denominations a, a, b, b, c, where a, b, and c are all distinct.)
    d. three of a kind? (This occurs when the cards have denominations a, a, a, b, c, where a, b, and c are all distinct.)
    e. four of a kind? (This occurs when the cards have denominations a, a, a, a, b.)

16. Poker dice is played by simultaneously rolling 5 dice. Show that
    a. P{no two alike}=.0926;
    b. P{one pair}=.4630;
    c. P{two pair}=.2315;
    d. P{three alike}=.1543;
    e. P{full house}=.0386;
    f. P{four alike}=.0193;
    g. P{five alike}=.0008.

17. Twenty five people, consisting of 15 women and 10 men are lined up in a random order. Find the probability that the ninth woman to appear is in position 17. That is, find the probability there are 8 women in positions 1 thru 16 and a woman in position 17.
18. Two cards are randomly selected from an ordinary playing deck. What is the probability that they form a blackjack? That is, what is the probability that one of the cards is an ace and the other one is either a ten, a jack, a queen, or a king?
19. Two symmetric dice have had two of their sides painted red, two painted black, one painted yellow, and the other painted white. When this pair of dice is rolled, what is the probability that both dice land with the same color face up?
20. Suppose that you are playing blackjack against a dealer. In a freshly shuffled deck, what is the probability that neither you nor the dealer is dealt a blackjack?
21. A small community organization consists of 20 families, of which 4 have one child, 8 have two children, 5 have three children, 2 have four children, and 1 has five children.
    a. If one of these families is chosen at random, what is the probability it has i children, i=1, 2, 3, 4, 5?
    b. If one of the children is randomly chosen, what is the probability that child comes from a family having i children, i=1, 2, 3, 4, 5?

22. Consider the following technique for shuffling a deck of n cards: For any initial ordering of the cards, go through the deck one card at a time and at each card, flip a fair coin. If the coin comes up heads, then leave the card where it is; if the coin comes up tails, then move that card to the end of the deck. After the coin has been flipped n times, say that one round has been completed. For instance, if n=4 and the initial ordering is 1, 2, 3, 4, then if the successive flips result in the outcome h,t,t,h, then the ordering at the end of the round is 1, 4, 2, 3. Assuming that all possible outcomes of the sequence of n coin flips are equally likely, what is the probability that the ordering after one round is the same as the initial ordering?
23. A pair of fair dice is rolled. What is the probability that the second die lands on a higher value than does the first?
24. If two dice are rolled, what is the probability that the sum of the upturned faces equals i? Find it for i=2, 3, ... , 11, 12.
25. A pair of dice is rolled until a sum of either 5 or 7 appears. Find the probability that a 5 occurs first.
Hint: Let En denote the event that a 5 occurs on the nth roll and no 5 or 7 occurs on the first n−1 rolls. Compute P(En) and argue that $\sum_{n=1}^{\infty} P(E_n)$ is the desired probability.
26. The game of craps is played as follows: A player rolls two dice. If the sum of the dice is either a 2,

3, or 12, the player loses; if the sum is either a 7 or an 11, the player wins. If the outcome is anything else, the player continues to roll the dice until she rolls either the initial outcome or a 7. If the 7 comes first, the player loses, whereas if the initial outcome reoccurs before the 7 appears, the player wins. Compute the probability of a player winning at craps.

Hint: Let Ei denote the event that the initial outcome is i and the player wins. The desired probability is $\sum_{i=2}^{12} P(E_i)$. To compute P(Ei), define the events Ei,n to be the event that the initial sum is *i* and the player wins on the nth roll. Argue that $P(E_i) = \sum_{n=1}^{\infty} P(E_{i,n})$.

27. An urn contains 3 red and 7 black balls. Players A and B withdraw balls from the urn consecutively until a red ball is selected. Find the probability that A selects the red ball. (A draws the first ball, then B, and so on. There is no replacement of the balls drawn.)

28. An urn contains 5 red, 6 blue, and 8 green balls. If a set of 3 balls is randomly selected, what is the probability that each of the balls will be (a) of the same color? (b) of different colors? Repeat under the assumption that whenever a ball is selected, its color is noted and it is then replaced in the urn before the next selection. This is known as *sampling with replacement*.

29. An urn contains n white and m black balls, where n and m are positive numbers.
    a. If two balls are randomly withdrawn, what is the probability that they are the same color?
    b. If a ball is randomly withdrawn and then replaced before the second one is drawn, what is the probability that the withdrawn balls are the same color?
    c. Show that the probability in part (b) is always larger than the one in part (a).

30. The chess clubs of two schools consist of, respectively, 8 and 9 players. Four members from each club are randomly chosen to participate in a contest between the two schools. The chosen players from one team are then randomly paired with those from the other team, and each pairing plays a game of chess. Suppose that Rebecca and her sister Elise are on the chess clubs at different schools. What is the probability that
    a. Rebecca and Elise will be paired?
    b. Rebecca and Elise will be chosen to represent their schools but will not play each other?
    c. either Rebecca or Elise will be chosen to represent her school?

31. A 3-person basketball team consists of a guard, a forward, and a center.
    a. If a person is chosen at random from each of three different such teams, what is the probability of selecting a complete team?
    b. What is the probability that all 3 players selected play the same position?

32. A group of individuals containing b boys and g girls is lined up in random order; that is, each of the (b+g)! permutations is assumed to be equally likely. What is the probability that the person in the ith position, $1 \le i \le b+g$, is a girl?

33. A forest contains 20 elk, of which 5 are captured, tagged, and then released. A certain time later, 4 of the 20 elk are captured. What is the probability that 2 of these 4 have been tagged? What assumptions are you making?

34. The second Earl of Yarborough is reported to have bet at odds of 1000 to 1 that a bridge hand of 13 cards would contain at least one card that is ten or higher. (By *ten or higher* we mean that a card is either a ten, a jack, a queen, a king, or an ace.) Nowadays, we call a hand that has no cards higher than 9 a *Yarborough*. What is the probability that a randomly selected bridge hand is a Yarborough?

35. Seven balls are randomly withdrawn from an urn that contains 12 red, 16 blue, and 18 green balls. Find the probability that
    a. 3 red, 2 blue, and 2 green balls are withdrawn;
    b. at least 2 red balls are withdrawn;
    c. all withdrawn balls are the same color;
    d. either exactly 3 red balls or exactly 3 blue balls are withdrawn.

36. Two cards are chosen at random from a deck of 52 playing cards. What is the probability that they
    a. are both aces?

b. have the same value?

37. An instructor gives her class a set of 10 problems with the information that the final exam will consist of a random selection of 5 of them. If a student has figured out how to do 7 of the problems, what is the probability that he or she will answer correctly
   a. all 5 problems?
   b. at least 4 of the problems?

38. There are *n* socks, 3 of which are red, in a drawer. What is the value of n if, when 2 of the socks are chosen randomly, the probability that they are both red is 12?
39. There are 5 hotels in a certain town. If 3 people check into hotels in a day, what is the probability that they each check into a different hotel? What assumptions are you making?
40. If 4 balls are randomly chosen from an urn containing 4 red, 5 white, 6 blue, and 7 green balls, find the probability that
   a. at least one of the 4 balls chosen is green;
   b. one ball of each color is chosen.

41. If a die is rolled 4 times, what is the probability that 6 comes up at least once?
42. Two dice are thrown *n* times in succession. Compute the probability that double 6 appears at least once. How large need n be to make this probability at least 12?
43.
   a. If N people, including A and B, are randomly arranged in a line, what is the probability that A and B are next to each other?
   b. What would the probability be if the people were randomly arranged in a circle?

44. Five people, designated as A, B, C, D, E, are arranged in linear order. Assuming that each possible order is equally likely, what is the probability that
   a. there is exactly one person between A and B?
   b. there are exactly two people between A and B?
   c. there are three people between A and B?

45. A woman has n keys, of which one will open her door.
   a. If she tries the keys at random, discarding those that do not work, what is the probability that she will open the door on her kth try?
   b. What if she does not discard previously tried keys?

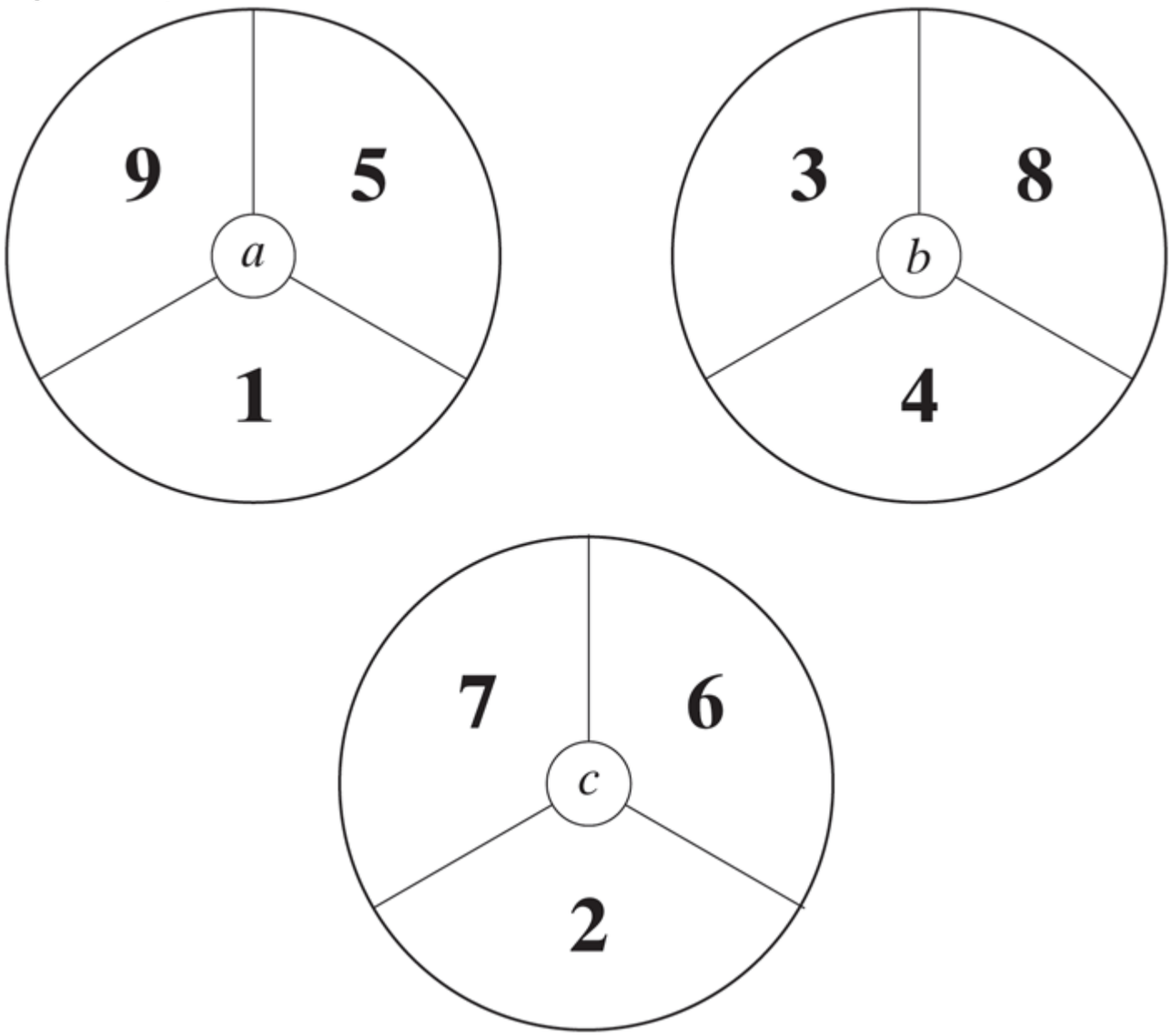
46. How many people have to be in a room in order that the probability that at least two of them celebrate their birthday in the same month is at least 12? Assume that all possible monthly outcomes are equally likely.
47. Suppose that 5 of the numbers 1, 2, ... , 14 are chosen. Find the probability that 9 is the third smallest value chosen.
48. Given 20 people, what is the probability that among the 12 months in the year, there are 4 months containing exactly 2 birthdays and 4 containing exactly 3 birthdays?
49. A group of 6 men and 6 women is randomly divided into 2 groups of size 6 each. What is the probability that both groups will have the same number of men?
50. In a hand of bridge, find the probability that you have 5 spades and your partner has the remaining 8.
51. Suppose that n balls are randomly distributed into N compartments. Find the probability that m balls will fall into the first compartment. Assume that all Nn arrangements are equally likely.
52. A closet contains 10 pairs of shoes. If 8 shoes are randomly selected, what is the probability that there will be
   a. no complete pair?
   b. exactly 1 complete pair?

53. If 8 people, consisting of 4 couples, are randomly arranged in a row, find the probability that no person is next to their partner.
54. Compute the probability that a bridge hand is void in at least one suit. Note that the answer is not (41)(3913)(5213)

(Why not?)
Hint: Use **Proposition 4.4**.
55. Compute the probability that a hand of 13 cards contains
    a. the ace and king of at least one suit;
    b. all 4 of at least 1 of the 13 denominations.

56. Two players play the following game: Player A chooses one of the three spinners pictured in **Figure 2.6**, and then player B chooses one of the remaining two spinners. Both players then spin their spinner, and the one that lands on the higher number is declared the winner. Assuming that each spinner is equally likely to land in any of its 3 regions, would you rather be player *A* or player B? Explain your answer!

Figure 2.6 Spinners



Theoretical Exercises

Prove the following relations:

1. $EF \subset E \subset E \cup F$.
2. If $E \subset F$, then $F^c \subset E^c$.
3. $F = FE \cup FE^c$ and $E \cup F = E \cup E^c F$.
4. $(\cup_{1}^{\infty} E_i)F = \cup_{1}^{\infty} E_i F$ and $(\cap_{1}^{\infty} E_i) \cup F = \cap_{1}^{\infty} (E_i \cup F)$.

5. For any sequence of events $E_1, E_2, \ldots$, define a new sequence $F_1, F_2, \ldots$ of disjoint events (that is, events such that $F_i F_j = \emptyset$ whenever $i \neq j$) such that for all $n \geq 1$, $\cup_{1}^{n} F_i = \cup_{1}^{n} E_i$

6. Let E, F, and G be three events. Find expressions for the events so that, of E, F, and G,
   a. only E occurs;
   b. both E and G, but not F, occur;
   c. at least one of the events occurs;
   d. at least two of the events occur;
   e. all three events occur;
   f. none of the events occurs;
   g. at most one of the events occurs;
   h. at most two of the events occur;
   i. exactly two of the events occur;
   j. at most three of the events occur.

7. Use Venn diagrams
   a. to simplify the expression $(E \cup F)(E \cup F^c)$;
   b. to prove DeMorgan's laws for events E and F. [That is, prove $(E \cup F)^c = E^c F^c$, and $(EF)^c = E^c \cup F^c$.]

8. Let S be a given set. If, for some $k > 0$, $S_1, S_2, \ldots, S_k$ are mutually exclusive nonempty subsets of S such that $\cup_{i=1}^{k} S_i = S$,, then we call the set $\{S_1, S_2, \ldots, S_k\}$ a *partition* of S. Let T_n denote the number of different partitions of $\{1, 2, \ldots, n\}$. Thus, $T_1 = 1$ (the only partition being $S_1 = \{1\}$) and $T_2 = 2$ (the two partitions being $\{\{1,2,\}\}$, $\{\{1\}, \{2\}\}$).
   a. Show, by computing all partitions, that $T_3 = 5, T_4 = 15$.
   b. Show that
      $T_{n+1} = 1 + \sum_{k=1}^{n} \binom{n}{k} T_k$

      and use this equation to compute T_{10}.
      Hint: One way of choosing a partition of n+1 items is to call one of the items *special*. Then we obtain different partitions by first choosing k, $k = 0, 1, \ldots, n$, then a subset of size $n-k$ of the nonspecial items, and then any of the T_k partitions of the remaining k nonspecial items. By adding the special item to the subset of size $n-k$, we obtain a partition of all n+1 items.

9. Suppose that an experiment is performed n times. For any event E of the sample space, let n(E) denote the number of times that event E occurs and define $f(E) = n(E)/n$. Show that $f(\cdot)$ satisfies Axioms 1, 2, and 3.
10. Prove that $P(E \cup F \cup G) = P(E) + P(F) + P(G) - P(E^c FG) - P(EF^c G) - P(EFG^c) - P(EFG^c) - 2P(EFG)$.
11. If $P(E) = .9$ and $P(F) = .8$, show that $P(EF) \geq .7$. In general, prove *Bonferroni's inequality*, namely, $P(EF) \geq P(E) + P(F) - 1$

12. Show that the probability that exactly one of the events E or F occurs equals $P(E)+P(F)-2P(EF)$.
13. Prove that $P(E\ F^c)=P(E)-P(EF)$.
14. Prove **Proposition 4.4** by mathematical induction.
15. An urn contains M white and N black balls. If a random sample of size r is chosen, what is the probability that it contains exactly k white balls?
16. Use induction to generalize *Bonferroni's inequality* to n events. That is, show that $P(E_1E_2\cdots E_n)\geq P(E_1)+\cdots+P(E_n)-(n-1)$

17. Consider the matching problem, **Example 5m**, and define A_N to be the number of ways in which the N men can select their hats so that no man selects his own. Argue that $A_N=(N-1)(A_{N-1}+A_{N-2})$

This formula, along with the boundary conditions $A_1=0$, $A_2=1$, can then be solved for A_N, and the desired probability of no matches would be $A_N/N!$.
Hint: After the first man selects a hat that is not his own, there remain N−1 men to select among a set of N−1 hats that does not contain the hat of one of these men. Thus, there is one extra man and one extra hat. Argue that we can get no matches either with the extra man selecting the extra hat or with the extra man not selecting the extra hat.
18. Let f_n denote the number of ways of tossing a coin n times such that successive heads never appear. Argue that
$f_n=f_{n-1}+f_{n-2}\qquad n\geq 2$, where $f_0\equiv 1$, $f_1\equiv 2$

Hint: How many outcomes are there that start with a head, and how many start with a tail? If P_n denotes the probability that successive heads never appear when a coin is tossed n times, find P_n (in terms of f_n) when all possible outcomes of the n tosses are assumed equally likely. Compute P_{10}.
19. An urn contains n red and m blue balls. They are withdrawn one at a time until a total of $r, r\leq n$, red balls have been withdrawn. Find the probability that a total of k balls are withdrawn.
Hint: A total of k balls will be withdrawn if there are r−1 red balls in the first k−1 withdrawals and the kth withdrawal is a red ball.
20. Consider an experiment whose sample space consists of a countably infinite number of points. Show that not all points can be equally likely. Can all points have a positive probability of occurring?
*21. Consider **Example 5o**, which is concerned with the number of runs of wins obtained when n wins and m losses are randomly permuted. Now consider the total number of runs–that is, win runs plus loss runs–and show that
$P\{2k \text{ runs}\}=2\binom{m-1}{k-1}\binom{n-1}{k-1}\binom{m+n}{n}$ $P\{2k+1 \text{ runs}\}=\binom{m-1}{k-1}\binom{n-1}{k}+\binom{m-1}{k}\binom{n-1}{k-1}\binom{m+n}{n}$

Self-Test Problems and Exercises

1. A cafeteria offers a three-course meal consisting of an entree, a starch, and a dessert. The possible choices are given in the following table:

Course	Choices
Entree	Chicken or roast beef
Starch	Pasta or rice or potatoes
Dessert	Ice cream or Jello or apple pie or a peach

A person is to choose one course from each category.

a. How many outcomes are in the sample space?
b. Let A be the event that ice cream is chosen. How many outcomes are in A?
c. Let B be the event that chicken is chosen. How many outcomes are in B?
d. List all the outcomes in the event *AB*.
e. Let C be the event that rice is chosen. How many outcomes are in C?
f. List all the outcomes in the event *ABC*.

2. A customer visiting the suit department of a certain store will purchase a suit with probability .22, a shirt with probability .30, and a tie with probability .28. The customer will purchase both a suit and a shirt with probability .11, both a suit and a tie with probability .14, and both a shirt and a tie with probability .10. A customer will purchase all 3 items with probability .06. What is the probability that a customer purchases

a. none of these items?
b. exactly 1 of these items?

3. A deck of cards is dealt out. What is the probability that the 14th card dealt is an ace? What is the probability that the first ace occurs on the 14th card?

4. Let A denote the event that the midtown temperature in Los Angeles is 70◦F, and let B denote the event that the midtown temperature in New York is 70 ◦ F . Also, let C denote the event that the maximum of the midtown temperatures in New York and in Los Angeles is 70◦F. If P(A)=.3,P(B)=.4, and P(C)=.2, find the probability that the minimum of the two midtown temperatures is 70◦F.

5. An ordinary deck of 52 cards is shuffled. What is the probability that the top four cards have

a. different denominations?
b. different suits?

6. Urn A contains 3 red and 3 black balls, whereas urn B contains 4 red and 6 black balls. If a ball is randomly selected from each urn, what is the probability that the balls will be the same color?

7. In a state lottery, a player must choose 8 of the numbers from 1 to 40. The lottery commission then performs an experiment that selects 8 of these 40 numbers. Assuming that the choice of the lottery commission is equally likely to be any of the (40 8) combinations, what is the probability that a player has

a. all 8 of the numbers selected by the lottery commission?
b. 7 of the numbers selected by the lottery commission?
c. at least 6 of the numbers selected by the lottery commission?

8. From a group of 3 first-year students, 4 sophomores, 4 juniors, and 3 seniors, a committee of size 4 is randomly selected. Find the probability that the committee will consist of

3.2 Conditional Probabilities

Suppose that we toss 2 dice, and suppose that each of the 36 possible outcomes is equally likely to occur and hence has probability 136. Suppose further that we observe that the first die is a 3. Then, given this information, what is the probability that the sum of the 2 dice equals 8? To calculate this probability, we reason as follows: Given that the initial die is a 3, there can be at most 6 possible outcomes of our experiment, namely, (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), and (3, 6). Since each of these outcomes originally had the same probability of occurring, the outcomes should still have equal probabilities. That is, given that the first die is a 3, the (conditional) probability of each of the outcomes (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), and (3, 6) is 16, whereas the (conditional) probability of the other 30 points in the sample space is 0. Hence, the desired probability will be 16.

If we let E and F denote, respectively, the event that the sum of the dice is 8 and the event that the first die is a 3, then the probability just obtained is called the *conditional probability that E occurs given that F has occurred* and is denoted by

$P(E|F)$

A general formula for $P(E|F)$ that is valid for all events E and F is derived in the same manner: If the event F occurs, then, in order for E to occur, it is necessary that the actual occurrence be a point both in E and in F; that is, it must be in EF. Now, since we know that F has occurred, it follows that F becomes our new, or reduced, sample space; hence, the probability that the event EF occurs will equal the probability of *EF* relative to the probability of F. That is, we have the following definition.

> Definition
>
> If $P(F)>0$, then
>
> $$P(E|F)=P(EF)P(F) \tag{2.1}$$

Example 2a

Joe is 80 percent certain that his missing key is in one of the two pockets of his hanging jacket, being 40 percent certain it is in the left-hand pocket and 40 percent certain it is in the right-hand pocket. If a search of the left-hand pocket does not find the key, what is the conditional probability that it is in the other pocket?

Solution

If we let L be the event that the key is in the left-hand pocket of the jacket, and R be the event that it is in the right-hand pocket, then the desired probability $P(R|Lc)$ can be obtained as follows:

$$P(R|Lc)=P(RLc)P(Lc)=P(R)1−P(L)=2/3$$

If each outcome of a finite sample space S is equally likely, then, conditional on the event that the outcome lies in a subset $F⊂S$, all outcomes in F become equally likely. In such cases, it is often convenient to compute conditional probabilities of the form $P(E|F)$ by using F as the sample space. Indeed, working with this reduced sample space often results in an easier and better understood solution. Our next two examples illustrate this point.

Example 2b

A coin is flipped twice. Assuming that all four points in the sample space $S={(h,h),(h,t),(t,h),(t,t)}$ are equally likely, what is the conditional probability that both flips land on heads, given that (a) the first flip lands on heads? (b) at least one flip lands on heads?

Solution

Let $B={(h,h)}$ be the event that both flips land on heads; let $F={(h,h),(h,t)}$ be the event that the first flip lands on heads; and let $A={(h,h),(h,t),(t,h)}$ be the event that at least one flip lands on heads. The probability for (a) can be obtained from

$$P(B|F)=P(BF)P(F)=P({(h,h)})P({(h,h),(h,t)})=1/42/4=1/2$$

For (b), we have

$$P(B|A)=P(BA)P(A)=P({(h,h)})P({(h,h),(h,t),(t,h)})=1/43/4=1/3$$

Thus, the conditional probabilityhat both flips land on heads given that the first one does is 1/2, whereas the conditional probability that both flips land on heads given that at least one does is only 1/3. Many students initially find this latter result surprising. They reason that given that at least one flip lands on heads, there are two possible results: Either they both land on heads or only one does. Their mistake, however, is in assuming that these two possibilities are equally likely. Initially there are 4 equally likely outcomes. Because the information that at least one flip lands on heads is equivalent to the information that the outcome is not (t,t), we are left with the 3 equally likely outcomes (h,h),(h,t),(t,h), only one of which results in both flips landing on heads.

Example 2c

In the card game bridge, the 52 cards are dealt out equally to 4 players–called East, West, North, and South. If North and South have a total of 8 spades among them, what is the probability that East has 3 of the remaining 5 spades?

Solution

Probably the easiest way to compute the desired probability is to work with the reduced sample space. That is, given that North–South have a total of 8 spades among their 26 cards, there remains a total of 26 cards, exactly 5 of them being spades, to be distributed among the East–West hands. Since each distribution is equally likely, it follows that the conditional probability that East will have exactly 3 spades among his or her 13 cards is

$$(53)(2110)(2613)≈.339$$

Multiplying both sides of **Equation (2.1)** by P(F), we obtain

$$P(EF)=P(F)P(E|F) \tag{2.2}$$

In words, **Equation (2.2)** states that the probability that both E and F occur is equal to the probability that F occurs multiplied by the conditional probability of E given that F occurred. **Equation (2.2)** is often quite useful in computing the probability of the intersection of events.

Example 2d

Celine is undecided as to whether to take a French course or a chemistry course. She estimates that her probability of receiving an A grade would be 12 in a French course and 23 in a chemistry course. If Celine decides to base her decision on the flip of a fair coin, what is the probability that she gets an A in chemistry?

Solution

Let *C* be the event that Celine takes chemistry and A denote the event that she receives an A in whatever course she takes, then the desired probability is $P(CA)$, which is calculated by using **Equation (2.2)** as follows:

$$P(CA)=P(C)P(A|C)=(12)(23)=13$$

Example 2e

Suppose that an urn contains 8 red balls and 4 white balls. We draw 2 balls from the urn without replacement. (a) If we assume that at each draw, each ball in the urn is equally likely to be chosen, what is the probability that both balls drawn are red? (b) Now suppose that the balls have different weights, with each red ball having weight r and each white ball having weight w. Suppose that the probability that a given ball in the urn is the next one selected is its weight divided by the sum of the weights of all balls currently in the urn. Now what is the probability that both balls are red?

Solution

Let R1 and R2 denote, respectively, the events that the first and second balls drawn are red. Now, given that the first ball selected is red, there are 7 remaining red balls and 4 white balls, so $P(R2|R1)=711$. As $P(R1)$ is clearly 812, the desired probability is

P(R1R2)=P(R1)P(R2|R1)=(23)(711)=1433

Of course, this probability could have been computed by P(R1R2)=(82)/(122).

For part (b), we again let Ri be the event that the ith ball chosen is red and use

P(R1R2)=P(R1)P(R2|R1)

Now, number the red balls, and let Bi,i=1,…,8 be the event that the first ball drawn is red ball number i. Then

P(R1)=P(∪i=18Bi)=∑i=18P(Bi)=8 r8r+4w

Moreover, given that the first ball is red, the urn then contains 7 red and 4 white balls. Thus, by an argument similar to the preceding one,

P(R2|R1)=7r7r+4w

Hence, the probability that both balls are red is

P(R1R2)=8r8r+4w 7r7r+4w

A generalization of **Equation (2.2)**, which provides an expression for the probability of the intersection of an arbitrary number of events, is sometimes referred to as the *multiplication rule*.

> The multiplication rule
>
> P(E1E2E3⋯En)=P(E1)P(E2|E1)P(E3|E1E2)⋯P(En|E1⋯En−1)

In words, the multiplication rule states that P(E1E2⋯En), the probability that all of the events E1,E2,…,En occur, is equal to P(E1), the probability that E1 occurs, multiplied by P(E2|E1), the conditional probability that E2 occurs given that E1 has occurred, multiplied by P(E3|E1E2), the conditional probability that E3 occurs given that both E1 and E2 have occurred, and so on.

To prove the multiplication rule, just apply the definition of conditional probability to its right-hand side, giving

P(E1)P(E1E2)P(E1)P(E1E2E3)P(E1E2)⋯P(E1E2⋯En)P(E1E2⋯En−1)=P(E1E2⋯En)

Example 2f

In the match problem stated in **Example 5m** of **Chapter 2**, it was shown that PN, the probability that there are no matches when N people randomly select from among their own N hats, is given by

PN=∑i=0N(−1)i/i!

What is the probability that exactly k of the N people have matches?

Solution

Let us fix our attention on a particular set of k people and determine the probability that these k individuals have matches and no one else does. Letting E denote the event that everyone in this set has a match, and letting G be the event that none of the other N−k people have a match, we have

P(EG)=P(E)P(G|E)

Now, let Fi,i=1,…,k, be the event that the ith member of the set has a match. Then

P(E)=P(F1F2⋯Fk)=P(F1)P(F2|F1)P(F3|F1F2)⋯P(Fk|F1⋯Fk−1)=1N 1N−1 1N−2⋯1N−k+1=(N−k)!N!

Given that everyone in the set of k has a match, the other N−k people will be randomly choosing among their own N−k hats, so the probability that none of them has a match is equal to the probability of no matches in a problem having N−k people choosing among their own N−k hats. Therefore,

P(G|E)=PN−k=∑i=0N−k(−1)i/i!

showing that the probability that a specified set of k people have matches and no one else does is

P(EG)=(N−k)!N! PN−k

Because there will be exactly k matches if the preceding is true for any of the (Nk) sets of k individuals, the desired probability is

P(exactly k matches)=(Nk)P(EG)=PN−K/K!≈e−1/k!when N is large

Example 2g

An ordinary deck of 52 playing cards is randomly divided into 4 piles of 13 cards each. Compute the probability that each pile has exactly 1 ace.

Solution

Define events Ei,i=1,2,3,4, as follows:

E1={the ace of spades is in any one of the piles}E2={the ace of spades and the ace of hearts are in different piles}E3={the aces of spades, hearts, and diamonds are all in different piles}E4={all 4 aces are in different piles}

The desired probability is P(E1E2E3E4), and by the multiplication rule,

P(E1E2E3E4)=P(E1)P(E2|E1)P(E3|E1E2)P(E4|E1E2E3)

Now,

P(E1)=1

since E1 is the sample space S. To determine P(E2|E1), consider the pile that contains the ace of spades. Because its remaining 12 cards are equally likely to be any 12 of the remaining 51 cards, the probability that the ace of hearts is among them is 12/51, giving that

P(E2|E1)=1−1251=3951

Also, given that the ace of spades and ace of hearts are in different piles, it follows that the set of the remaining 24 cards of these two piles is equally likely to be any set of 24 of the remaining 50 cards. As the probability that the ace of diamonds is one of these 24 is 24/50, we see that

P(E3|E1E2)=1−2450=2650

Because the same logic as used in the preceding yields that

P(E4|E1E2E3)=1−3649=1349

the probability that each pile has exactly 1 ace is

P(E1E2E3E4)=39·26·1351·50·49≈.105

That is, there is approximately a 10.5 percent chance that each pile will contain an ace. (**Problem 13** gives another way of using the multiplication rule to solve this problem.)

Example 2h

Four of the eight teams in the quarterfinal round of the 2016 European Champions League Football (soccer) tournament were the acknowledged strong teams Barcelona, Bayern Munich, Real Madrid, and Paris St-Germain. The pairings in this round are supposed to be totally random, in the sense that all possible pairings are equally likely. Assuming this is so, find the probability that none of the strong teams play each other in this round. (Surprisingly, it seems to be a common occurrence in this tournament that, even though the pairings are supposedly random, the very strong teams are rarely matched against each other in this round.)

Solution

If we number the four strong teams 1 through 4, and then let Wi,i=1,2,3,4, be the event that team i plays one of the four weak teams, then the desired probability is P(W1W2W3W4). By the multiplication rule

P(W1W2W3W4)=P(W1)P(W2|W1)P(W3|W1W2)P(W4|W1W2W3)=(4/7)(3/5)(2/3)(1)=8/35

The preceding follows by first noting that because team 1 is equally likely to be matched with any of the other 7 teams, we have that P(W1)=4/7. Now, given that W1 occurs, team 2 is equally likely to be matched with any of five teams: namely, teams 3, 4, or any of the three weak teams not matched with team 1. As three of these five teams are weak, we see that P(W2|W1)=3/5. Similarly, given that events W1 and W2 have occurred, team 3 is equally likely to be matched with any from a set of three teams, consisting of team 4 and the remaining two weaker teams not matched with 1 or 2. Hence, P(W3|W1W2)=2/3. Finally, given that W1, W2, and W3 all occur, team 4 will be matched with the remaining weak team not matched with any of 1,2,3, giving that P(W4|W1W2W3)=1.

Remarks Our definition of P(E|F) is consistent with the interpretation of probability as being a long-run relative frequency. To see this, suppose that n repetitions of the experiment are to be performed, where n is large. We claim that if we consider only those experiments in which F occurs, then P(E|F) will equal the long-run proportion of them in which E also occurs. To verify this statement, note that since P(F) is the long-run proportion of experiments in which F occurs, it follows that in the n repetitions of the experiment, F will occur approximately nP(F) times. Similarly, in approximately nP(EF) of these experiments, both E and F will occur. Hence, out of the approximately nP(F) experiments in which F occurs, the proportion of them in which E also occurs is approximately equal to

nP(EF)nP(F)=P(EF)P(F)

Because this approximation becomes exact as n becomes larger and larger, we have the appropriate definition of P(E|F).

3.3 Bayes's Formula

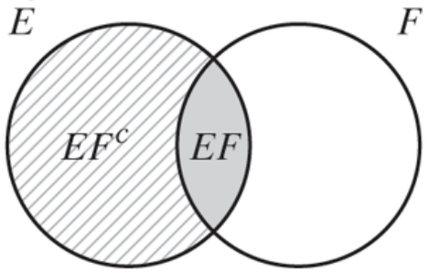
Let E and F be events. We may express E as

$$E=EF\cup EFc$$

for, in order for an outcome to be in E, it must either be in both E and F or be in E but not in F. (See **Figure 3.1**.) As *EF* and EFc are clearly mutually exclusive, we have, by Axiom 3,

$$P(E)=P(EF)+P(EFc)=P(E|F)P(F)+P(E|Fc)P(Fc)=P(E|F)P(F)+P(E|Fc)[1-P(F)] \quad (3.1)$$

Figure 3.1 E=EF∪EFc. EF=Shaded Area; EFc=Striped Area.



Equation (3.1) states that the probability of the event E is a weighted average of the conditional probability of E given that F has occurred and the conditional probability of E given that F has not occurred—each conditional probability being given as much weight as the event on which it is conditioned has of occurring. This is an extremely useful formula, because its use often enables us to determine the probability of an event by first "conditioning" upon whether or not some second event has occurred. That is, there are many instances in which it is difficult to compute the probability of an event directly, but it is straightforward to compute it once we know whether or not some second event has occurred. We illustrate this idea with some examples.

Example 3a

(Part 1)

An insurance company believes that people can be divided into two classes: those who are accident prone and those who are not. The company's statistics show that an accident-prone person will have an accident at some time within a fixed 1-year period with probability .4, whereas this probability decreases to .2 for a person who is not accident prone. If we assume that 30 percent of the population is accident prone, what is the probability that a new policyholder will have an accident within a year of purchasing a policy?

Solution

We shall obtain the desired probability by first conditioning upon whether or not the policyholder is accident prone. Let A1 denote the event that the policyholder will have an accident within a year of purchasing the policy, and let A denote the event that the policyholder is accident prone. Hence, the desired probability is given by

$$P(A1)=P(A1|A)P(A)+P(A1|Ac)P(Ac)=(.4)(.3)+(.2)(.7)=.26$$

Example 3a

(Part 2)

Suppose that a new policyholder has an accident within a year of purchasing a policy. What is the probability that he or she is accident prone?

Solution

The desired probability is

$$P(A|A1)=P(AA1)P(A1)=P(A)P(A1|A)P(A1)=(.3)(.4).26=613$$

Example 3b

Consider the following game played with an ordinary deck of 52 playing cards: The cards are shuffled and then turned over one at a time. At any time, the player can guess that the next card to be turned over will be the ace of spades; if it is, then the player wins. In addition, the player is said to win if the ace of spades has not yet appeared when only one card remains and no guess has yet been made. What is a good strategy? What is a bad strategy?

Solution

Every strategy has probability 1/52 of winning! To show this, we will use induction to prove the stronger result that for an n card deck, one of whose cards is the ace of spades, the probability of winning is 1/n, no matter what strategy is employed. Since this is clearly true for n=1, assume it to be true for an n−1 card deck, and now consider an n card deck. Fix any strategy, and let p denote the probability that the strategy guesses that the first card is the ace of spades. Given that it does, the player's probability of winning is 1/n. If, however, the strategy does not guess that the first card is the ace of spades, then the probability that the player wins is the probability that the first card is not the ace of spades, namely, (n−1)/n, multiplied by the conditional probability of winning given that the first card is not the ace of spades. But this latter conditional probability is equal to the probability of winning when using an n−1 card deck containing a single ace of spades; it is thus, by the induction hypothesis, 1/(n−1). Hence, given that the strategy does not guess the first card, the probability of winning is

$$\frac{n-1}{n}\frac{1}{n-1}=\frac{1}{n}$$

Thus, letting G be the event that the first card is guessed, we obtain

$$P\{\text{win}\}=P\{\text{win}|G\}P(G)+P\{\text{win}|G^c\}(1-P(G))=\frac{1}{n}p+\frac{1}{n}(1-p)=\frac{1}{n}$$

Example 3c

In answering a question on a multiple-choice test, a student either knows the answer or guesses. Let p be the probability that the student knows the answer and 1−p be the probability that the student guesses. Assume that a student who guesses at the answer will be correct with probability 1/m, where m is the number of multiple-choice alternatives. What is the conditional probability that a student knew the answer to a question given that he or she answered it correctly?

Solution

Let C and K denote, respectively, the events that the student answers the question correctly and the event that he or she actually knows the answer. Now,

$$P(K|C)=\frac{P(KC)}{P(C)}=\frac{P(C|K)P(K)}{P(C|K)P(K)+P(C|K^c)P(K^c)}=\frac{p}{p+(1/m)(1-p)}=\frac{mp}{1+(m-1)p}$$

For example, if m=5,p=12, then the probability that the student knew the answer to a question he or she answered correctly is 56.

Example 3d

A laboratory blood test is 95 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a "false positive" result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then, with probability .01, the test result will imply that he or she has the disease.) If .5 percent of the population actually has the disease, what is the probability that a person has the disease given that the test result is positive?

Solution

Let D be the event that the person tested has the disease and E the event that the test result is positive. Then the desired probability is

$$P(D|E)=\frac{P(DE)}{P(E)}=\frac{P(E|D)P(D)}{P(E|D)P(D)+P(E|D^c)P(D^c)}=\frac{(.95)(.005)}{(.95)(.005)+(.01)(.995)}=\frac{95}{294}\approx.323$$

Thus, only 32 percent of those persons whose test results are positive actually have the disease. Many students are often surprised at this result (they expect the percentage to be much higher, since the blood test seems to be a good one), so it is probably worthwhile to present a second argument that, although less rigorous than the preceding one, is probably more revealing. We now do so.

Since .5 percent of the population actually has the disease, it follows that, on the average, 1 person out of every 200 tested will have it. The test will correctly confirm that this person has the disease with probability .95. Thus, on the average, out of every 200 persons tested, the test will correctly confirm that .95 person has the disease. On the other hand, out of the (on the average) 199 healthy people, the test will incorrectly state that (199)(.01) of these people have the disease. Hence, for every .95 diseased persons that the test correctly states is ill, there are (on the average) (199)(.01) healthy persons who the test incorrectly states are ill. Thus, the proportion of time that the test result is correct when it states that a person is ill is

$$\frac{.95}{.95+(199)(.01)}=\frac{95}{294}\approx.323$$

Equation (3.1) is also useful when one has to reassess one's personal probabilities in the light of additional information. For instance, consider the examples that follow.

Example 3e

Consider a medical practitioner pondering the following dilemma: "If I'm at least 80 percent certain that my patient has this disease, then I always recommend surgery, whereas if I'm not quite as certain, then I recommend additional tests that are expensive and sometimes painful. Now, initially I was only 60 percent certain that Jones had the disease, so I ordered the series A test, which always gives a positive result when the patient has the disease and almost never does when he is healthy. The test result was positive, and I was all set to

recommend surgery when Jones informed me, for the first time, that he was diabetic. This information complicates matters because, although it doesn't change my original 60 percent estimate of his chances of having the disease in question, it does affect the interpretation of the results of the A test. This is so because the A test, while never yielding a positive result when the patient is healthy, does unfortunately yield a positive result 30 percent of the time in the case of *diabetic* patients who are not suffering from the disease. Now what do I do? More tests or immediate surgery?"

Solution

In order to decide whether or not to recommend surgery, the doctor should first compute her updated probability that Jones has the disease given that the A test result was positive. Let D denote the event that Jones has the disease and E the event that the A test result is positive. The desired conditional probability is then

$$P(D|E)=P(DE)P(E)=P(D)P(E|D)P(E|D)P(D)+P(E|Dc)P(Dc)=(.6)11(.6)+(.3)(.4)=.833$$

Note that we have computed the probability of a positive test result by conditioning on whether or not Jones has the disease and then using the fact that because Jones is a diabetic, his conditional probability of a positive result given that he does not have the disease, P(E|Dc), equals .3. Hence, as the doctor should now be more than 80 percent certain that Jones has the disease, she should recommend surgery.

Example 3f

At a certain stage of a criminal investigation, the inspector in charge is 60 percent convinced of the guilt of a certain suspect. Suppose, however, that a *new* piece of evidence which shows that the criminal has a certain characteristic (such as left-handedness, baldness, or brown hair) is uncovered. If 20 percent of the population possesses this characteristic, how certain of the guilt of the suspect should the inspector now be if it turns out that the suspect has the characteristic?

Solution

Letting G denote the event that the suspect is guilty and C the event that he possesses the characteristic of the criminal, we have

$$P(G|C)=P(GC)P(C)=P(C|G)P(G)P(C|G)P(G)+P(C|Gc)P(Gc)=1(.6)1(.6)+(.2)(.4)\approx.882$$

where we have supposed that the probability of the suspect having the characteristic if he is, in fact, innocent is equal to .2, the proportion of the population possessing the characteristic.

Example 3g

In the world bridge championships held in Buenos Aires in May 1965, the famous British bridge partnership of Terrence Reese and Boris Schapiro was accused of cheating by using a system of finger signals that could indicate the number of hearts held by the players. Reese and Schapiro denied the accusation, and eventually a hearing was held by the British bridge league. The hearing was in the form of a legal proceeding with prosecution and defense teams, both having the power to call and cross-examine witnesses. During the course of the proceeding, the prosecutor examined specific hands played by Reese and Schapiro and claimed that their playing these hands was consistent with the hypothesis that they were guilty of having illicit knowledge of the heart suit. At this point, the defense attorney pointed out that their play of these hands was also perfectly consistent with their standard line of play. However, the prosecution then argued that as long as their play was consistent with the hypothesis of guilt, it must be counted as evidence toward that hypothesis. What do you think of the reasoning of the prosecution?

Solution

The problem is basically one of determining how the introduction of new evidence (in this example, the playing of the hands) affects the probability of a particular hypothesis. If we let H denote a particular hypothesis (such as the hypothesis that Reese and Schapiro are guilty) and E the new evidence, then

$$P(H|E)=P(HE)P(E)=P(E|H)P(H)P(E|H)P(H)+P(E|Hc)[1-P(H)] \tag{3.2}$$

where *P(H)* is our evaluation of the likelihood of the hypothesis before the introduction of the new evidence. The new evidence will be in support of the hypothesis whenever it makes the hypothesis more likely—that is, whenever $P(H|E)\geq P(H)$. From **Equation (3.2)**, this will be the case whenever

$$P(E|H)\geq P(E|H)P(H)+P(E|Hc)[1-P(H)]$$

or, equivalently, whenever

$$P(E|H)\geq P(E|Hc)$$

In other words, any new evidence can be considered to be in support of a particular hypothesis only if its occurrence is more likely when the hypothesis is true than when it is false. In fact, the new probability of the hypothesis depends on its initial probability and the ratio of these conditional probabilities, since, from **Equation (3.2)**,

$$P(H|E)=P(H)P(H)+[1-P(H)]P(E|Hc)P(E|H)$$

Hence, in the problem under consideration, the play of the cards can be considered to support the hypothesis of guilt only if such play would have been more likely if the partnership were cheating than if it were not. As the prosecutor never made this claim, his assertion that the evidence is in support of the guilt hypothesis is invalid.

Example 3h

Twins can be either identical or fraternal. Identical, also called monozygotic, twins form when a single fertilized egg splits into two genetically identical parts. Consequently, identical twins always have the same set of genes. Fraternal, also called dizygotic, twins develop when two eggs are fertilized and implant in the uterus. The genetic connection of fraternal twins is no more or less the same as siblings born at separate times. A Los Angeles County, California, scientist wishing to know the current fraction of twin pairs born in the county that are identical twins has assigned a county statistician to study this issue. The statistician initially requested each hospital in the county to record all twin births, indicating whether or not the resulting twins were identical. The hospitals, however, told her that to determine whether newborn twins were identical was not a simple task, as it involved the permission of the twins' parents to perform complicated and expensive DNA studies that the hospitals could not afford. After some deliberation, the statistician just asked the hospitals for data listing all twin births along with an indication as to whether the twins were of the same sex. When such data indicated that approximately 64 percent of twin births were same-sexed, the statistician declared that approximately 28 percent of all twins were identical. How did she come to this conclusion?

Solution

The statistician reasoned that identical twins are always of the same sex, whereas fraternal twins, having the same relationship to each other as any pair of siblings, will have probability 1/2 of being of the same sex. Letting I be the event that a pair of twins is identical, and SS be the event that a pair of twins is of the same sex, she computed the probability P(SS) by conditioning on whether the twin pair was identical. This gave

$$P(SS)=P(SS|I)P(I)+P(SS|Ic)P(Ic)$$

or

$$P(SS)=1\times P(I)+12\times[1-P(I)]=12+12\,P(I)$$

which, using that $P(SS)\approx.64$ yielded the result

$$P(I)\approx.28$$

The change in the probability of a hypothesis when new evidence is introduced can be expressed compactly in terms of the change in the *odds* of that hypothesis, where the concept of odds is defined as follows.

> Definition
>
> The odds of an event A are defined by
>
> $$P(A)P(Ac)=P(A)1-P(A)$$
>
> That is, the odds of an event A tell how much more likely it is that the event A occurs than it is that it does not occur. For instance, if $P(A)=23$, then $P(A)=2P(Ac)$, so the odds are 2. If the odds are equal to α, then it is common to say that the odds are "α to 1" in favor of the hypothesis.

Consider now a hypothesis H that is true with probability P(H), and suppose that new evidence E is introduced. Then, the conditional probabilities, given the evidence E, that H is true and that H is not true are respectively given by

$$P(H|E)=P(E|H)P(H)P(E)\quad P(Hc|E)=P(E|Hc)P(Hc)P(E)$$

Therefore, the new odds after the evidence E has been introduced are

$$P(H|E)P(Hc|E)=P(H)P(Hc)P(E|H)P(E|Hc) \tag{3.3}$$

That is, the new value of the odds of H is the old value multiplied by the ratio of the conditional probability of the new evidence given that H is true to the conditional probability given that H is not true. Thus, **Equation (3.3)** verifies the result of **Example 3f**, since the odds, and thus the probability of H, increase whenever the new evidence is more likely when H is true than when it is false. Similarly, the odds decrease whenever the new evidence is more likely when H is false than when it is true.

Example 3i

An urn contains two type A coins and one type B coin. When a type A coin is flipped, it comes up heads with probability 1/4, whereas when a type B coin is flipped, it comes up heads with probability 3/4. A coin is randomly chosen from the urn and flipped. Given that the flip landed on heads, what is the probability that it was a type A coin?

Solution

Let A be the event that a type A coin was flipped, and let B=Ac be the event that a type B coin was flipped. We want P(A| heads), where heads is the event that the flip landed on heads. From **Equation (3.3)**, we see that

$$P(A|heads)P(Ac|heads)=P(A)P(B)P(\ heads|A)P(\ heads|B)=2/31/3\ 1/43/4=2/3$$

Hence, the odds are 2/3:1, or, equivalently, the probability is 2/5 that a type A coin was flipped.

Equation (3.1) may be generalized as follows: Suppose that F1,F2,…,Fn are mutually exclusive events such that

$$\cup i=1nFi=S$$

In other words, exactly one of the events F1,F2,…,Fn must occur. By writing

$$E=\cup i=1nEFi$$

and using the fact that the events EFi,i=1,…,n are mutually exclusive, we obtain

$$P(E)=\sum i=1nP(EFi)=\sum i=1nP(E|Fi)P(Fi) \tag{3.4}$$

Thus, **Equation (3.4)**, often referred to as the *law of total probability*, shows how, for given events F1,F2,…,Fn, of which one and only one must occur, we can compute P(E) by first conditioning on which one of the Fi occurs. That is, **Equation (3.4)** states that P(E) is equal to a weighted average of P(E|Fi), each term being weighted by the probability of the event on which it is conditioned.

Example 3j

In **Example 5j** of **Chapter 2**, we considered the probability that, for a randomly shuffled deck, the card following the first ace is some specified card, and we gave a combinatorial argument to show that this probability is 152. Here is a probabilistic argument based on conditioning: Let E be the event that the card following the first ace is some specified card, say, card x. To compute P(E), we ignore card x and condition on the relative ordering of the other *51* cards in the deck. Letting O be the ordering gives

$$P(E)=\sum OP(E|O)P(O)$$

Now, given O, there are *52* possible orderings of the cards, corresponding to having card x being the ith card in the deck, i=1,…,52. But because all 52! possible orderings were initially equally likely, it follows that, conditional on O, each of the *52* remaining possible orderings is equally likely. Because card x will follow the first ace for only one of these orderings, we have $P(E|O)=1/52$, implying that $P(E)=1/52$.

Again, let F1,…,Fn be a set of mutually exclusive and exhaustive events (meaning that exactly one of these events must occur).

Suppose now that E has occurred and we are interested in determining which one of the Fj also occurred. Then, by **Equation (3.4)**, we have the following proposition.

Proposition 3.1

$$P(Fj|E)=P(EFj)P(E)=P(E|Fj)P(Fj)\sum i=1nP(E|Fi)P(Fi) \tag{3.5}$$

Equation (3.5) is known as *Bayes's formula*, after the English philosopher Thomas Bayes. If we think of the events Fj as being possible "hypotheses" about some subject matter, then Bayes's formula may be interpreted as showing us how opinions about these hypotheses held before the experiment was carried out [that is, the P(Fj)] should be modified by the evidence produced by the experiment.

Example 3k

A plane is missing, and it is presumed that it was equally likely to have gone down in any of 3 possible regions. Let $1-\beta i, i=1,2,3$, denote the probability that the plane will be found upon a search of the ith region when the plane is, in fact, in that region. (The constants βi are called *overlook probabilities*, because they represent the probability of overlooking the plane; they are generally attributable to the geographical and environmental conditions of the regions.) What is the conditional probability that the plane is in the ith region given that a search of region 1 is unsuccessful?

Solution

Let Ri,i=1,2,3, be the event that the plane is in region i, and let E be the event that a search of region 1 is unsuccessful. From Bayes's formula, we obtain

$$P(R1|E)=P(ER1)P(E)=P(E|R1)P(R1)\sum i=13P(E|Ri)P(Ri)=(\beta 1)13(\beta 1)13+(1)13+(1)13=\beta 1\beta 1+2$$

For j=2, 3,

$$P(Rj|E)=P(E|Rj)P(Rj)P(E)=(1)13(\beta 1)13+13+13=1\beta 1+2 \quad j=2,3$$

Note that the updated (that is, the conditional) probability that the plane is in region j, given the information that a search of region 1 did not find it, is greater than the initial probability that it was in region j when j≠1 and is less than the initial probability when j=1. This statement is certainly intuitive, since not finding the plane in region 1 would seem to decrease its chance of being in that region and increase its chance of being elsewhere. Further, the conditional probability that the plane is in region 1 given an unsuccessful search of that region is an increasing function of the overlook probability β1. This statement is also intuitive, since the larger β1 is, the more it is reasonable to attribute the unsuccessful search to “bad luck” as opposed to the plane’s not being there. Similarly, P(Rj|E),j≠1, is a decreasing function of β1.

The next example has often been used by unscrupulous probability students to win money from their less enlightened friends.

Example 3l

Suppose that we have 3 cards that are identical in form, except that both sides of the first card are colored red, both sides of the second card are colored black, and one side of the third card is colored red and the other side black. The 3 cards are mixed up in a hat, and 1 card is randomly selected and put down on the ground. If the upper side of the chosen card is colored red, what is the probability that the other side is colored black?

Solution

Let *RR, BB*, and *RB* denote, respectively, the events that the chosen card is all red, all black, or the red–black card. Also, let R be the event that the upturned side of the chosen card is red. Then, the desired probability is obtained by

P(RB|R)=P(RB∩R)P(R)=P(R|RB)P(RB)P(R|RR)P(RR)+P(R|RB)P(RB)+P(R|BB)P(BB)=(12)(13)(1)(13)+(12)(13)+0(13)=13

Hence, the answer is 13. Some students guess 12 as the answer by incorrectly reasoning that given that a red side appears, there are two equally likely possibilities: that the card is the all-red card or the red–black card. Their mistake, however, is in assuming that these two possibilities are equally likely. For, if we think of each card as consisting of two distinct sides, then we see that there are 6 equally likely outcomes of the experiment—namely, R1,R2,B1,B2,R3,B3—where the outcome is R1 if the first side of the all-red card is turned face up, R2 if the second side of the all-red card is turned face up, R3 if the red side of the red–black card is turned face up, and so on. Since the other side of the upturned red side will be black only if the outcome is R3, we see that the desired probability is the conditional probability of R3 given that either R1 or R2 or R3 occurred, which obviously equals 13.

Example 3m

A new couple, known to have two children, has just moved into town. Suppose that the mother is encountered walking with one of her children. If this child is a girl, what is the probability that both children are girls?

Solution

Let us start by defining the following events:

G_1: The first (that is, the oldest) child is a girl.

G_2: The second child is a girl.

G: The child seen with the mother is a girl.

Also, let B1,B2, and B denote similar events, except that “girl” is replaced by “boy.” Now, the desired probability is P(G1G2|G), which can be expressed as follows:

P(G1G2|G)=P(G1G2G)P(G)=P(G1G2)P(G)

Also,

P(G)=P(G|G1G2)P(G1G2)+P(G|G1B2)P(G1B2)+P(G|B1G2)P(B1G2)+P(G|B1B2)P(B1B2)=P(G1G2)+P(G|G1B2)P(G1B2)+P(G|B1G2)P(B1G2)

where the final equation used the results P(G|G1G2)=1 and P(G|B1B2)=0. If we now make the usual assumption that all 4 gender possibilities are equally likely, then we see that

P(G1G2|G)=1414+P(G|G1B2)/4+P(G|B1G2)/4=11+P(G|G1B2)+P(G|B1G2)

Thus, the answer depends on whatever assumptions we want to make about the conditional probabilities that the child seen with the mother is a girl given the event G1B2 and that the child seen with the mother is a girl given the event G2B1. For instance, if we want to assume, on the one hand, that, independently of the genders of the children, the child walking with the mother is the elder child with some probability p, then it would follow that

P(G|G1B2)=p=1−P(G|B1G2)

implying under this scenario that

P(G1G2|G)=12

If, on the other hand, we were to assume that if the children are of different genders, then the mother would choose to walk with the girl with probability q, independently of the birth order of the children, then we would have

$$P(G|G1B2)=P(G|B1G2)=q$$

implying that

$$P(G1G2|G)=11+2q$$

For instance, if we took q=1, meaning that the mother would always choose to walk with a daughter, then the conditional probability that she has two daughters would be 13, which is in accord with **Example 2b** because seeing the mother with a daughter is now equivalent to the event that she has at least one daughter.

Hence, as stated, the problem is incapable of solution. Indeed, even when the usual assumption about equally likely gender probabilities is made, we still need to make additional assumptions before a solution can be given. This is because the sample space of the experiment consists of vectors of the form s1,s2, i, where s1 is the gender of the older child, s2 is the gender of the younger child, and i identifies the birth order of the child seen with the mother. As a result, to specify the probabilities of the events of the sample space, it is not enough to make assumptions only about the genders of the children; it is also necessary to assume something about the conditional probabilities as to which child is with the mother given the genders of the children.

Example 3n

A bin contains 3 types of disposable flashlights. The probability that a type 1 flashlight will give more than 100 hours of use is .7, with the corresponding probabilities for type 2 and type 3 flashlights being .4 and .3, respectively. Suppose that 20 percent of the flashlights in the bin are type 1, 30 percent are type 2, and 50 percent are type 3.

a. What is the probability that a randomly chosen flashlight will give more than 100 hours of use?
b. Given that a flashlight lasted more than 100 hours, what is the conditional probability that it was a type j flashlight, j=1,2,3

Solution

a. Let A denote the event that the flashlight chosen will give more than 100 hours of use, and let Fj be the event that a type j flashlight is chosen, j=1,2,3. To compute P(A), we condition on the type of the flashlight, to obtain
$P(A)=P(A|F1)P(F1)+P(A|F2)P(F2)+P(A|F3)P(F3)=(.7)(.2)+(.4)(.3)+(.3)(.5)=.41$

   There is a 41 percent chance that the flashlight will last for more than 100 hours.
b. The probability is obtained by using Bayes's formula:
$P(Fj|A)=P(AFj)P(A)=P(A|Fj)P(Fj).41$

   Thus,
$P(F1|A)=(.7)(.2)/.41=14/41P(F2|A)=(.4)(.3)/.41=12/41P(F3|A)=(.3)(.5)/.41=15/41$

   For instance, whereas the initial probability that a type 1 flashlight is chosen is only .2, the information that the flashlight has lasted more than 100 hours raises the probability of this event to $14/41\approx.341$.

Example 3o

A crime has been committed by a solitary individual, who left some DNA at the scene of the crime. Forensic scientists who studied the recovered DNA noted that only five strands could be identified and that each innocent person, independently, would have a probability of 10^{-5} of having his or her DNA match on all five strands. The district attorney supposes that the perpetrator of the crime could be any of the 1 million residents of the town. Ten thousand of these residents have been released from prison within the past 10 years; consequently, a sample of their DNA is on file. Before any checking of the DNA file, the district attorney thinks that each of the 10,000 ex-criminals has probability α of being guilty of the new crime, whereas each of the remaining 990,000 residents has probability β, where $\alpha=c\beta$. (That is, the district attorney supposes that each recently released convict is c times as likely to be the crime's perpetrator as is each town member who is not a recently released convict.) When the DNA that is analyzed is compared against the database of the 10,000 ex-convicts, it turns out that A. J. Jones is the only one whose DNA matches the profile. Assuming that the district attorney's estimate of the relationship between α and β is accurate, what is the probability that A. J. is guilty?

Solution

To begin, note that because probabilities must sum to 1, we have

$$1=10{,}000\alpha+990{,}000\beta=(10{,}000c+990{,}000)\beta$$

Thus,

$$\beta=110{,}000c+990{,}000, \quad \alpha=c10{,}000c+990{,}000$$

Now, let G be the event that A. J. is guilty, and let M denote the event that A. J. is the only one of the 10,000 on file to have a match. Then,

$$P(G|M)=P(GM)P(M)=P(G)P(M|G)P(M|G)P(G)+P(M|Gc)P(Gc)$$

On the one hand, if A. J. is guilty, then he will be the only one to have a DNA match if none of the others on file have a match. Therefore,

$$P(M|G)=(1-10-5)9999$$

On the other hand, if A. J. is innocent, then in order for him to be the only match, his DNA must match (which will occur with probability 10−5), all others in the database must be innocent, and none of these others can have a match. Now, given that A. J. is innocent, the conditional probability that all the others in the database are also innocent is

$$P(\text{ all others innocent}|AJ \text{ innocent})=P(\text{ all in database innocent})P(AJ \text{ innocent})=1-10,000\alpha 1-\alpha$$

Also, the conditional probability, given their innocence, that none of the others in the database will have a match is (1−10−5)9999. Therefore,

$$P(M|Gc)=10-5(1-10,000\alpha 1-\alpha)(1-10-5)9999$$

Because P(G)=α, the preceding formula gives

$$P(G|M)=\alpha\alpha+10-5(1-10,000\alpha)=1.9+10-5\alpha$$

Thus, if the district attorney's initial thoughts were that an arbitrary ex-convict was 100 times more likely to have committed the crime than was a nonconvict (that is, c=100), then α=119,900 and

$$P(G|M)=11.099\approx 0.9099$$

If the district attorney initially thought that the appropriate ratio was c=10, then α=1109,000 and

$$P(G|M)=11.99\approx 0.5025$$

If the district attorney initially thought that the criminal was equally likely to be any of the members of the town(c=1), then α=10−6 and

$$P(G|M)=110.9\approx 0.0917$$

Thus, the probability ranges from approximately 9 percent when the district attorney's initial assumption is that all the members of the population have the same chance of being the perpetrator to approximately 91 percent when she assumes that each ex-convict is 100 times more likely to be the criminal than is a specified townsperson who is not an ex-convict.

3.4 Independent Events

The previous examples in this chapter show that P(E|F), the conditional probability of E given F, is not generally equal to P(E), the unconditional probability of E. In other words, knowing that F has occurred generally changes the chances of E's occurrence. In the special cases where P(E|F) does in fact equal P(E), we say that E is independent of F. That is, E is independent of F if knowledge that F has occurred does not change the probability that E occurs.

Since P(E|F)=P(EF)/P(F), it follows that E is independent of F if

$$P(EF)=P(E)P(F) \tag{4.1}$$

The fact that **Equation (4.1)** is symmetric in E and F shows that whenever E is independent of F, F is also independent of E. We thus have the following definition.

> Definition
>
> Two events E and F are said to be *independent* if **Equation (4.1)** holds.
>
> Two events E and F that are not independent are said to be *dependent*.

Example 4a

A card is selected at random from an ordinary deck of 52 playing cards. If E is the event that the selected card is an ace and F is the event that it is a spade, then E and F are independent. This follows because P(EF)=152, whereas P(E)=452 and P(F)=1352.

Example 4b

Two coins are flipped, and all 4 outcomes are assumed to be equally likely. If E is the event that the first coin lands on heads and F the event that the second lands on tails, then E and F are independent, since P(EF)=P({(H,T)})=14, whereas P(E)=P({(H,H),(H,T)})=12 and P(F)=P({(H,T),(T,T)})=12.

Example 4c

Suppose that we toss 2 fair dice. Let E1 denote the event that the sum of the dice is 6 and F denote the event that the first die equals 4. Then

$$P(E1F)=P(\{(4,2)\})=136$$

whereas

$$P(E1)P(F)=(536)(16)=5216$$

Hence, E1 and F are not independent. Intuitively, the reason for this is clear because if we are interested in the possibility of throwing a 6 (with 2 dice), we shall be quite happy if the first die lands on 4 (or, indeed, on any of the numbers 1, 2, 3, 4, and 5), for then we shall still have a possibility of getting a total of 6. If, however, the first die landed on 6, we would be unhappy because we would no longer have a chance of getting a total of 6. In other words, our chance of getting a total of 6 depends on the outcome of the first die; thus, E1 and F cannot be independent.

Now, suppose that we let E2 be the event that the sum of the dice equals 7. Is E2 independent of F? The answer is yes, since

$$P(E2F)=P(\{(4,3)\})=136$$

whereas

$$P(E2)P(F)=(16)(16)=136$$

We leave it for the reader to present the intuitive argument why the event that the sum of the dice equals 7 is independent of the outcome on the first die.

Example 4d

If we let E denote the event that the next president is a Republican and F the event that there will be a major earthquake within the next year, then most people would probably be willing to assume that E and F are independent. However, there would probably be some controversy over whether it is reasonable to assume that E is independent of G, where G is the event that there will be a recession within two years after the election.

We now show that if E is independent of F, then E is also independent of Fc.

Proposition 4.1

If E and F are independent, then so are E and Fc.

Proof. Assume that E and F are independent. Since E=EF∪EFc and *EF* and EFc are obviously mutually exclusive, we have

$$P(E)=P(EF)+P(EFc)=P(E)P(F)+P(EFc)$$

or, equivalently,

$$P(EFc)=P(E)[1-P(F)]=P(E)P(Fc)$$

and the result is proved.

Thus, if E is independent of F, then the probability of E's occurrence is unchanged by information as to whether or not F has occurred.

Suppose now that E is independent of F and is also independent of G. Is E then necessarily independent of *FG*? The answer, somewhat surprisingly, is no, as the following example demonstrates.

Example 4e

Two fair dice are thrown. Let E denote the event that the sum of the dice is 7. Let F denote the event that the first die equals 4 and G denote the event that the second die equals 3. From **Example 4c**, we know that E is independent of F, and the same reasoning as applied there shows that E is also independent of G; but clearly, E is not independent of *FG* [since P(E|FG)=1].

It would appear to follow from **Example 4e** that an appropriate definition of the independence of three events E, F, and G would have to go further than merely assuming that all of the (32) pairs of events are independent. We are thus led to the following definition.

Definition

Three events E, F, and G are said to be independent if

$$P(EFG)=P(E)P(F)P(G)P(EF)=P(E)P(F)P(EG)=P(E)P(G)P(FG)=P(F)P(G)$$

Note that if E, F, and G are independent, then E will be independent of any event formed from F and G. For instance, E is independent of F∪G, since

$$P[E(F\cup G)]=P(EF\cup EG)=P(EF)+P(EG)-P(EFG)=P(E)P(F)+P(E)P(G)-P(E)P(FG)=P(E)[P(F)+P(G)-P(FG)]=P(E)P(F\cup G)$$

Of course, we may also extend the definition of independence to more than three events. The events E1,E2,…,En are said to be independent if for every subset E1′,E2′,…,Er′,r≤n of these events,

$$P(E1'E2'\cdots Er')=P(E1')P(E2')\cdots P(Er')$$

Finally, we define an infinite set of events to be independent if every finite subset of those events is independent.

Sometimes, a probability experiment under consideration consists of performing a sequence of subexperiments. For instance, if the experiment consists of continually tossing a coin, we may think of each toss as being a subexperiment. In many cases, it is reasonable to assume that the outcomes of any group of the subexperiments have no effect on the probabilities of the outcomes of the other subexperiments. If such is the case, we say that the subexperiments are independent. More formally, we say that the subexperiments are independent if E1,E2,…,En,… is necessarily an independent sequence of events whenever Ei is an event whose occurrence is completely determined by the outcome of the ith subexperiment.

If each subexperiment has the same set of possible outcomes, then the subexperiments are often called *trials*.

Example 4f

An infinite sequence of independent trials is to be performed. Each trial results in a success with probability p and a failure with probability 1−p. What is the probability that

a. at least 1 success occurs in the first n trials;
b. exactly k successes occur in the first n trials;
c. all trials result in successes?

Solution

In order to determine the probability of at least 1 success in the first n trials, it is easiest to compute first the probability of the complementary event: that of no successes in the first n trials. If we let Ei denote the event of a failure on the ith trial, then the probability of no successes is, by independence,

$$P(E1E2\cdots En)=P(E1)P(E2)\cdots P(En)=(1-p)n$$

Hence, the answer to part (a) is 1−(1−p)n.

To compute the answer to part (b), consider any particular sequence of the first n outcomes containing k successes and n−k failures. Each one of these sequences will, by the assumed independence of trials, occur with probability pk(1−p)n−k. Since there are (nk) such sequences [there are n!/k!(n−k)! permutations of k successes and n−k failures], the desired probability in part (b) is

$$P\{\text{exactly k successes}\}=(nk)pk(1-p)n-k$$

To answer part (c), we note that, by part (a), the probability of the first n trials all resulting in success is given by

$$P(E1cE2c\cdots Enc)=pn$$

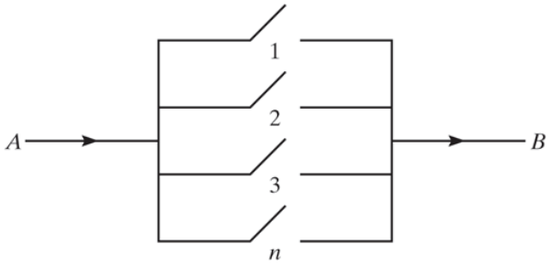
Thus, using the continuity property of probabilities (**Section 2.6**), we see that the desired probability is given by

$$P(\cap i=1\infty Eic)=P(limn\to\infty\cap i=1\infty Eic)=limn\to\infty P(\cap i=1\infty Eic)=limnpn=\{\ 0if\ p<11if\ p=1$$

Example 4g

A system composed of n separate components is said to be a parallel system if it functions when at least one of the components functions. (See **Figure 3.2**.) For such a system, if component i, which is independent of the other components, functions with probability pi,i=1,…,n, what is the probability that the system functions?

Figure 3.2 Parallel system: functions if current flows from *A* to *B*.



Solution

Let Ai denote the event that component i functions. Then,

P{ system functions }=1−P{ system does not function }=1−P{ all components do not function }=1−P(∩iAic)=1−∏i=1i(1−pi) by independence

Example 4h

Independent trials consisting of rolling a pair of fair dice are performed. What is the probability that an outcome of 5 appears before an outcome of 7 when the outcome of a roll is the sum of the dice?

Solution

If we let En denote the event that no 5 or 7 appears on the first n−1 trials and a 5 appears on the nth trial, then the desired probability is

P(∪n=1∞En)=∑n=1∞P(En)

Now, since P{5 on any trial}=436 and P{7 on any trial}=636, we obtain, by the independence of trials,

P(En)=(1−1036)n−1436

Thus,

P(∪n=1∞En)=19∑n=1∞(1318)n−1=1911−1318=25

This result could also have been obtained by the use of conditional probabilities. If we let E be the event that a 5 occurs before a 7, then we can obtain the desired probability, P(E), by conditioning on the outcome of the first trial, as follows: Let F be the event that the first trial results in a 5, let G be the event that it results in a 7, and let H be the event that the first trial results in neither a 5 nor a 7. Then, conditioning on which one of these events occurs gives

P(E)=P(E|F)P(F)+P(E|G)P(G)+P(E|H)P(H)

However,

P(E|F)=1P(E|G)=0P(E|H)=P(E)

The first two equalities are obvious. The third follows because if the first outcome results in neither a 5 nor a 7, then at that point the situation is exactly as it was when the problem first started—namely, the experimenter will continually roll a pair of fair dice until either a 5 or 7 appears. Furthermore, the trials are independent; therefore, the outcome of the first trial will have no effect on subsequent rolls of the dice. Since P(F)=436,P(G)=636, and P(H)=2636, it follows that

P(E)=19+P(E)1318

or

P(E)=25

The reader should note that the answer is quite intuitive. That is, because a 5 occurs on any roll with probability 436 and a 7 with probability 636, it seems intuitive that the odds that a 5 appears before a 7 should be 6 to 4 against. The probability should then be 410, as indeed it is.

The same argument shows that if E and F are mutually exclusive events of an experiment, then, when independent trials of the experiment are performed, the event E will occur before the event F with probability

P(E)P(E)+P(F)

Example 4i

Suppose there are n types of coupons and that each new coupon collected is, independent of previous selections, a type i coupon with probability pi, ∑i=1npi=1. Suppose k coupons are to be collected. If Ai is the event that there is at least one type i coupon among those collected, then, for i≠j, find

a. P(Ai)
b. P(Ai∪Aj)
c. P(Ai|Aj)

Solution

$P(A_i)=1-P(A_i^c)=1-P\{\text{no coupon is type } i\}=1-(1-p_i)^k$

where the preceding used that each coupon is, independently, not of type i with probability $1-p_i$. Similarly,

$P(A_i \cup A_j)=1-P((A_i \cup A_j)^c)=1-P\{\text{no coupon is either type } i \text{ or type } j\}=1-(1-p_i-p_j)^k$

where the preceding used that each coupon is, independently, neither of type i nor type j with probability $1-p_i-p_j$.

To determine $P(A_i|A_j)$, we will use the identity

$P(A_i \cup A_j)=P(A_i)+P(A_j)-P(A_iA_j)$

which, in conjunction with parts (a) and (b), yields

$P(A_iA_j)=1-(1-p_i)^k+1-(1-p_j)^k-[1-(1-p_i-p_j)^k]=1-(1-p_i)^k-(1-p_j)^k+(1-p_i-p_j)^k$

Consequently,

$P(A_i|A_j)=\frac{P(A_iA_j)}{P(A_j)}=\frac{1-(1-p_i)^k-(1-p_j)^k+(1-p_i-p_j)^k}{1-(1-p_j)^k}$

The next example presents a problem that occupies an honored place in the history of probability theory. This is the famous *problem of the points*. In general terms, the problem is this: Two players put up stakes and play some game, with the stakes to go to the winner of the game. An interruption requires them to stop before either has won and when each has some sort of a "partial score." How should the stakes be divided?

This problem was posed to the French mathematician Blaise Pascal in 1654 by the Chevalier de Méré, who was a professional gambler at that time. In attacking the problem, Pascal introduced the important idea that the proportion of the prize deserved by the competitors should depend on their respective probabilities of winning if the game were to be continued at that point. Pascal worked out some special cases and, more importantly, initiated a correspondence with the famous Frenchman Pierre de Fermat, who had a reputation as a great mathematician. The resulting exchange of letters not only led to a complete solution to the problem of the points, but also laid the framework for the solution to many other problems connected with games of chance. This celebrated correspondence, considered by some as the birth date of probability theory, was also important in stimulating interest in probability among the mathematicians in Europe, for Pascal and Fermat were both recognized as being among the foremost mathematicians of the time. For instance, within a short time of their correspondence, the young Dutch mathematician Christiaan Huygens came to Paris to discuss these Problems and solutions, and interest and activity in this new field grew rapidly.

Example 4j The problem of the points

Independent trials resulting in a success with probability p and a failure with probability $1-p$ are performed. What is the probability that n successes occur before m failures? If we think of A and B as playing a game such that A gains 1 point when a success occurs and B gains 1 point when a failure occurs, then the desired probability is the probability that A would win if the game were to be continued in a position where A needed n and B needed m more points to win.

Solution

We shall present two solutions. The first comes from Pascal and the second from Fermat.

Let us denote by $P_{n,m}$ the probability that n successes occur before m failures. By conditioning on the outcome of the first trial, we obtain

$P_{n,m}=pP_{n-1,m}+(1-p)P_{n,m-1} n \geq 1, m \geq 1$

(Why? Reason it out.) Using the obvious boundary conditions $P_{n,0}=0, P_{0,m}=1$, we can solve these equations for $P_{n,m}$. Rather than go through the tedious details, let us instead consider Fermat's solution.

Fermat argued that in order for n successes to occur before m failures, it is necessary and sufficient that there be at least n successes in the first $m+n-1$ trials. (Even if the game were to end before a total of $m+n-1$ trials were completed, we could still imagine that the necessary additional trials were performed.) This is true, for if there are at least n successes in the first $m+n-1$ trials, there could be at most $m-1$ failures in those $m+n-1$ trials; thus, n successes would occur before m failures. If, however, there were fewer than n successes in the first $m+n-1$ trials, there would have to be at least m failures in that same number of trials; thus, n successes would not occur before m failures.

Hence, since, as shown in **Example 4f**, the probability of exactly k successes in $m+n-1$ trials is $\binom{m+n-1}{k}p^k(1-p)^{m+n-1-k}$, it follows that the desired probability of n successes before m failures is

$P_{n,m}=\sum_{k=n}^{m+n-1}\binom{m+n-1}{k}p^k(1-p)^{m+n-1-k}$

The following example gives another instance where determining the probability that a player wins a match is made easier by assuming that the play continues even after the match winner has been determined.

Example 4k Service protocol in a serve and rally game

Consider a serve and rally match (such as volleyball, badminton, or squash) between two players, A and B. The match consists of a sequence of rallies, with each rally beginning with a serve by one of the players and continuing until one of the players has won the rally. The winner of the rally receives a point, and the match ends when one of the players has won a total of n points, with that player being declared the winner of the match. Suppose whenever a rally begins with A as the server, that A wins that rally with probability p_A and that B wins it with probability $q_A=1-p_A$, and that a rally that begins with B as the server is won by A with probability p_B and by B with probability $q_B=1-p_B$. Player A is to be the initial server. There are two possible server protocols that are under consideration: "winner serves," which means that the winner of a rally is the server for the next rally, or "alternating serve," which means that the server alternates from rally to rally, so that no two consecutive rallies have the same server. Thus, for instance, if n=3, then the successive servers under the "winner serves" protocol would be A,A,B,A,A if A wins the first point, then B the next, then A wins the next two. On the other hand, the sequence of servers under the "alternating serve" protocol will always be A,B,A,B,A,... until the match winner is decided. If you were player A, which protocol would you prefer?

Solution

Surprisingly, it turns out that it makes no difference, in that the probability that A is the match winner is the same under either protocol. To show that this is the case, it is advantageous to suppose that the players continue to play until a

total of 2n−1 rallies have been completed. The first player to win n rallies would then be the one who has won at least n of the 2n−1 rallies. To begin, note that if the alternating serve protocol is being used, then player A will serve exactly n times and player B will serve exactly n−1 times in the 2n−1 rallies.

Now consider the winner serve protocol, again assuming that the players continue to play until 2n−1 rallies have been completed. Because it makes no difference who serves the “extra rallies” after the match winner has been decided, suppose that at the point at which the match has been decided (because one of the players has won n points), the remainder (if there are any) of the 2n−1 rallies are all served by the player who lost the match. Note that this modified service protocol does not change the fact that the winner of the match will still be the player who wins at least n of the 2n−1 rallies. We claim that under this modified service protocol, A will always serve n times and B will always serve n−1 times. Two cases show this.

Case 1: A wins the match.

Because A serves first, it follows that A’s second serve will immediately follow A’s first point; A’s third serve will immediately follow A’s second point; and, in particular, A’s nth serve will immediately follow A’s (n−1) point. But this will be the last serve of A before the match result is decided. This is so because either A will win the point on that serve and so have n points, or A will lose the point and so the serve will switch to, and remain with, B until A wins point number n. Thus, provided that A wins the match, it follows that A would have served a total of n times at the moment the match is decided. Because, by the modified service protocol, A will never again serve, it follows in this case that A serves exactly n times.

Case 2: B wins the match.

Because A serves first, B’s first serve will come immediately after B’s first point; B’s second serve will come immediately after B’s second point; and, in particular, B’s (n−1) serve will come immediately after B’s (n−1) point. But that will be the last serve of B before the match is decided because either B will win the point on that serve and so have n points, or B will lose the point and so the serve will switch to, and remain with, A until B wins point number n. Thus, provided that B wins the match, we see that B would have served a total of n−1 times at the moment the match is decided. Because, by the modified service protocol, B will never again serve, it follows in this case that B serves exactly n−1 times, and, as there are a total of 2n−1 rallies, that A serves exactly n times.

Thus, we see that under either protocol, A will always serve n times and B will serve n−1 times and the winner of the match will be the one who wins at least n points. But since A wins each rally that he serves with probability pA and wins each rally that B serves with probability pB it follows that the probability that A is the match winner is, under either protocol, equal to the probability that there are at least n successes in 2n−1 independent trials, when n of these trials result in a success with probability pA and the other n−1 trials result in a success with probability pB. Consequently, the win probabilities for both protocols are the same.

Our next two examples deal with gambling problems, with the first having a surprisingly elegant analysis.*

*The remainder of this section should be considered optional.

Example 4l

Suppose that initially there are r players, with player i having ni units, ni>0,i=1,…,r. At each stage, two of the players are chosen to play a game, with the winner of the game receiving 1 unit from the loser. Any player whose fortune drops to 0 is eliminated, and this continues until a single player has all n≡∑i=1rni units, with that player designated as the victor. Assuming that the results of successive games are independent and that each game is equally likely to be won by either of its two players, find Pi, the probability that player i is the victor.

Solution

To begin, suppose that there are n players, with each player initially having 1 unit. Consider player i. Each stage she plays will be equally likely to result in her either winning or losing 1 unit, with the results from each stage being independent.

In addition, she will continue to play stages until her fortune becomes either 0 or n. Because this is the same for all n players, it follows that each player has the same chance of being the victor, implying that each player has probability 1/n of being the victor. Now, suppose these n players are divided into r teams, with team i containing ni players, i=1,…,r. Then, the probability that the victor is a member of team i is ni/n. But because

a. team i initially has a total fortune of ni units, i=1,…,r, and
b. each game played by members of different teams is equally likely to be won by either player and results in the fortune of members of the winning team increasing by 1 and the fortune of the members of the losing team decreasing by *1*,

it is easy to see that the probability that the victor is from team i is exactly the probability we desire. Thus, Pi=ni/n. Interestingly, our argument shows that this result does *not* depend on how the players in each stage are chosen.

In the *gambler’s ruin problem*, there are only *2* gamblers, but they are not assumed to be of equal skill.

Example 4m The gambler’s ruin problem

Two gamblers, A and B, bet on the outcomes of successive flips of a coin. On each flip, if the coin comes up heads, A collects 1 unit from B, whereas if it comes up tails, A pays 1 unit to B. They continue to do this until one of them runs out of money. If it is assumed that the successive flips of the coin are independent and each flip results in a head with probability p, what is the probability that A ends up with all the money if he starts with i units and B starts with N−i units?

Solution

Let E denote the event that A ends up with all the money when he starts with i and B starts with N−i, and to make clear the dependence on the initial fortune of A, let Pi=P(E). We shall obtain an expression for *P(E)* by conditioning on the outcome of the first flip as follows: Let H denote the event that the first flip lands on heads; then

Pi=P(E)=P(E|H)P(H)+P(E|Hc)P(Hc)=pP(E|H)+(1−p)P(E|Hc)

Now, given that the first flip lands on heads, the situation after the first bet is that A has i+1 units and B has N−(i+1). Since the successive flips are assumed to be independent with a common probability p of heads, it follows that from that point on, A’s probability of winning all the money is exactly the same as if the game were just starting with A having an initial fortune of i+1 and B having an initial fortune of N−(i+1). Therefore,

P(E|H)=Pi+1

and similarly,

$$P(E|Hc)=Pi-1$$

Hence, letting q=1−p, we obtain

$$Pi=pPi+1+qPi-1i=1,2,\ldots,N-1 \tag{4.2}$$

By making use of the obvious boundary conditions P0=0 and PN=1, we shall now solve **Equation (4.2)**. Since p+q=1, these equations are equivalent to

$$pPi+qPi=pPi+1+qPi-1$$

or

$$Pi+1-Pi=qp(Pi-Pi-1)i=1,2,\ldots,N-1 \tag{4.3}$$

Hence, since P0=0, we obtain, from **Equation (4.3)**,

$$P2-P1=qp(P1-P0)=qpP1P3-P2=qp(P2-P1)=(qp)2P1...Pi-Pi-1=qp(Pi-1-Pi-2)=(qp)i-1P1...PN-PN-1=qp(PN-1-PN-2)=(qp)N-1P1 \tag{4.4}$$

Adding the first i−1 equations of (**4.4**) yields

$$Pi-P1=P1[(qp)+(qp)2+\cdots+(qp)i-1]$$

or

$$Pi=\{1-(q/p)i1-(q/p)P1if\ qp\neq1iP1if\ qp=1$$

Using the fact that PN=1, we obtain

$$P1=\{1-(q/p)1-(q/p)Nif\ p\neq121N \quad if\ p=12$$

Hence,

$$Pi=\{1-(q/p)i1-(q/p)Nif\ p\neq12iNif\ p=12 \tag{4.5}$$

Let Qi denote the probability that B winds up with all the money when A starts with i and B starts with N−i. Then, by symmetry to the situation described, and on replacing p by q and i by N−i, it follows that

$$Qi=\{1-(p/q)N-i1-(p/q)Nif\ q\neq12N-iNif\ q=12$$

Moreover, since q=12 is equivalent to p=12, we have, when q≠12,

$$Pi+Qi=1-(q/p)i1-(q/p)N+1-(p/q)N-i1-(p/q)N=pN-pN(q/p)ipN-qN+qN-qN(p/q)N-iqN-pN=pN-pN-iqi-qN+qipN-ipN-qN=1$$

This result also holds when p=q=12, so

$$Pi+Qi=1$$

In words, this equation states that with probability 1, either A or B will wind up with all of the money; in other words, the probability that the game continues indefinitely with A's fortune always being between 1 and N−1 is zero. (The reader must be careful because, a priori, there are three possible outcomes of this gambling game, not two: Either A wins, or B wins, or the game goes on forever with nobody winning. We have just shown that this last event has probability 0.)

As a numerical illustration of the preceding result, if A were to start with 5 units and B with 10, then the probability of A's winning would be 13 if p were 12, whereas it would jump to

$$1-(23)51-(23)15\approx.87$$

if p were .6.

A special case of the gambler's ruin problem, which is also known as the problem of *duration of play*, was proposed to Huygens by Fermat in 1657. The version Huygens proposed, which he himself solved, was that A and B have 12 coins each. They play for these coins in a game with 3 dice as follows: Whenever 11 is thrown (by either—it makes no difference who rolls the dice), A gives a coin to B. Whenever 14 is thrown, B gives a coin to A. The person who first wins all the coins wins the game. Since P{roll 11}=27216 and P{roll 14}=15216, we see from **Example 4h** that, for A, this is just the gambler's ruin problem with p=1542,i=12, and N=24. The general form of the gambler's ruin problem was solved by the mathematician James Bernoulli and published 8 years after his death in 1713.

For an Application of the gambler's ruin problem to drug testing, suppose that two new drugs have been developed for treating a certain disease. Drug i has a cure rate pi,i=1,2, in the sense that each patient treated with drug i will be cured with probability pi. These cure rates are, however, not known, and we are interested in finding a method for deciding whether p1>p2 or p2>p1. To decide on one of these alternatives, consider the following test: Pairs of patients are to be treated sequentially, with one member of the pair receiving drug 1 and the other drug 2. The results for each pair are determined, and the testing stops when the cumulative number of cures from one of the drugs exceeds the cumulative number of cures from the other by some fixed, predetermined number. More formally, let

$$Xj=\{1if\ the\ patient\ in\ the\ jth\ pair\ that\ receives\ drug\ 1\ is\ cured0otherwiseYj=\{1if\ the\ patient\ in\ the\ jth\ pair\ that\ receives\ drug\ 2\ is\ cured0otherwise$$

For a predetermined positive integer M, the test stops after pair N, where N is the first value of n such that either

$$X1+\cdots+Xn-(Y1+\cdots+Yn)=M$$

or

$$X_1+\cdots+X_n-(Y_1+\cdots+Y_n)=-M$$

In the former case, we assert that $p_1>p_2$ and in the latter that $p_2>p_1$.

In order to help ascertain whether the foregoing is a good test, one thing we would like to know is the probability that it leads to an incorrect decision. That is, for given p_1 and p_2, where $p_1>p_2$, what is the probability that the test will incorrectly assert that $p_2>p_1$? To determine this probability, note that after each pair is checked, the cumulative difference of cures using drug 1 versus drug 2 will go up by 1 with probability $p_1(1-p_2)$—since this is the probability that drug 1 leads to a cure and drug 2 does not—or go down by 1 with probability $(1-p_1)p_2$, or remain the same with probability $p_1p_2+(1-p_1)(1-p_2)$. Hence, if we consider only those pairs in which the cumulative difference changes, then the difference will go up by 1 with probability

$$p=P\{\text{up 1} \mid \text{up 1 or down 1}\}=\frac{p_1(1-p_2)}{p_1(1-p_2)+(1-p_1)p_2}$$

and down by 1 with probability

$$1-p=\frac{p_2(1-p_1)}{p_1(1-p_2)+(1-p_1)p_2}$$

Thus, the probability that the test will assert that $p_2>p_1$ is equal to the probability that a gambler who wins each (one-unit) bet with probability p will go down M before going up M. But **Equation (4.5)**, with $i=M, N=2M$, shows that this probability is given by

$$p=P\{\text{ test asserts that } p_2>p_1 \}=\frac{1-1-(1-pp)^M}{1-(1-pp)^{2M}}=\frac{1-1}{1+(1-pp)^M}=\frac{1}{1+\gamma^M}$$

where

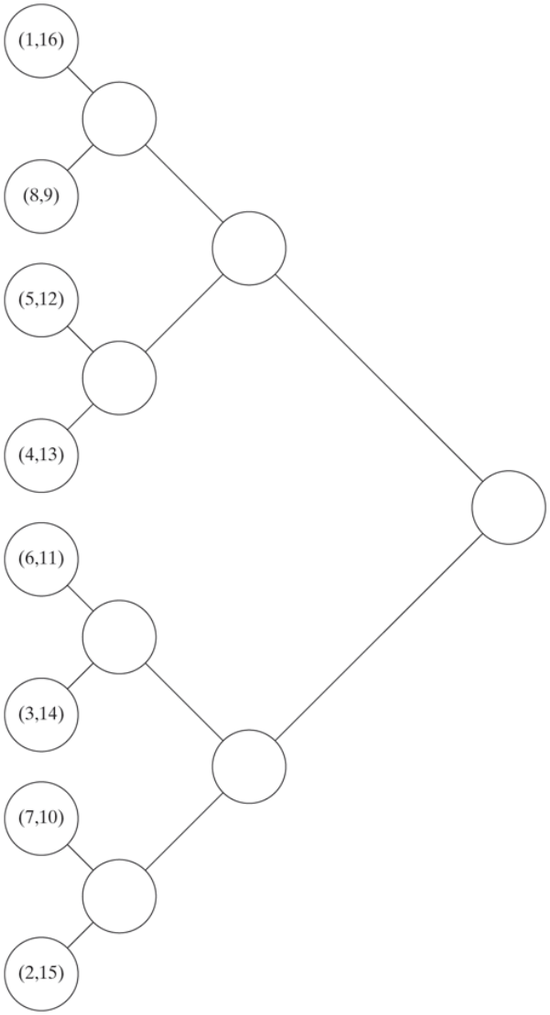
$$\gamma=\frac{1-p}{p}=\frac{p_1(1-p_2)}{p_2(1-p_1)}$$

For instance, if $p_1=.6$ and $p_2=.4$, then the probability of an incorrect decision is .017 when $M=5$ and reduces to .0003 when $M=10$.

Example 4n

A total of 64 teams are selected to play in the end of season NCAA college basketball tournament. These 64 are divided into four groups, called brackets, of size 16 each, with the teams in each bracket being given seedings ranging from 1 (the top rated team in the bracket) to 16 (the lowest rated team in the bracket). The teams in each bracket play each other in a knockout style tournament, meaning a loss knocks a team out of the tournament. Naming a team by its seeding, the schedule of games to be played by the teams in a bracket is as given by the following graph:

Figure 3.3 Ncaa tournament bracket format



Thus, for instance, teams 1 and 16 play a game in round one, as do teams 8 and 9, with the winners of these games then playing each other in round two. Let r(i,j)=r(j,i), i≠j denote the round in which i and j would play if both teams win up to that point. That is, r(i,j)=k if i and j would play in round k if each won its first k−1 games. For instance, r(1,16)=1, r(1,8)=2, r(1,5)=3, r(1,6)=4.

Let us focus on a single one of the brackets, and let us suppose that, no matter what has previously occurred, if i and j ever play each other then i will win with probability pi,j=1−pj,i. Let Pi be the probability that team i is the winner of the bracket, i=1,…,16. Because Pi is the probability that i wins 4 games, we will compute the values P1,…,P16 by determining the quantities Pi(k), i=1,…,16, where Pi(k) is defined to be the probability that i wins its first k games. The probabilities Pi(k) will be determined recursively, first for k=1, then for k=2, then for k=3, and finally for k=4 which will yield Pi=Pi(4).

Let Oi(k)={j:(r(i,j)=k} be the set of possible opponents of i in round k. To compute Pi(k), we will condition on which of the teams in Oi(k) reaches round k. Because a team will reach round k if that team wins its first k−1 games, this gives

$$P_i(k)=\sum_{j\in O_i(k)} P(i \text{ win its first } k \text{ games} \mid j \text{ reaches round } k)\, P_j(k-1) \tag{4.6}$$

Now, because any team that plays a team in Oi(k) in any of rounds 1,…,k−1 is a possible opponent of team i in round k, it follows that all games in rounds 1,…,k−1 involving a team in Oi(k) will be against another team in that set, and thus the results of these games do not affect which teams i would play in its first k−1 games. Consequently, whether team i reaches round k is independent of which team in Oi(k) reaches round k. Hence, for j∈Oi(k)

$$P(i \text{ win its first } k \text{ games} \mid j \text{ reaches round } k)=P(i \text{ win its first } k-1 \text{ games}, i \text{ beats } j \mid j \text{ reaches round } k)=P(i \text{ win its first } k-1 \text{ games})P(i \text{ beats } j \mid i \text{ and } j \text{ reach round } k)=P_i(k-1)\, p_{i,j}$$

where the next to last equality follows because whether i wins its first k−1 games is independent of the event that j wins its first k−1 games. Hence, from (**4.6**) and the preceding equation, we have that

$$P_i(k)=\sum_{j\in O_i(k)} P_i(k-1)\, p_{i,j}\, P_j(k-1)=P_i(k-1)\sum_{j\in O_i(k)} P_j(k-1)\, p_{i,j} \tag{4.7}$$

Starting with Pi(0)=1, the preceding enables us to determine Pi(1) for all i, which then enables us to determine Pi(2) for all i, and so on, up to Pi=Pi(4).

To indicate how to apply the recursive **equations (4.7)**, suppose that pi,j=ji+j. Thus, for instance, the probability that team 2 (the second seed) beats team 7 (the seventh seed) is p2,7=7/9. To compute, Pi=Pi(4), the probability that i wins the bracket, start with the quantities Pi(1), i=1,…,16, equal to the probabilities that each team wins its first game.

$$P1(1)=p1,16=16/17=1-P16(1)P2(1)=p2,15=15/17=1-P15(1)P3(1)=p3,14=14/17=1-P14(1)P4(1)=p4,13=13/17=1-P13(1)P5(1)=p5,12=12/17=1-P12(1)P6(1)=p6,11=11/17=1-P11(1)P7(1)=p7,10=10/17=1-P10(1)P8(1)=p8,9=9/17=1-P9(1)$$

The quantities Pi(2) are then obtained by using the preceding along with the recursion (**4.7**). For instance, because the set of possible opponents of team 1 in round 2 is O1(2)={8,9}, we have that

$$P1(2)=P1(1)(P8(1)p1,8+P9(1)p1,9)=1617(91789+817910)\approx.8415$$

The other quantities Pi(2),…,P16(2) are obtained similarly, and are used to obtain the quantities Pi(3), i=1,…,16, which are then used to obtain Pi=Pi(4), i= 1,…,16.

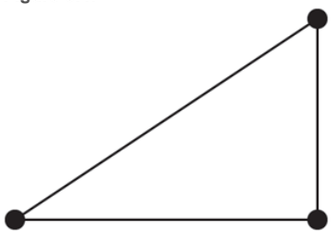
Suppose that we are presented with a set of elements and we want to determine whether at least one member of the set has a certain property. We can attack this question probabilistically by randomly choosing an element of the set in such a way that each element has a positive probability of being selected. Then the original question can be answered by a consideration of the probability that the randomly selected element does not have the property of interest. If this probability is equal to 1, then none of the elements of the set has the property; if it is less than 1, then at least one element of the set has the property.

The final example of this section illustrates this technique.

Example 4o

The complete graph having n vertices is defined to be a set of n points (called vertices) in the plane and the (n2) lines (called edges) connecting each pair of vertices. The complete graph having 3 vertices is shown in **Figure 3.4**. Suppose now that each edge in a complete graph having n vertices is to be colored either red or blue. For a fixed integer k, a question of interest is, Is there a way of coloring the edges so that no set of k vertices has all of its (k2) connecting edges the same color? It can be shown by a probabilistic argument that if n is not too large, then the answer is yes.

Figure 3.4:



The argument runs as follows: Suppose that each edge is, independently, equally likely to be colored either red or blue. That is, each edge is red with probability 12. Number the (nk) sets of k vertices and define the events Ei,i=1,…,(nk) as follows:

Ei={all of the connecting edges of the ith set of k vertices are the same color}

Now, since each of the (k2) connecting edges of a set of k vertices is equally likely to be either red or blue, it follows that the probability that they are all the same color is

$$P(Ei)=2(12)k(k-1)/2$$

Therefore, because

$$P(\cup iEi)\leq\sum iP(Ei) \quad (\text{Boole's inequality})$$

we find that P(∪iEi), the probability that there is a set of k vertices all of whose connecting edges are similarly colored, satisfies

$$P(\cup iEi)\leq(nk)(12)k(k-1)/2-1$$

Hence, if

$$(nk)(12)k(k-1)/2-1<1$$

or, equivalently, if

$$(nk)<2k(k-1)/2-1$$

then the probability that at least one of the (nk) sets of k vertices has all of its connecting edges the same color is less than 1. Consequently, under the preceding condition on n and k, it follows that there is a positive probability that no set of k vertices has all of its connecting edges the same color. But this conclusion implies that there is at least one way of coloring the edges for which no set of k vertices has all of its connecting edges the same color.

Remarks

a. Whereas the preceding argument established a condition on n and k that guarantees the existence of a coloring scheme satisfying the desired property, it gives no information about how to obtain such a scheme (although one possibility would be simply to choose the colors at random, check to see if the resulting coloring satisfies the property, and repeat the procedure until it does).
b. The method of introducing probability into a problem whose statement is purely deterministic has been called the *probabilistic method*.† Other examples of this method are given in Theoretical **Exercise 24** and Examples 2t and 2u of **Chapter 7**.

†See N. Alon, J. Spencer, and P. Erdos, *The Probabilistic Method* (New York: John Wiley & Sons, Inc., 1992).

Because the independence of trials implies that F and H are independent, and because P(F)=pn−1, it follows from **Equation (5.3)** that

$$P(E|H)=p^{n-1}+(1-p^{n-1})P(E|H^c) \tag{5.4}$$

We now obtain an expression for P(E|Hc) in a similar manner. That is, we let G denote the event that trials 2 through m are all *failures*. Then,

$$P(E|H^c)=P(E|GH^c)P(G|H^c)+P(E|GcH^c)P(G^c|H^c) \tag{5.5}$$

Now, GHc is the event that the first m trials all result in failures, so P(E|GHc)=0. Also, if GcHc occurs, then the first trial is a failure, but there is at least one success in the next m−1 trials. Hence, since this success wipes out all previous failures, we see that

$$P(E|GcH^c)=P(E|H)$$

Thus, because P(Gc|Hc)=P(Gc)=1−qm−1, we obtain, from (**5.5**),

$$P(E|H^c)=(1-q^{m-1})P(E|H) \tag{5.6}$$

Solving **Equations (5.4)** and (**5.6**) yields

$$P(E|H)=p^{n-1}p^{n-1}+q^{m-1}-p^{n-1}q^{m-1}$$

and

$$P(E|H^c)=(1-q^{m-1})p^{n-1}p^{n-1}+q^{m-1}-p^{n-1}q^{m-1}$$

Thus,

$$P(E)=pP(E|H)+qP(E|H^c)=p^n+qp^{n-1}(1-q^{m-1})p^{n-1}+q^{m-1}-p^{n-1}q^{m-1}=p^{n-1}(1-q^m)p^{n-1}+q^{m-1}-p^{n-1}q^{m-1} \tag{5.7}$$

It is interesting to note that by the symmetry of the problem, the probability of obtaining a run of m failures before a run of n successes would be given by **Equation (5.7)** with p and q interchanged and n and m interchanged. Hence, this probability would equal

$$P\{\text{run of m failures before a run of n successes}\}=q^{m-1}(1-p^n)q^{m-1}+p^{n-1}-q^{m-1}p^{n-1} \tag{5.8}$$

Since **Equations (5.7)** and (**5.8**) sum to 1, it follows that, with probability 1, either a run of n successes or a run of m failures will eventually occur.

As an example of **Equation (5.7)**, we note that, in tossing a fair coin, the probability that a run of 2 heads will precede a run of 3 tails is 710. For 2 consecutive heads before 4 consecutive tails, the probability rises to 56.

In our next example, we return to the matching problem and obtain a solution by using conditional probabilities.

Example 5d

At a party, n people take off their hats. The hats are then mixed up, and each person randomly selects one. We say that a match occurs if a person selects his or her own hat. What is the probability of

a. no matches?
b. exactly k matches?

Solution

a. Let E denote the event that no matches occur, and to make explicit the dependence on n, write Pn=P(E). We start by conditioning on whether or not the first person selects his or her own hat—call these events M and Mc, respectively. Then,
$P_n=P(E)=P(E|M)P(M)+P(E|M^c)P(M^c)$

   Clearly, P(E|M)=0, so
$P_n=P(E|M^c)p^{n-1}_n$ (5.9)

Now, P(E|Mc) is the probability of no matches when n−1 people select from a set of n−1 hats, when one person, called the “extra” person, does not have their hat in the collection, and one hat, called the “extra” hat, does not belong to any of the people. This can happen in either of two mutually exclusive ways: Either there are no matches and the extra person does not select the extra hat (this being the hat of the person who chose first) or there are no matches and the extra person does select the extra hat. The probability of the first of these events is just Pn−1, which is seen by regarding the extra hat as “belonging” to the extra person. Because the second event has probability [1/(n−1)]Pn−2, we have
P(E|Mc)=Pn−1+1n−1Pn−2

Thus, from **Equation (5.9)**,
Pn=n−1nPn−1+1nPn−2

or, equivalently,
Pn−Pn−1=−1n(Pn−1−Pn−2) (5.10)

However, since Pn is the probability of no matches when n people select among their own hats, we have
P1=0 P2=12

So, from **Equation (5.10)**,
P3−P2=−(P2−P1)3=−13! or P3=12!−13!P4−P3=−(P3−P2)4=14! or P4=12!−13!+14!

and, in general,
Pn=12!−13!+14!−⋯+(−1)nn!

b. To obtain the probability of exactly k matches, we consider any fixed group of k people. The probability that they, and only they, select their own hats is
1n1n−1⋯1n−(k−1)Pn−k=(n−k)!n!Pn−k

where Pn−k is the conditional probability that the other n−k people, selecting among their own hats, have no matches. Since there are (nk) choices of a set of k people, the desired probability of exactly k matches is
Pn−kk!=12!−13!+⋯+(−1)n−k(n−k)!k!

An important concept in probability theory is that of the conditional independence of events. We say that the events E1 and E2 are *conditionally independent* given F if given that F occurs, the conditional probability that E1 occurs is unchanged by information as to whether or not E2 occurs. More formally, E1 and E2 are said to be conditionally independent given F if

P(E1|E2F)=P(E1|F) (5.11)

or, equivalently,

P(E1E2|F)=P(E1|F)P(E2|F) (5.12)

The notion of conditional independence can easily be extended to more than two events, and this extension is left as an exercise.

The reader should note that the concept of conditional independence was implicitly employed in **Example 5a**, where it was assumed that the events that a policyholder had an accident in his or her ith year, i=1,2,…, were conditionally independent given whether or not the person was accident prone. The following example, sometimes referred to as *Laplace’s rule of succession*, further illustrates the concept of conditional independence.

Example 5e Laplace’s rule of succession

There are k+1 coins in a box. When flipped, the ith coin will turn up heads with probability i/k,i=0,1,…,k. A coin is randomly selected from the box and is then repeatedly flipped. If the first n flips all result in heads, what is the conditional probability that the (n+1) flip will do likewise?

Solution

Letting Hn denote the event that the first n flips all land heads, the desired probability is

P(Hn+1|Hn)=P(Hn+1Hn)P(Hn)=P(Hn+1)P(Hn)

To compute P(Hn), we condition on which coin is chosen. That is, letting Ci denote the event that coin i is selected, we have that

$$P(Hn)=\sum_{i=0}^{k}P(Hn|Ci)P(Ci)$$

Now, given that coin i is selected, it is reasonable to assume that the outcomes will be conditionally independent, with each one resulting in a head with probability i/k. Hence,

$$P(Hn|Ci)=(i/k)n$$

As P(Ci)=1k+1, this yields that

$$P(Hn)=1k+1\sum_{i=0}^{k}(i/k)n$$

Thus,

$$P(Hn+1|Hn)=\sum_{i=0}^{k}(i/k)n+1\sum_{i=0}^{k}(i/k)n$$

If k is large, we can use the integral approximations

$$1k\sum_{i=0}^{k}(ik)n+1\approx\int_{0}^{1}xn+1dx=1n+21k\sum_{j=0}^{k}(jk)n\approx\int_{0}^{1}xndx=1n+1$$

So, for k large,

$$P(Hn+1|Hn)\approx n+1n+2$$

Example 5f Updating information sequentially

Suppose there are n mutually exclusive and exhaustive possible hypotheses, with initial (sometimes referred to as *prior*) probabilities P(Hi), $\sum_{i=1}^{n}P(Hi)=1$. Now, if information that the event E has occurred is received, then the conditional probability that Hi is the true hypothesis (sometimes referred to as the *updated* or *posterior* probability of Hi) is

$$P(Hi|E)=P(E|Hi)P(Hi)\sum_{j}P(E|Hj)P(Hj) \tag{5.13}$$

Suppose now that we learn first that E1 has occurred and then that E2 has occurred. Then, given only the first piece of information, the conditional probability that Hi is the true hypothesis is

$$P(Hi|E1)=P(E1|Hi)P(Hi)P(E1)=P(E1|Hi)P(Hi)\sum_{j}P(E1|Hj)P(Hj)$$

whereas given both pieces of information, the conditional probability that Hi is the true hypothesis is P(Hi|E1E2), which can be computed by

$$P(Hi|E1E2)=P(E1E2|Hi)P(Hi)\sum_{j}P(E1E2|Hj)P(Hj)$$

One might wonder, however, when one can compute P(Hi|E1E2) by using the right side of **Equation (5.13)** with E=E2 and with P(Hj) replaced by P(Hj|E1), j=1,…,n. That is, when is it legitimate to regard P(Hj|E1), j≥1, as the prior probabilities and then use (**5.13**) to compute the posterior probabilities?

Solution

The answer is that the preceding is legitimate, provided that for each j=1,…,n, the events E1 and E2 are conditionally independent, given Hj. For if this is the case, then

$$P(E1E2|Hj)=P(E2|Hj)P(E1|Hj),\quad j=1,\ldots,n$$

Therefore,

$$P(Hi|E1E2)=P(E2|Hi)P(E1|Hi)P(Hi)P(E1E2)=P(E2|Hi)P(E1Hi)P(E1E2)=P(E2|Hi)P(Hi|E1)P(E1)P(E1E2)=P(E2|Hi)P(Hi|E1)Q(1,2)$$

where Q(1,2)=P(E1E2)P(E1). Since the preceding equation is valid for all i, we obtain, upon summing,

$$1=\sum_{i=1}^{n}P(Hi|E1E2)=\sum_{i=1}^{n}P(E2|Hi)P(Hi|E1)Q(1,2)$$

showing that

$$Q(1,2)=\sum_{i=1}^{n}P(E2|Hi)P(Hi|E1)$$

and yielding the result

$$P(Hi|E1E2)=P(E2|Hi)P(Hi|E1)\sum_{i=1}^{n}P(E2|Hi)P(Hi|E1)$$

For instance, suppose that one of two coins is chosen to be flipped. Let Hi be the event that coin i, i=1,2, is chosen, and suppose that when coin i is flipped, it lands on heads with probability pi, i=1,2. Then, the preceding equations show that to sequentially update the probability that coin 1 is the one being flipped, given the results of the previous flips, all that must be saved after each new flip is the conditional probability that coin 1 is the coin being used. That is, it is not necessary to keep track of all earlier results.

Summary

For events E and F, the conditional probability of E given that F has occurred is denoted by P(E|F) and is defined by

P(E|F)=P(EF)P(F)

The identity

P(E1E2⋯En)=P(E1)P(E2|E1)⋯P(En|E1⋯En−1)

is known as the *multiplication rule* of probability.

A valuable identity is

P(E)=P(E|F)P(F)+P(E|Fc)P(Fc)

which can be used to compute P(E) by “conditioning” on whether F occurs.

P(H)/P(Hc) is called the *odds* of the event H. The identity

P(H|E)P(Hc|E)=P(H) P(E|H)P(Hc)P(E|Hc)

shows that when new evidence E is obtained, the value of the odds of H becomes its old value multiplied by the ratio of the conditional probability of the new evidence when H is true to the conditional probability when H is not true.

Let Fi, i=1,…,n, be mutually exclusive events whose union is the entire sample space. The identity

P(Fj|E)=P(E|Fj)P(Fj)∑i=1nP(E|Fi)P(Fi)

is known as *Bayes’s formula*. If the events Fi, i=1,…,n, are competing hypotheses, then Bayes’s formula shows how to compute the conditional probabilities of these hypotheses when additional evidence E becomes available.

The denominator of Bayes’s formula uses that

P(E)=∑i=1nP(E|Fi)P(Fi)

which is called the law of total probability.

If P(EF)=P(E)P(F), then we say that the events E and F are *independent*. This condition is equivalent to P(E|F)=P(E) and to P(F|E)=P(F). Thus, the events E and F are independent if knowledge of the occurrence of one of them does not affect the probability of the other.

The events E1,…,En are said to be independent if, for any subset Ei1,…,Eir of them,

P(Ei1⋯Eir)=P(Ei1)⋯P(Eir)

For a fixed event F, P(E|F) can be considered to be a probability function on the events E of the sample space.

Problems

3.1. Two fair dice are rolled. What is the conditional probability that at least one lands on 6 given that the dice land on different numbers?
3.2. If two fair dice are rolled, what is the conditional probability that the first one lands on 6 given that the sum of the dice is i? Compute for all values of i between 2 and 12.
3.3. Use **Equation (2.1)** to compute in a hand of bridge the conditional probability that East has 3 spades given that North and South have a combined total of 8 spades.
3.4. What is the probability that at least one of a pair of fair dice lands on 6, given that the sum of the dice is i, i=2,3,…,12?
3.5. An urn contains 6 white and 9 black balls. If 4 balls are to be randomly selected without replacement, what is the probability that the first 2 selected are white and the last 2 black?
3.6. Consider an urn containing 12 balls, of which 8 are white. A sample of size 4 is to be drawn with replacement (without replacement). What is the conditional probability (in each case) that the first and third balls drawn will be white given that the sample drawn contains exactly 3 white balls?
3.7. The king comes from a family of 2 children. What is the probability that the other child is his sister?
3.8. A couple has 2 children. What is the probability that both are girls if the older of the two is a girl?
3.9. Consider 3 urns. Urn A contains 2 white and 4 red balls, urn B contains 8 white and 4 red balls, and urn C contains 1 white and 3 red balls. If 1 ball is selected from each urn, what is the probability that the ball chosen from urn A was white given that exactly 2 white balls were selected?
3.10. Three cards are randomly selected, without replacement, from an ordinary deck of 52 playing cards. Compute the conditional probability that the first card selected is a spade given that the second and third cards are spades.
3.11. Two cards are randomly chosen without replacement from an ordinary deck of 52 cards. Let B be the event that both cards are aces, let As be the event that the ace of spades is chosen, and let A be the event that at least one ace is chosen. Find

a. P(B|As)
b. P(B|A)

3.12. Suppose distinct values are written on each of 3 cards, which are then randomly given the designations A, B, and C. Given that card A's value is less than card B's value, find the probability it is also less than card C's value.
3.13. A recent college graduate is planning to take the first three actuarial examinations in the coming summer. She will take the first actuarial exam in June. If she passes that exam, then she will take the second exam in July, and if she also passes that one, then she will take the third exam in September. If she fails an exam, then she is not allowed to take any others. The probability that she passes the first exam is .9. If she passes the first exam, then the conditional probability that she passes the second one is .8, and if she passes both the first and the second exams, then the conditional probability that she passes the third exam is .7.

a. What is the probability that she passes all three exams?
b. Given that she did not pass all three exams, what is the conditional probability that she failed the second exam?

3.14. Suppose that an ordinary deck of 52 cards (which contains 4 aces) is randomly divided into 4 hands of 13 cards each. We are interested in determining p, the probability that each hand has an ace. Let Ei be the event that the ith hand has exactly one ace. Determine p=P(E1E2E3E4) by using the multiplication rule.
3.15. An urn initially contains 5 white and 7 black balls. Each time a ball is selected, its color is noted and it is replaced in the urn along with 2 other balls of the same color. Compute the probability that

a. the first 2 balls selected are black and the next 2 are white;
b. of the first 4 balls selected, exactly 2 are black.

3.16. An ectopic pregnancy is twice as likely to develop when the pregnant woman is a smoker as it is when she is a nonsmoker. If 32 percent of women of childbearing age are smokers, what percentage of women having ectopic pregnancies are smokers?
3.17. Ninety-eight percent of all babies survive delivery. However, 15 percent of all births involve Cesarean (C) sections, and when a C section is performed, the baby survives 96 percent of the time. If a randomly chosen pregnant woman does not have a C section, what is the probability that her baby survives?
3.18. In a certain community, 36 percent of the families own a dog and 22 percent of the families that own a dog also own a cat. In addition, 30 percent of the families own a cat. What is

a. the probability that a randomly selected family owns both a dog and a cat?
b. the conditional probability that a randomly selected family owns a dog given that it owns a cat?

3.19. A total of 46 percent of the voters in a certain city classify themselves as Independents, whereas 30 percent classify themselves as Liberals and 24 percent say that they are Conservatives. In a recent local election, 35 percent of the Independents, 62 percent of the Liberals, and 58 percent of the Conservatives voted. A voter is chosen at random. Given that this person voted in the local election, what is the probability that he or she is

a. an Independent?
b. a Liberal?
c. a Conservative?
d. What percent of voters participated in the local election?

3.20. A total of 48 percent of the women and 37 percent of the men who took a certain “quit smoking” class remained nonsmokers for at least one year after completing the class. These people then attended a success party at the end of a year. If 62 percent of the original class was male,

a. what percentage of those attending the party were women?
b. what percentage of the original class attended the party?

3.21. Fifty-two percent of the students at a certain college are females. Five percent of the students in this college are majoring in computer science. Two percent of the students are women majoring in computer science. If a student is selected at random, find the conditional probability that

a. the student is female given that the student is majoring in computer science;
b. this student is majoring in computer science given that the student is female.

3.22. A total of 500 married working couples were polled about their annual salaries, with the following information resulting:

Wife	Husband	
	Less than $125,000	More than $125,000
Less than $125,000	212	198
More than $125,000	36	54

For instance, in 36 of the couples, the wife earned more and the husband earned less than $125,000. If one of the couples is randomly chosen, what is

a. the probability that the husband earns less than $125,000?
b. the conditional probability that the wife earns more than $125,000 given that the husband earns more than this amount?
c. the conditional probability that the wife earns more than $125,000 given that the husband earns less than this amount?

3.23. A red die, a blue die, and a yellow die (all six sided) are rolled. We are interested in the probability that the number appearing on the blue die is less than that appearing on the yellow die, which is less than that appearing on the red die. That is, with B,Y, and R denoting, respectively, the number appearing on the blue, yellow, and red die, we are interested in P(B<Y<R).

a. What is the probability that no two of the dice land on the same number?
b. Given that no two of the dice land on the same number, what is the conditional probability that B<Y<R?
c. What is P(B<Y<R)

3.24. Urn I contains 2 white and 4 red balls, whereas urn II contains 1 white and 1 red ball. A ball is randomly chosen from urn I and put into urn II, and a ball is then randomly selected from urn II. What is

a. the probability that the ball selected from urn II is white?
b. the conditional probability that the transferred ball was white given that a white ball is selected from urn II?

3.25. Twenty percent of B's phone calls are with her daughter. Sixty five percent of the time that B speaks with her daughter she hangs up the phone with a smile on her face. Given that B has just hung up the phone with a smile on her face, we are interested in the conditional probability that the phone call was with her daughter. Do we have enough information to determine this probability. If yes, what is it; if no, what additional information is needed.

3.26. Each of 2 balls is painted either black or gold and then placed in an urn. Suppose that each ball is colored black with probability 12 and that these events are independent.

a. Suppose that you obtain information that the gold paint has been used (and thus at least one of the balls is painted gold). Compute the conditional probability that both balls are painted gold.
b. Suppose now that the urn tips over and 1 ball falls out. It is painted gold. What is the probability that both balls are gold in this case? Explain.

3.27. The following method was proposed to estimate the number of people over the age of 50 who reside in a town of known population 100,000: "As you walk along the streets, keep a running count of the percentage of people you encounter who are over 50. Do this for a few days; then multiply the percentage you obtain by 100,000 to obtain the estimate." Comment on this method.

Hint: Let p denote the proportion of people in the town who are over 50. Furthermore, let $\alpha 1$ denote the proportion of time that a person under the age of 50 spends in the streets, and let $\alpha 2$ be the corresponding value for those over 50. What quantity does the method suggested estimate? When is the estimate approximately equal to p?

3.28. Suppose that 5 percent of men and 0.25 percent of women are color blind. A color-blind person is chosen at random. What is the probability of this person being male? Assume that there are an equal number of males and females. What if the population consisted of twice as many males as females?

3.29. All the workers at a certain company drive to work and park in the company's lot. The company is interested in estimating the average number of workers in a car. Which of the following methods will enable the company to estimate this quantity? Explain your answer.

A. Randomly choose n workers, find out how many were in the cars in which they were driven, and take the average of the n values.
B. Randomly choose n cars in the lot, find out how many were driven in those cars, and take the average of the n values.

3.30 Suppose that an ordinary deck of 52 cards is shuffled and the cards are then turned over one at a time until the first ace appears. Given that the first ace is the 20th card to appear, what is the conditional probability that the card following it is the

a. ace of spades?

b. two of clubs?

3.31. There are 15 tennis balls in a box, of which 9 have not previously been used. Three of the balls are randomly chosen, played with, and then returned to the box. Later, another 3 balls are randomly chosen from the box. Find the probability that none of these balls has ever been used.
3.32. Consider two boxes, one containing 1 black and 1 white marble, the other 2 black and 1 white marble. A box is selected at random, and a marble is drawn from it at random. What is the probability that the marble is black? What is the probability that the first box was the one selected given that the marble is white?
3.33. Ms. Aquina has just had a biopsy on a possibly cancerous tumor. Not wanting to spoil a weekend family event, she does not want to hear any bad news in the next few days. But if she tells the doctor to call only if the news is good, then if the doctor does not call, Ms. Aquina can conclude that the news is bad. So, being a student of probability, Ms. Aquina instructs the doctor to flip a coin. If it comes up heads, the doctor is to call if the news is good and not call if the news is bad. If the coin comes up tails, the doctor is not to call. In this way, even if the doctor doesn't call, the news is not necessarily bad. Let α be the probability that the tumor is cancerous; let β be the conditional probability that the tumor is cancerous given that the doctor does not call.

a. Which should be larger, α or β
b. Find β in terms of α, and prove your answer in part (a).

3.34. A family has j children with probability pj, where p1=.1,p2=.25,p3=.35,p4=.3. A child from this family is randomly chosen. Given that this child is the eldest child in the family, find the conditional probability that the family has

a. only 1 child;
b. 4 children.

Redo (a) and (b) when the randomly selected child is the youngest child of the family.
3.35. On rainy days, Joe is late to work with probability .3; on nonrainy days, he is late with probability .1. With probability .7, it will rain tomorrow.

a. Find the probability that Joe is early tomorrow.
b. Given that Joe was early, what is the conditional probability that it rained?

3.36. In **Example 3f**, suppose that the new evidence is subject to different possible interpretations and in fact shows only that it is 90 percent likely that the criminal possesses the characteristic in question. In this case, how likely would it be that the suspect is guilty (assuming, as before, that he has the characteristic)?
3.37. With probability .6, the present was hidden by mom; with probability .4, it was hidden by dad. When mom hides the present, she hides it upstairs 70 percent of the time and downstairs 30 percent of the time. Dad is equally likely to hide it upstairs or downstairs.

a. What is the probability that the present is upstairs?
b. Given that it is downstairs, what is the probability it was hidden by dad?

3.38. Stores A, B, and C have 50, 75, and 100 employees, respectively, and 50, 60, and 70 percent of them respectively are women. Resignations are equally likely among all employees, regardless of sex. One woman employee resigns. What is the probability that she works in store C?
3.39.

a. A gambler has a fair coin and a two-headed coin in his pocket. He selects one of the coins at random; when he flips it, it shows heads. What is the probability that it is the fair coin?
b. Suppose that he flips the same coin a second time and, again, it shows heads. Now what is the probability that it is the fair coin?
c. Suppose that he flips the same coin a third time and it shows tails. Now what is the probability that it is the fair coin?

3.40. Urn A has 5 white and 7 black balls. Urn B has 3 white and 12 black balls. We flip a fair coin. If the outcome is heads, then a ball from urn A is selected, whereas if the outcome is tails, then a ball from urn B is selected. Suppose that a white ball is selected. What is the probability that the coin landed tails?
3.41. In **Example 3a**, what is the probability that someone has an accident in the second year given that he or she had no accidents in the first year?
3.42. Consider a sample of size 3 drawn in the following manner: We start with an urn containing 5 white and 7 red balls. At each stage, a ball is drawn and its color is noted. The ball is then returned to the urn, along with an additional ball of the same color. Find the probability that the sample will contain exactly

a. 0 white balls;
b. 1 white ball;
c. 3 white balls;
d. 2 white balls.

3.43. A deck of cards is shuffled and then divided into two halves of 26 cards each. A card is drawn from one of the halves; it turns out to be an ace. The ace is then placed in the second half-deck. The half is then shuffled, and a card is drawn from it. Compute the probability that this drawn card is an ace.
Hint: Condition on whether or not the interchanged card is selected.
3.44. Twelve percent of all U.S. households are in California. A total of 1.3 percent of all U.S. households earn more than $250,000 per year, while a total of 3.3 percent of all California households earn more than $250,000 per year.

a. What proportion of all non-California households earn more than $250,000 per year?
b. Given that a randomly chosen U.S. household earns more than $250,000 per year, what is the probability it is a California household?

3.45. There are 3 coins in a box. One is a two-headed coin, another is a fair coin, and the third is a biased coin that comes up heads 75 percent of the time. When one of the 3 coins is selected at random and flipped, it shows heads. What is the probability that it was the two-headed coin?
3.46. Three prisoners are informed by their jailer that one of them has been chosen at random to be executed and the other two are to be freed. Prisoner A asks the jailer to tell him privately which of his fellow prisoners will be set free, claiming that there would be no harm in divulging this information because he already knows that at least one of the two will go free. The jailer refuses to answer the question, pointing out that if A knew which of his fellow prisoners were to be set free, then his own probability of being executed would rise from 13 to 12 because he would then be one of two prisoners. What do you think of the jailer's reasoning?
3.47. There is a 30 percent chance that A can fix her busted computer. If A cannot, then there is a 40 percent chance that her friend B can fix it.

a. Find the probability it will be fixed by either A or B.
b. If it is fixed, what is the probability it will be fixed by B.

3.48. In any given year, a male automobile policyholder will make a claim with probability pm and a female policyholder will make a claim with probability pf, where pf≠pm. The fraction of the policyholders that are male is α,0<α<1. A policyholder is randomly chosen. If Ai denotes the event that this policyholder will make a claim in year i, show that
P(A2|A1)P(A1)

Give an intuitive explanation of why the preceding inequality is true.
3.49. An urn contains 5 white and 10 black balls. A fair die is rolled and that number of balls is randomly chosen from the urn. What is the probability that all of the balls selected are white? What is the conditional probability that the die landed on 3 if all the balls selected are white?
3.50. Each of 2 cabinets identical in appearance has 2 drawers. Cabinet A contains a silver coin in each drawer, and cabinet B contains a silver coin in one of its drawers and a gold coin in the other. A

cabinet is randomly selected, one of its drawers is opened, and a silver coin is found. What is the probability that there is a silver coin in the other drawer?

3.51. Prostate cancer is the most common type of cancer found in males. As an indicator of whether a male has prostate cancer, doctors often perform a test that measures the level of the prostate-specific antigen (PSA) that is produced only by the prostate gland. Although PSA levels are indicative of cancer, the test is notoriously unreliable. Indeed, the probability that a noncancerous man will have an elevated PSA level is approximately .135, increasing to approximately .268 if the man does have cancer. If, on the basis of other factors, a physician is 70 percent certain that a male has prostate cancer, what is the conditional probability that he has the cancer given that

a. the test indicated an elevated PSA level?
b. the test did not indicate an elevated PSA level?

Repeat the preceding calculation, this time assuming that the physician initially believes that there is a 30 percent chance that the man has prostate cancer.

3.52. Suppose that an insurance company classifies people into one of three classes: good risks, average risks, and bad risks. The company's records indicate that the probabilities that good-, average-, and bad-risk persons will be involved in an accident over a 1-year span are, respectively, .05, .15, and .30. If 20 percent of the population is a good risk, 50 percent an average risk, and 30 percent a bad risk, what proportion of people have accidents in a fixed year? If policyholder A had no accidents in 2012, what is the probability that he or she is a good risk? is an average risk?

3.53. A worker has asked her supervisor for a letter of recommendation for a new job. She estimates that there is an 80 percent chance that she will get the job if she receives a strong recommendation, a 40 percent chance if she receives a moderately good recommendation, and a 10 percent chance if she receives a weak recommendation. She further estimates that the probabilities that the recommendation will be strong, moderate, and weak are .7, .2, and .1, respectively.

a. How certain is she that she will receive the new job offer?
b. Given that she does receive the offer, how likely should she feel that she received a strong recommendation? a moderate recommendation? a weak recommendation?
c. Given that she does not receive the job offer, how likely should she feel that she received a strong recommendation? a moderate recommendation? a weak recommendation?

3.54. Players A, B, C, D are randomly lined up. The first two players in line then play a game; the winner of that game then plays a game with the person who is third in line; the winner of that game then plays a game with the person who is fourth in line. The winner of that last game is considered the winner of the tournament. If A wins each game it plays with probability p, determine the probability that A is the winner of the tournament.

3.55. Players 1,2,3 are playing a tournament. Two of these three players are randomly chosen to play a game in round one, with the winner then playing the remaining player in round two. The winner of round two is the tournament victor. Assume that all games are independent and that i wins when playing against j with probability ii+j.

a. Find the probability that 1 is the tournament victor.
b. If 1 is the tournament victor, find the conditional probability that 1 did not play in round one.

3.56. Suppose there are two coins, with coin 1 landing heads when flipped with probability .3 and coin 2 with probability .5. Suppose also that we randomly select one of these coins and then continually flip it. Let Hj denote the event that flip j,j≥1, lands heads. Also, let Ci be the event that coin i was chosen, i=1,2.

a. Find P(H1).
b. Find P(H2|H1).
c. Find P(C1|H1).
d. Find P(H2H3H4|H1).

3.57. In a 7 game series played with two teams, the first team to win a total of 4 games is the winner. Suppose that each game played is independently won by team A with probability p.

a. Given that one team leads 3 to 0, what is the probability that it is team A that leads.
b. Given that one team leads 3 to 0, what is the probability that team wins the series.

3.58. A parallel system functions whenever at least one of its components works. Consider a parallel system of n components, and suppose that each component works independently with probability 12. Find the conditional probability that component 1 works given that the system is functioning.
3.59. If you had to construct a mathematical model for events E and F, as described in parts (a) through (e), would you assume that they were independent events? Explain your reasoning.

a. E is the event that a businesswoman has blue eyes, and F is the event that her secretary has blue eyes.
b. E is the event that a professor owns a car, and F is the event that he is listed in the telephone book.
c. E is the event that a man is under 6 feet tall, and F is the event that he weighs more than 200 pounds.
d. E is the event that a woman lives in the United States, and F is the event that she lives in the Western Hemisphere.
e. E is the event that it will rain tomorrow, and F is the event that it will rain the day after tomorrow.

3.60. In a class, there are 4 first-year boys, 6 first-year girls, and 6 sophomore boys. How many sophomore girls must be present if sex and class are to be independent when a student is selected at random?
3.61. Suppose that you continually collect coupons and that there are m different types. Suppose also that each time a new coupon is obtained, it is a type i coupon with probability pi,i=1,…,m. Suppose that you have just collected your nth coupon. What is the probability that it is a new type?
Hint: Condition on the type of this coupon.
3.62. A simplified model for the movement of the price of a stock supposes that on each day the stock's price either moves up 1 unit with probability p or moves down 1 unit with probability 1−p. The changes on different days are assumed to be independent.

a. What is the probability that after 2 days the stock will be at its original price?
b. What is the probability that after 3 days the stock's price will have increased by 1 unit?
c. Given that after 3 days the stock's price has increased by 1 unit, what is the probability that it went up on the first day?

3.63. Suppose that we want to generate the outcome of the flip of a fair coin, but that all we have at our disposal is a biased coin that lands on heads with some unknown probability p that need not be equal to 12. Consider the following procedure for accomplishing our task:

1. Flip the coin.
2. Flip the coin again.
3. If both flips land on heads or both land on tails, return to step 1.
4. Let the result of the last flip be the result of the experiment.

a. Show that the result is equally likely to be either heads or tails.
b. Could we use a simpler procedure that continues to flip the coin until the last two flips are different and then lets the result be the outcome of the final flip?

3.64. Independent flips of a coin that lands on heads with probability p are made. What is the probability that the first four outcomes are

a. H, H, H, H?
b. T, H, H, H?
c. What is the probability that the pattern H, H, H, H? occurs before the pattern T, H, H, H?

Hint for part (c): How can the pattern H, H, H, H? occur first?

3.65. The color of a person's eyes is determined by a single pair of genes. If they are both blue-eyed genes, then the person will have blue eyes; if they are both brown-eyed genes, then the person will have brown eyes; and if one of them is a blue-eyed gene and the other a brown-eyed gene, then the person will have brown eyes. (Because of the latter fact, we say that the brown-eyed gene is *dominant* over the blue-eyed one.) A newborn child independently receives one eye gene from each of its parents, and the gene it receives from a parent is equally likely to be either of the two eye genes of that parent. Suppose that Smith and both of his parents have brown eyes, but Smith's sister has blue eyes.

a. What is the probability that Smith possesses a blueeyed gene?
b. Suppose that Smith's wife has blue eyes. What is the probability that their first child will have blue eyes?
c. If their first child has brown eyes, what is the probability that their next child will also have brown eyes?

3.66. Genes relating to albinism are denoted by A and a. Only those people who receive the a gene from both parents will be albino. Persons having the gene pair A, a are normal in appearance and, because they can pass on the trait to their offspring, are called carriers. Suppose that a normal couple has two children, exactly one of whom is an albino. Suppose that the nonalbino child mates with a person who is known to be a carrier for albinism.

a. What is the probability that their first offspring is an albino?
b. What is the conditional probability that their second offspring is an albino given that their firstborn is not?

3.67. Barbara and Dianne go target shooting. Suppose that each of Barbara's shots hits a wooden duck target with probability p1, while each shot of Dianne's hits it with probability p2. Suppose that they shoot simultaneously at the same target. If the wooden duck is knocked over (indicating that it was hit), what is the probability that

a. both shots hit the duck?
b. Barbara's shot hit the duck?

What independence assumptions have you made?

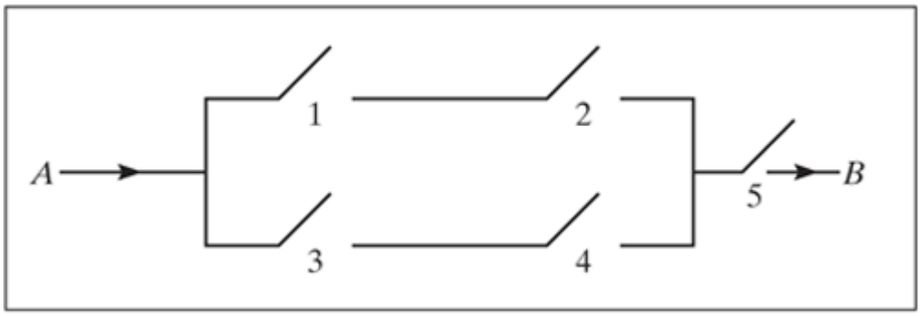
3.68. A and B are involved in a duel. The rules of the duel are that they are to pick up their guns and shoot at each other simultaneously. If one or both are hit, then the duel is over. If both shots miss, then they repeat the process. Suppose that the results of the shots are independent and that each shot of A will hit B with probability pA, and each shot of B will hit A with probability pB. What is

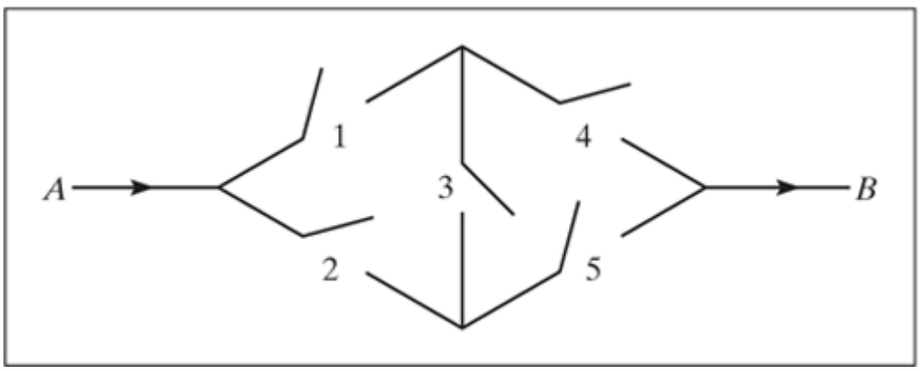
a. the probability that A is not hit?
b. the probability that both duelists are hit?
c. the probability that the duel ends after the nth round of shots?
d. the conditional probability that the duel ends after the nth round of shots given that A is not hit?
e. the conditional probability that the duel ends after the nth round of shots given that both duelists are hit?

3.69. Assume, as in**Example 3h**, that 64 percent of twins are of the same sex. Given that a newborn set of twins is of the same sex, what is the conditional probability that the twins are identical?

3.70. The probability of the closing of the ith relay in the circuits shown in **Figure 3.5** is given by pi,i=1, 2, 3, 4, 5. If all relays function independently, what is the probability that a current flows between A and B for the respective circuits?

Figure 3.5 Circuits for problem 3.70





Hint for (b): Condition on whether relay 3 closes.

3.71. An engineering system consisting of n components is said to be a k-out-of-n system (k≤n) if the system functions if and only if at least k of the n components function. Suppose that all components function independently of one another.

a. If the ith component functions with probability Pi,i=1, 2, 3, 4, compute the probability that a 2-out-of-4 system functions.
b. Repeat part (a) for a 3-out-of-5 system.
c. Repeat for a k-out-of-n system when all the Pi equal p (that is, Pi=p,i=1,2,…,n).

3.72. In **Problem 3.70a**, find the conditional probability that relays 1 and 2 are both closed given that a current flows from A to B.

3.73. A certain organism possesses a pair of each of 5 different genes (which we will designate by the first 5 letters of the English alphabet). Each gene appears in 2 forms (which we designate by lowercase and capital letters). The capital letter will be assumed to be the dominant gene, in the sense that if an organism possesses the gene pair *xX*, then it will outwardly have the appearance of the X gene. For instance, if X stands for brown eyes and x for blue eyes, then an individual having either gene pair *XX* or *xX* will have brown eyes, whereas one having gene pair *xx* will have blue eyes. The characteristic appearance of an organism is called its phenotype, whereas its genetic constitution is called its genotype. (Thus, 2 organisms with respective genotypes *aA, bB, cc, dD, ee* and *AA, BB, cc, DD, ee* would have different genotypes but the same phenotype.) In a mating between 2 organisms, each one contributes, at random, one of its gene pairs of each type. The 5 contributions of an organism (one of each of the 5 types) are assumed to be independent and are also independent of the contributions of the organism's mate. In a mating between organisms having genotypes *aA, bB, cC, dD, eE* and *aa, bB, cc, Dd, ee* what is the probability that the progeny will (i) phenotypically and (ii) genotypically resemble

a. the first parent?
b. the second parent?

c. either parent?
d. neither parent?

3.74. There is a 50–50 chance that the queen carries the gene for hemophilia. If she is a carrier, then each prince has a 50–50 chance of having hemophilia. If the queen has had three princes without the disease, what is the probability that the queen is a carrier? If there is a fourth prince, what is the probability that he will have hemophilia?

3.75. A town council of 7 members contains a steering committee of size 3. New ideas for legislation go first to the steering committee and then on to the council as a whole if at least 2 of the 3 committee members approve the legislation. Once at the full council, the legislation requires a majority vote (of at least 4) to pass. Consider a new piece of legislation, and suppose that each town council member will approve it, independently, with probability p. What is the probability that a given steering committee member's vote is decisive in the sense that if that person's vote were reversed, then the final fate of the legislation would be reversed? What is the corresponding probability for a given council member not on the steering committee?

3.76.Suppose that each child born to a couple is equally likely to be a boy or a girl, independently of the sex distribution of the other children in the family. For a couple having 5 children, compute the probabilities of the following events:

a. All children are of the same sex.
b. The 3 eldest are boys and the others girls.
c. Exactly 3 are boys.
d. The 2 oldest are girls.
e. There is at least 1 girl.

3.77. A and B alternate rolling a pair of dice, stopping either when A rolls the sum 9 or when B rolls the sum 6. Assuming that A rolls first, find the probability that the final roll is made by A.

3.78. In a certain village, it is traditional for the eldest son (or the older son in a two-son family) and his wife to be responsible for taking care of his parents as they age. In recent years, however, the women of this village, not wanting that responsibility, have not looked favorably upon marrying an eldest son.

a. If every family in the village has two children, what proportion of all sons are older sons?
b. If every family in the village has three children, what proportion of all sons are eldest sons? Assume that each child is, independently, equally likely to be either a boy or a girl.

3.79. Suppose that E and F are mutually exclusive events of an experiment. Show that if independent trials of this experiment are performed, then E will occur before F with probability $P(E)/[P(E)+P(F)]$.

3.80. Consider an unending sequence of independent trials, where each trial is equally likely to result in any of the outcomes 1, 2, or 3. Given that outcome 3 is the last of the three outcomes to occur, find the conditional probability that

a. the first trial results in outcome 1;
b. the first two trials both result in outcome 1.

3.81. A and B play a series of games. Each game is independently won by A with probability p and by B with probability $1-p$. They stop when the total number of wins of one of the players is two greater than that of the other player. The player with the greater number of total wins is declared the winner of the series.

a. Find the probability that a total of 4 games are played.
b. Find the probability that A is the winner of the series.

3.82. In successive rolls of a pair of fair dice, what is the probability of getting 2 sevens before 6 even numbers?

3.83. In a certain contest, the players are of equal skill and the probability is 12 that a specified one of the two contestants will be the victor. In a group of 2n players, the players are paired off against each

other at random. The 2n−1 winners are again paired off randomly, and so on, until a single winner remains. Consider two specified contestants, A and B, and define the events Ai,i≤n,E by
Ai:A plays in exactly i contestsE:A and B never play each other

a. Find P(Ai),i=1,…,n.
b. Find P(E).
c. Let Pn=P(E). Show that
   Pn=12n−1+2n−22n−1(12)2Pn−1

   and use this formula to check the answer you obtained in part (b).
   Hint: Find P(E) by conditioning on which of the events Ai,i=1,…,n occur. In simplifying your answer, use the algebraic identity
   ∑i=1n−1ixi−1=1−nxn−1+(n−1)xn(1−x)2

   For another approach to solving this problem, note that there are a total of 2n−1 games played.
d. Explain why 2n−1 games are played.
   Number these games, and let Bi denote the event that A and B play each other in game i,i=1,…,2n−1.
e. What is P(Bi)
f. Use part (e) to find P(E).

3.84. An investor owns shares in a stock whose present value is 25. She has decided that she must sell her stock if it goes either down to 10 or up to 40. If each change of price is either up 1 point with probability .55 or down 1 point with probability .45, and the successive changes are independent, what is the probability that the investor retires a winner?
3.85. A and B flip coins. A starts and continues flipping until a tail occurs, at which point B starts flipping and continues until there is a tail. Then A takes over, and so on. Let P1 be the probability of the coin landing on heads when A flips and P2 when B flips. The winner of the game is the first one to get

a. 2 heads in a row;
b. a total of 2 heads;
c. 3 heads in a row;
d. a total of 3 heads.

In each case, find the probability that A wins.
3.86. Die A has 4 red and 2 white faces, whereas die B has 2 red and 4 white faces. A fair coin is flipped once. If it lands on heads, the game continues with die A; if it lands on tails, then die B is to be used.

a. Show that the probability of red at any throw is 12.
b. If the first two throws result in red, what is the probability of red at the third throw?
c. If red turns up at the first two throws, what is the probability that it is die A that is being used?

3.87. An urn contains 12 balls, of which 4 are white. Three players—A, B, and C successively draw from the urn, A first, then B, then C, then A, and so on. The winner is the first one to draw a white ball. Find the probability of winning for each player if

a. each ball is replaced after it is drawn;
b. the balls that are withdrawn are not replaced.

3.88. Repeat **Problem 3.87** when each of the 3 players selects from his own urn. That is, suppose that there are 3 different urns of 12 balls with 4 white balls in each urn.
3.89. Let S={1,2,…,n} and suppose that A and B are, independently, equally likely to be any of the 2n subsets (including the null set and S itself) of S.

a. Show that

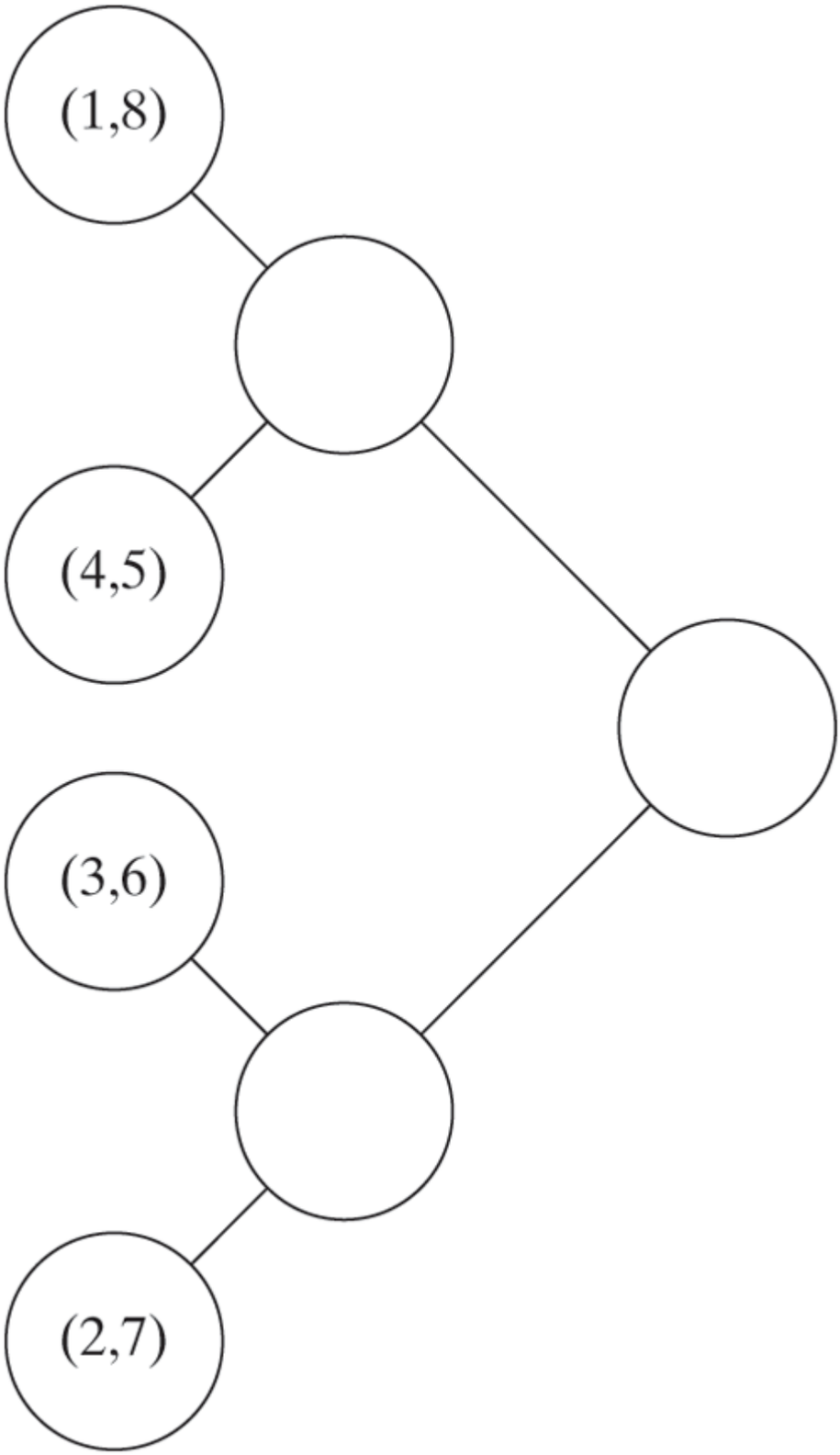
$P\{A \subset B\} = (34)^n$

Hint: Let N(B) denote the number of elements in B. Use
$P\{A \subset B\} = \sum_{i=0}^{n} P\{A \subset B | N(B) = i\} P\{N(B) = i\}$

Show that $P\{AB = \emptyset\} = (34)^n$.

3.90. Consider an eight team tournament with the format given in **Figure 3.6**. If the probability that team i beats team j if they play is $\frac{i}{i+j}$, find the probability that team 1 wins the tournament.

Figure 3.6



3.91. Consider**Example 2a**, but now suppose that when the key is in a certain pocket, there is a 10 percent chance that a search of that pocket will not find the key. Let R and L be, respectively, the events that the key is in the right-hand pocket of the jacket and that it is in the left-hand pocket. Also, let SR be the event that a search of the right-hand jacket pocket will be successful in finding the key, and let UL be the event that a search of the left-hand jacket pocket will be unsuccessful and, thus, not find the key. Find P(SR|UL), the conditional probability that a search of the right-hand pocket will find the key given that a search of the left-hand pocket did not, by

a. using the identity
   $P(SR|UL)=P(SRUL)P(UL)$

   determining P(SRUL) by conditioning on whether or not the key is in the right-hand pocket, and determining P(UL) by conditioning on whether or not the key is in the left-hand pocket;
b. using the identity
   $P(SR|UL)=P(SR|RUL)P(R|UL)+ P(SR|RcUL)P(Rc|UL)$

3.92. In**Example 5e**, what is the conditional probability that the ith coin was selected given that the first n trials all result in heads?

3.93. In Laplace's rule of succession (**Example 5e**), are the outcomes of the successive flips independent? Explain.

3.94. A person tried by a 3-judge panel is declared guilty if at least 2 judges cast votes of guilty. Suppose that when the defendant is in fact guilty, each judge will independently vote guilty with probability .7, whereas when the defendant is in fact innocent, this probability drops to .2. If 70 percent of defendants are guilty, compute the conditional probability that judge number 3 votes guilty given that

a. judges 1 and 2 vote guilty;
b. judges 1 and 2 cast 1 guilty and 1 not guilty vote;
c. judges 1 and 2 both cast not guilty votes.

Let Ei,i=1,2,3 denote the event that judge i casts a guilty vote. Are these events independent? Are they conditionally independent? Explain.

3.95. Each of n workers is independently qualified to do an incoming job with probability p. If none of them is qualified then the job is rejected; otherwise the job is assigned to a randomly chosen one of the qualified workers. Find the probability that worker 1 is assigned to the first incoming job. **Hint:** Condition on whether or not at least one worker is qualified.

3.96. Suppose in the preceding problem that n=2 and that worker i is qualified with probability pi, i=1,2.

a. Find the probability that worker 1 is assigned to the first incoming job.
b. Given that worker 1 is assigned to the first job, find the conditional probability that worker 2 was qualified for that job.

3.97. Each member of a population of size n is, independently of other members, female with probability p or male with probability 1−p. Two individuals of the same sex will, independently of other pairs, be friends with probability α; whereas two individuals of opposite sex will be friends with probability β. Let Ak,r be the event that persons k and r are friends.

a. Find P(A1,2).
b. Are A1,2 and A1,3 independent.
c. Are A1,2 and A1,3 conditionally independent given the sex of person 1.
d. Find P(A1,2A1,3).

Theoretical Exercises

3.1. Show that if $P(A)>0$, then
$P(AB|A) \geq P(AB|A \cup B)$

3.2. Let $A \subset B$. Express the following probabilities as simply as possible:
$P(A|B)$, $P(A|Bc)$, $P(B|A)$, $P(B|Ac)$

3.3. Consider a school community of m families, with n_i of them having i children, $i=1,\ldots,k, \sum_{i=1}^{k} n_i = m$. Consider the following two methods for choosing a child:

A. Choose one of the m families at random and then randomly choose a child from that family.
B. Choose one of the $\sum_{i=1}^{k} i n_i$ children at random.

Show that method 1 is more likely than method 2 to result in the choice of a firstborn child.
Hint: In solving this problem, you will need to show that
$\sum_{i=1}^{k} i n_i \sum_{j=1}^{k} n_j j \geq \sum_{i=1}^{k} n_i \sum_{j=1}^{k} n_j$

To do so, multiply the sums and show that for all pairs i, j, the coefficient of the term $n_i n_j$ is greater in the expression on the left than in the one on the right.
3.4. A ball is in any one of n boxes and is in the ith box with probability p_i. If the ball is in box i, a search of that box will uncover it with probability α_i. Show that the conditional probability that the ball is in box j, given that a search of box i did not uncover it, is
$p_j 1-\alpha_i p_i \text{if } j \neq i \quad (1-\alpha_i) p_i 1-\alpha_i p_i \text{if } j=i$

3.5.

a. Prove that if E and F are mutually exclusive, then
$P(E|E \cup F) = P(E) P(E) + P(F)$

b. Prove that if $E_i, i \geq 1$ are mutually exclusive, then
$P(E_j | \cup_{i=1}^{\infty} E_i) = P(E_j) \sum_{i=1}^{\infty} P(E_i)$

3.6. Prove that if $E_1, E_2, \ldots, E_n$ are independent events, then
$P(E_1 \cup E_2 \cup \cdots \cup E_n) = 1 - \prod_{i=1}^{n} [1 - P(E_i)]$

3.7.

a. An urn contains n white and m black balls. The balls are withdrawn one at a time until only those of the same color are left. Show that with probability $n/(n+m)$, they are all white.
Hint: Imagine that the experiment continues until all the balls are removed, and consider the last ball withdrawn.
b. A pond contains 3 distinct species of fish, which we will call the Red, Blue, and Green fish. There are r Red, b Blue, and g Green fish. Suppose that the fish are removed from the pond in a random order. (That is, each selection is equally likely to be any of the remaining fish.) What is the probability that the Red fish are the first species to become extinct in the pond?
Hint: Write $P\{R\} = P\{RBG\} + P\{RGB\}$, and compute the probabilities on the right by first conditioning on the last species to be removed.

3.8. Let A,B, and C be events relating to the experiment of rolling a pair of dice.

a. If
$P(A|C) > P(B|C)$ and $P(A|Cc) > P(B|Cc)$

either prove that $P(A) > P(B)$ or give a counterexample by defining events A,B, and C for which

that relationship is not true.

b. If
$P(A|C)>P(A|Cc)$ and $P(B|C)>P(B|Cc)$

either prove that $P(AB|C)>P(AB|Cc)$ or give a counterexample by defining events A,B, and C for which that relationship is not true.

Hint: Let C be the event that the sum of a pair of dice is 10; let A be the event that the first die lands on 6; let B be the event that the second die lands on 6.

3.9. Consider two independent tosses of a fair coin. Let A be the event that the first toss results in heads, let B be the event that the second toss results in heads, and let C be the event that in both tosses the coin lands on the same side. Show that the events A, B, and C are pairwise independent —that is, A and B are independent, A and C are independent, and B and C are independent—but not independent.

3.10. Two percent of women age 45 who participate in routine screening have breast cancer. Ninety percent of those with breast cancer have positive mammographies. Eight percent of the women who do not have breast cancer will also have positive mammographies. Given that a woman has a positive mammography, what is the probability she has breast cancer?

3.11. In each of n independent tosses of a coin, the coin lands on heads with probability p. How large need n be so that the probability of obtaining at least one head is at least 12?

3.12. Show that $0\leq ai\leq 1, i=1,2,\ldots$, then
$\sum i=1\infty[ai\prod j=1i-1(1-aj)]+\prod i=1\infty(1-ai)=1$

Hint: Suppose that an infinite number of coins are to be flipped. Let ai be the probability that the ith coin lands on heads, and consider when the first head occurs.

3.13. The probability of getting a head on a single toss of a coin is p. Suppose that A starts and continues to flip the coin until a tail shows up, at which point B starts flipping. Then B continues to flip until a tail comes up, at which point A takes over, and so on. Let Pn,m denote the probability that A accumulates a total of n heads before B accumulates m. Show that
$Pn,m=pPn-1,m+(1-p)(1-Pm,n)$

* 3.14. Suppose that you are gambling against an infinitely rich adversary and at each stage you either win or lose 1 unit with respective probabilities p and 1−p. Show that the probability that you eventually go broke is
1 if $p\leq 123pt(q/p)i$ if $p>12$

where $q=1-p$ and where i is your initial fortune.

3.15. Independent trials that result in a success with probability p are successively performed until a total of r successes is obtained. Show that the probability that exactly n trials are required is
$(n-1r-1)pr(1-p)n-r$

Use this result to solve the problem of the points (**Example 4j**.

Hint: In order for it to take n trials to obtain r successes, how many successes must occur in the first n−1 trials?

3.16. Independent trials that result in a success with probability p and a failure with probability 1−p are called *Bernoulli trials*. Let Pn denote the probability that n Bernoulli trials result in an even number of successes (0 being considered an even number). Show that
$Pn=p(1-Pn-1)+(1-p)Pn-1 \quad n\geq 1$

and use this formula to prove (by induction) that
$Pn=1+(1-2p)n2$

3.17. Suppose that n independent trials are performed, with trial i being a success with probability

1/(2i+1). Let Pn denote the probability that the total number of successes that result is an odd number.

a. Find Pn for n=1,2,3,4,5.
b. Conjecture a general formula for Pn.
c. Derive a formula for Pn in terms of Pn−1.
d. Verify that your conjecture in part (b) satisfies the recursive formula in part (c). Because the recursive formula has a unique solution, this then proves that your conjecture is correct.

3.18. Let Qn denote the probability that no run of 3 consecutive heads appears in n tosses of a fair coin. Show that

$$Q_n=\frac{1}{2}Q_{n-1}+\frac{1}{4}Q_{n-2}+\frac{1}{8}Q_{n-3} \quad Q_0=Q_1=Q_2=1$$

Find Q8.

Hint: Condition on the first tail.

3.19. Consider the gambler's ruin problem, with the exception that A and B agree to play no more than n games. Let Pn,i denote the probability that A winds up with all the money when A starts with i and B starts with N−i. Derive an equation for Pn,i in terms of Pn−1, i+1 and Pn−1, i−1, and compute P7,3, N=5.

3.20. Consider two urns, each containing both white and black balls. The probabilities of drawing white balls from the first and second urns are, respectively, p and p′. Balls are sequentially selected with replacement as follows: With probability α, a ball is initially chosen from the first urn, and with probability 1−α, it is chosen from the second urn. The subsequent selections are then made according to the rule that whenever a white ball is drawn (and replaced), the next ball is drawn from the same urn, but when a black ball is drawn, the next ball is taken from the other urn. Let αn denote the probability that the nth ball is chosen from the first urn. Show that

$$\alpha_{n+1}=\alpha_n(p+p'-1)+1-p' \quad n\geq 1$$

and use this formula to prove that

$$\alpha_n=\frac{1-p'}{2-p-p'}+\left(\alpha-\frac{1-p'}{2-p-p'}\right)\times(p+p'-1)^{n-1}$$

Let Pn denote the probability that the nth ball selected is white. Find Pn. Also, compute $\lim_{n\to\infty}\alpha_n$ and $\lim_{n\to\infty}P_n$.

3.21. *The Ballot Problem*. In an election, candidate A receives n votes and candidate B receives m votes, where n>m. Assuming that all of the (n+m)!/n! m! orderings of the votes are equally likely, let Pn,m denote the probability that A is always ahead in the counting of the votes.

a. Compute P2,1,P3,1,P3,2,P4,1,P4,2,P4,3.
b. Find Pn,1,Pn,2.
c. On the basis of your results in parts (a) and (b), conjecture the value of Pn,m.
d. Derive a recursion for Pn,m in terms of Pn−1,m and Pn,m−1 by conditioning on who receives the last vote.
e. Use part (d) to verify your conjecture in part (c) by an induction proof on n+ m.

3.22. As a simplified model for weather forecasting, suppose that the weather (either wet or dry) tomorrow will be the same as the weather today with probability p. Show that the weather is dry on January 1, then Pn, the probability that it will be dry n days later, satisfies

$$P_n=(2p-1)P_{n-1}+(1-p) \quad n\geq 1 \quad P_0=1$$

Prove that

$$P_n=\frac{1}{2}+\frac{1}{2}(2p-1)^n \quad n\geq 0$$

3.23. A bag contains a white and b black balls. Balls are chosen from the bag according to the following method:

A. A ball is chosen at random and is discarded.

B. A second ball is then chosen. If its color is different from that of the preceding ball, it is replaced in the bag and the process is repeated from the beginning. If its color is the same, it is discarded and we start from step 2.

In other words, balls are sampled and discarded until a change in color occurs, at which point the last ball is returned to the urn and the process starts anew. Let Pa,b denote the probability that the last ball in the bag is white. Prove that
Pa,b=12

Hint: Use induction on k≡a+ b.
3.24. A round-robin tournament of n contestants is a tournament in which each of the (n2) pairs of contestants play each other exactly once, with the outcome of any play being that one of the contestants wins and the other loses. For a fixed integer k, k<n, a question of interest is whether it is possible that the tournament outcome is such that for every set of k players, there is a player who beat each member of that set. Show that if
(nk)[1−(12)k]n−k< 1

then such an outcome is possible.
Hint: Suppose that the results of the games are independent and that each game is equally likely to be won by either contestant. Number the (nk) sets of k contestants, and let Bi denote the event that no contestant beat all of the k players in the ith set. Then use Boole's inequality to bound P(∪iBi).
3.25. Prove directly that
P(E|F)=P(E|FG)P(G|F)+P(E|FGc)P(Gc|F)

3.26. Prove the equivalence of **Equations (5.11)** and (**5.12**).
3.27. Extend the definition of conditional independence to more than 2 events.
3.28. Prove or give a counterexample. If E1 and E2 are independent, then they are conditionally independent given F.
3.29. In Laplace's rule of succession (**Example 5e**), show that if the first n flips all result in heads, then the conditional probability that the next m flips also result in all heads is approximately (n+1)/(n+m−1) when k is large.
3.30. In Laplace's rule of succession (**Example 5e**), suppose that the first n flips resulted in r heads and n−r tails. Show that the probability that the (n+1) flip turns up heads is (r+1)/(n+2). To do so, you will have to prove and use the identity
∫01yn(1−y)mdy=n!m!(n+m+1)!

Hint: To prove the identity, let C(n,m)=∫01yn(1−y)mdy. Integrating by parts yields
C(n,m)=mn+1C(n+1,m−1)

Starting with C(n,0)=1/(n+1), prove the identity by induction on m.
3.31. Suppose that a nonmathematical, but philosophically minded, friend of yours claims that Laplace's rule of succession must be incorrect because it can lead to ridiculous conclusions. "For instance," says he, "the rule states that if a boy is 10 years old, having lived 10 years, the boy has probability 1112 of living another year. On the other hand, if the boy has an 80-year-old grandfather, then, by Laplace's rule, the grandfather has probability 8182 of surviving another year. However, this is ridiculous. Clearly, the boy is more likely to survive an additional year than the grandfather is." How would you answer your friend?

Self-Test Problems and Exercises

3.1. In a game of bridge, West has no aces. What is the probability of his partner's having (a) no aces? (b) 2 or more aces? (c) What would the probabilities be if West had exactly 1 ace?
3.2. The probability that a new car battery functions for more than 10,000 miles is .8, the probability that it functions for more than 20,000 miles is .4, and the probability that it functions for more than 30,000 miles is .1. If a new car battery is still working after 10,000 miles, what is the probability that

a. its total life will exceed 20,000 miles?
b. its additional life will exceed 20,000 miles?

3.3. How can 20 balls, 10 white and 10 black, be put into two urns so as to maximize the probability of drawing a white ball if an urn is selected at random and a ball is drawn at random from it?
3.4. Urn A contains 2 white balls and 1 black ball, whereas urn B contains 1 white ball and 5 black balls. A ball is drawn at random from urn A and placed in urn B. A ball is then drawn from urn B. It happens to be white. What is the probability that the ball transferred was white?
3.5. An urn has r red and w white balls that are randomly removed one at a time. Let Ri be the event that the ith ball removed is red. Find

a. P(Ri)
b. P(R5|R3)
c. P(R3|R5)

3.6. An urn contains b black balls and r red balls. One of the balls is drawn at random, but when it is put back in the urn, c additional balls of the same color are put in with it. Now, suppose that we draw another ball. Show that the probability that the first ball was black, given that the second ball drawn was red, is b/(b+r+c).
3.7. A friend randomly chooses two cards, without replacement, from an ordinary deck of 52 playing cards. In each of the following situations, determine the conditional probability that both cards are aces.

a. You ask your friend if one of the cards is the ace of spades, and your friend answers in the affirmative.
b. You ask your friend if the first card selected is an ace, and your friend answers in the affirmative.
c. You ask your friend if the second card selected is an ace, and your friend answers in the affirmative.
d. You ask your friend if either of the cards selected is an ace, and your friend answers in the affirmative.

3.8. Show that
$P(H|E)P(G|E)=P(H)P(G)P(E|H)P(E|G)$

Suppose that, before new evidence is observed, the hypothesis H is three times as likely to be true as is the hypothesis G. If the new evidence is twice as likely when G is true than it is when H is true, which hypothesis is more likely after the evidence has been observed?
3.9. You ask your neighbor to water a sickly plant while you are on vacation. Without water, it will die with probability .8; with water, it will die with probability .15. You are 90 percent certain that your neighbor will remember to water the plant.

a. What is the probability that the plant will be alive when you return?
b. If the plant is dead upon your return, what is the probability that your neighbor forgot to water it?

3.10. Six balls are to be randomly chosen from an urn containing 8 red, 10 green, and 12 blue balls.

a. What is the probability at least one red ball is chosen?

b. Given that no red balls are chosen, what is the conditional probability that there are exactly 2 green balls among the 6 chosen?

3.11. A type C battery is in working condition with probability .7, whereas a type D battery is in working condition with probability .4. A battery is randomly chosen from a bin consisting of 8 type C and 6 type D batteries.

a. What is the probability that the battery works?
b. Given that the battery does not work, what is the conditional probability that it was a type C battery?

3.12. Maria will take two books with her on a trip. Suppose that the probability that she will like book 1 is .6, the probability that she will like book 2 is .5, and the probability that she will like both books is .4. Find the conditional probability that she will like book 2 given that she did not like book 1.

3.13. Balls are randomly removed from an urn that initially contains 20 red and 10 blue balls.

a. What is the probability that all of the red balls are removed before all of the blue ones have been removed? Now suppose that the urn initially contains 20 red, 10 blue, and 8 green balls.
b. Now what is the probability that all of the red balls are removed before all of the blue ones have been removed?
c. What is the probability that the colors are depleted in the order blue, red, green?
d. What is the probability that the group of blue balls is the first of the three groups to be removed?

3.14. A coin having probability .8 of landing on heads is flipped. A observes the result—either heads or tails—and rushes off to tell B. However, with probability .4, A will have forgotten the result by the time he reaches B. If A has forgotten, then, rather than admitting this to B, he is equally likely to tell B that the coin landed on heads or that it landed tails. (If he does remember, then he tells B the correct result.)

a. What is the probability that B is told that the coin landed on heads?
b. What is the probability that B is told the correct result?
c. Given that B is told that the coin landed on heads, what is the probability that it did in fact land on heads?

3.15. In a certain species of rats, black dominates over brown. Suppose that a black rat with two black parents has a brown sibling.

a. What is the probability that this rat is a pure black rat (as opposed to being a hybrid with one black and one brown gene)?
b. Suppose that when the black rat is mated with a brown rat, all 5 of their offspring are black. Now what is the probability that the rat is a pure black rat?

3.16.

a. In **Problem 3.70b**, find the probability that a current flows from A to B, by conditioning on whether relay 1 closes.
b. Find the conditional probability that relay 3 is closed given that a current flows from A to B.

3.17. For the k-out-of-n system described in **Problem 3.71**, assume that each component independently works with probability 12. Find the conditional probability that component 1 is working, given that the system works, when

a. k=1,n=2;
b. k=2,n=3.

3.18. Mr. Jones has devised a gambling system for winning at roulette. When he bets, he bets on red and places a bet only when the 10 previous spins of the roulette have landed on a black number. He reasons that his chance of winning is quite large because the probability of 11 consecutive spins

resulting in black is quite small. What do you think of this system?

3.19. Three players simultaneously toss coins. The coin tossed by A(B)[C] turns up heads with probability P1(P2)[P3]. If one person gets an outcome different from those of the other two, then he is the odd man out. If there is no odd man out, the players flip again and continue to do so until they get an odd man out. What is the probability that A will be the odd man?

3.20. Suppose that there are n possible outcomes of a trial, with outcome i resulting with probability pi,i=1,…,n,∑i=1npi=1. If two independent trials are observed, what is the probability that the result of the second trial is larger than that of the first?

3.21. If A flips n+1 and B flips n fair coins, show that the probability that A gets more heads than B is 12

Hint: Condition on which player has more heads after each has flipped n coins. (There are three possibilities.)

3.22. Prove or give counterexamples to the following statements:

a. If E is independent of F and E is independent of G, then E is independent of F∪G.
b. If E is independent of F, and E is independent of G, and FG= Ø , then E is independent of F∪G.
c. If E is independent of F, and F is independent of G, and E is independent of *FG*, then G is independent of *EF*.

3.23. Let A and B be events having positive probability. State whether each of the following statements is (i) necessarily true, (ii) necessarily false, or (iii) possibly true.

a. If A and B are mutually exclusive, then they are independent.
b. If A and B are independent, then they are mutually exclusive.
c. P(A)=P(B)=.6, and A and B are mutually exclusive.
d. P(A)=P(B)=.6, and A and B are independent.

3.24. Rank the following from most likely to least likely to occur:

A. A fair coin lands on heads.
B. Three independent trials, each of which is a success with probability .8, all result in successes.
C. Seven independent trials, each of which is a success with probability .9, all result in successes.

3.25. Two local factories, A and B, produce radios. Each radio produced at factory A is defective with probability .05, whereas each one produced at factory B is defective with probability .01. Suppose you purchase two radios that were produced at the same factory, which is equally likely to have been either factory A or factory B. If the first radio that you check is defective, what is the conditional probability that the other one is also defective?

3.26. Show that if P(A|B)=1, then P(Bc|Ac)=1.

3.27. An urn initially contains 1 red and 1 blue ball. At each stage, a ball is randomly withdrawn and replaced by two other balls of the same color. (For instance, if the red ball is initially chosen, then there would be 2 red and 1 blue balls in the urn when the next selection occurs.) Show by mathematical induction that the probability that there are exactly i red balls in the urn after n stages have been completed is 1n+1,1≤i≤n+ 1.

3.28. A total of 2n cards, of which 2 are aces, are to be randomly divided among two players, with each player receiving n cards. Each player is then to declare, in sequence, whether he or she has received any aces. What is the conditional probability that the second player has no aces, given that the first player declares in the affirmative, when (a) n=2? (b) n=10? (c) n=100? To what does the probability converge as n goes to infinity? Why?

3.29. There are n distinct types of coupons, and each coupon obtained is, independently of prior types collected, of type i with probability pi, ∑i=1npi=1.

a. If n coupons are collected, what is the probability that one of each type is obtained?
b. Now suppose that p1=p2=…=pn=1/n. Let Ei be the event that there are no type i coupons among the n collected. Apply the inclusion–exclusion identity for the probability of the union of events to P(∪iEi) to prove the identity
n!=∑k=0n(−1)k(nk)(n−k)n